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Diseases of the Kidney and Upper Urinary Tract in Children

Todd Alexander | Christoph Licht | William E. Smoyer | Norman D. Rosenblum

CHAPTER OUTLINE

THE SPECTRUM OF KIDNEY AND URINARY TRACT DISORDERS IN CHILDREN

The spectrum of chronic pediatric kidney disorders is distinct from that observed in adults (Fig. 72.1). Congenital abnormalities of the kidneys and/or lower urinary tract are most frequent, together accounting for just less than 50% of all kidney–urinary tract disorders. Glomerular disorders constitute the next largest group, constituting approximately 25% of all disorders. Cystic disorders (autosomal dominant and recessive forms) are then most frequent followed by systemic disorders, such as vasculitis, and metabolic disorders. The prevalence of end-stage renal disease (ESRD) among children younger than 15 years in Europe is currently 34 per million age-related population (pmarp) or 5.4 per million population (pmp).[1] The annual incidence is 6.5 pmarp or 1.0 pmp. When the pediatric population is defined as those up to age 19 years, the prevalence of ESRD in children and adolescents is 80 to 90 pmp and the annual incidence is 15 pmp. This is only 10% of the incidence in young adults and fewer than 2% of that observed in the older population. Because of the preponderance of males among those with certain urinary tract abnormalities (e.g., urethral valves), renal replacement therapy (RRT) occurs almost 50% more often in males than in females. ESRD incidence is highest in adolescence (8 pmarp), lowest in middle childhood (4.6 pmarp), and intermediate in children younger than 5 years of age (6.7 pmarp). Because of the predominantly genetic origin of pediatric nephropathies, chronic kidney disease (CKD) in children is frequently associated with a variety of extrarenal abnormalities. Impaired neurodevelopment and sensory dysfunctions are among the most severe general disabilities interfering with psychosocial adjustment and integration in children with CKD.

EVALUATION OF KIDNEY FUNCTION IN CHILDREN

During the period from birth to adulthood, body weight increases 20-fold, body length threefold to fourfold, and body surface area (BSA), the closest correlate of basal metabolic rate, eightfold. By convention, glomerular filtration rate (GFR) is normalized to the BSA of an average-sized adult—that is, 1.73 m^2. Although nephron formation is complete by the 34th gestational week, nephrons continue to grow in size and functional capacity after birth. In early postnatal life, nephron size and capacity increase not only in absolute terms but also relative to BSA, which results in a significant physiologic gain in normalized global renal function. The average GFR in the neonate is 20 to 30 mL/min/1.73 m^2 and increases rapidly in the first few months of life. From 18 months of age, the absolute increase in GFR precisely matches the growth in BSA, which results in a constant normal range of BSA-indexed GFR throughout childhood and adolescence. After 2 years of age, the conventional CKD staging system, which assigns stages of CKD based on estimated GFR (eGFR) in multiples of 15 and 30 mL/min/1.73 m^2, can be used.

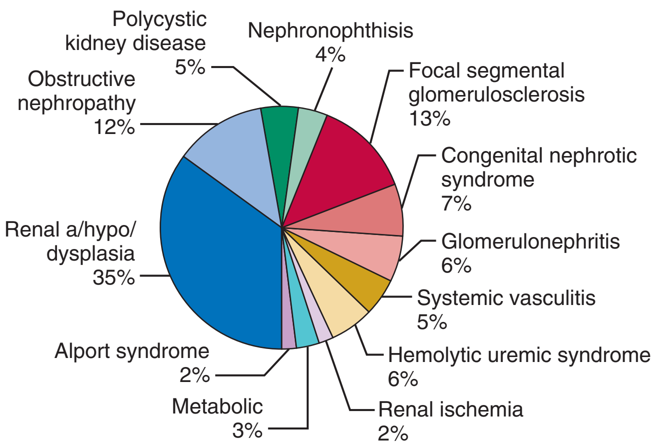

Fig. 72.1 Spectrum of underlying renal diagnoses in 1367 children with end-stage kidney disease. (From the International Pediatric Peritoneal Dialysis Network Registry.)

The direct measurement of GFR is a challenging task in children. The most commonly used measurement in children is endogenous creatinine clearance, which requires that the child be able to provide urine samples during a defined period. Creatinine clearance accurately reflects a reasonable estimate of GFR in patients with normal or mildly impaired renal function. As GFR declines, tubular creatinine secretion increases, which results in an overestimation of true GFR. In clinical practice, prediction equations that estimate GFR from one or more serum markers are commonly used. Although many such equations exist, the most commonly used equation uses creatinine as a marker. Because creatinine generation, and consequently steady-state serum creatinine levels, depends on muscle mass, which in turn is dependent on age and gender, this equation estimates GFR from the ratio of height to serum creatinine concentration multiplied by a constant, K.[2] In the original "Schwartz" equation, K was age and gender specific. In a more recent equation, Schwartz and colleagues use a uniform K value for both genders.[3] This equation is valid in children from 1 to 16 years of age:

$$\text{eGFR} = \text{Height (cm)} \times 0.413/\text{serum creatinine (mg/dL)}$$

Most healthy children excrete small amounts of protein in their urine. Physiologic proteinuria varies with the age and size of the child. When corrected for BSA, the upper normal limit of protein excretion is 300 mg/m^2/day at age 1 month in full-term infants, 250 mg at 1 year, 200 mg at 10 years, and 150 mg in late adolescence. Urinary protein excretion exceeding 1.0 g/m^2/day is strongly indicative of inherent kidney pathophysiology. The upper limit of normal for the urine protein-to-creatinine ratio (as grams of protein/gram of creatinine) is 0.5 at age 6 months to 2 years and 0.2 in older children and adolescents.[4] A urine protein-to-creatinine ratio above 3.0 is consistent with nephrotic-range proteinuria, thought to reflect glomerular disease, and is often associated with systemic manifestations, such as salt and fluid retention and edema. Estimates of the prevalence of isolated asymptomatic proteinuria in children range between 0.6% and 6%. Orthostatic proteinuria accounts for up to 60% of all cases of asymptomatic proteinuria reported in children, with an even higher incidence in adolescents. Children with orthostatic proteinuria usually excrete less than 1 g of protein/day (urine protein-to-creatinine ratio <1.0). Although patients with orthostatic proteinuria have an excellent prognosis, the long-term prognosis for children with isolated fixed proteinuria remains unknown.

CONGENITAL ANOMALIES OF THE KIDNEY AND URINARY TRACT

Developmental disorders of the kidney and urinary tract comprise a spectrum of malformations ranging from complete absence of kidney tissue to minor structural abnormalities. Together, these malformations are classified within the rubric Congenital Anomalies of the Kidney and Urinary Tract (CAKUT). Kidney–urinary tract structural abnormalities are an important cause of human disease; they are the most common cause of all birth defects, comprising 23% of all such defects[5] and cause 30% to 50% of all cases of ESRD in children.[6] Within the adult population, CAKUT is the cause of 2.2% of all ESRDs. Adult patients with CAKUT differ from adults with other causes of CKD in their requirement for RRT at a mean age of 30 years compared with 61 years.[7] This section is focused on the classification, epidemiology, pathogenesis, and clinical management of kidney malformations.

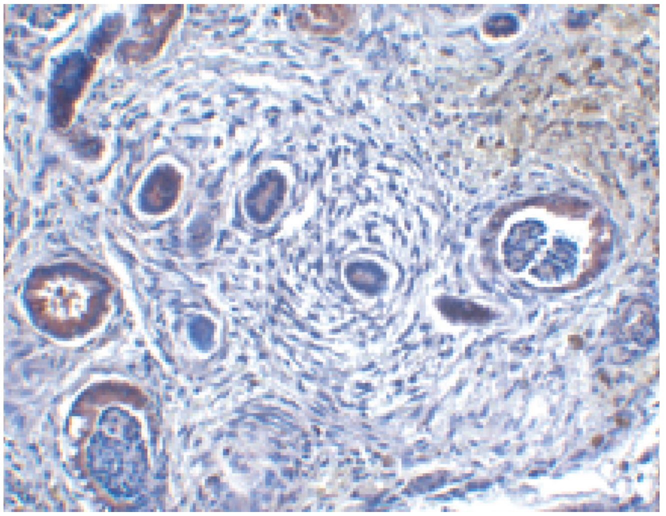

Fig. 72.2 Histologic section of renal dysplastic tissue. Dysplastic renal tissue demonstrates a paucity of glomerular and tubular elements, disorganization of tissue elements, an abundance of stroma, and thickened and dilated renal tubules.

CLINICAL CLASSIFICATION OF KIDNEY MALFORMATION

As a group, renal–urinary tract malformations have been grouped together under the rubric CAKUT. In support of this rubric: (1) multiple structures within one or both kidney–urinary tract units may be affected within any given affected individual, (2) mutation in a particular gene is associated with different urinary tract anomalies in different affected individuals, and (3) mutations in different genes give rise to similar renal and lower urinary tract phenotypes. A classification of kidney and urinary tract malformations within the CAKUT rubric is as follows:

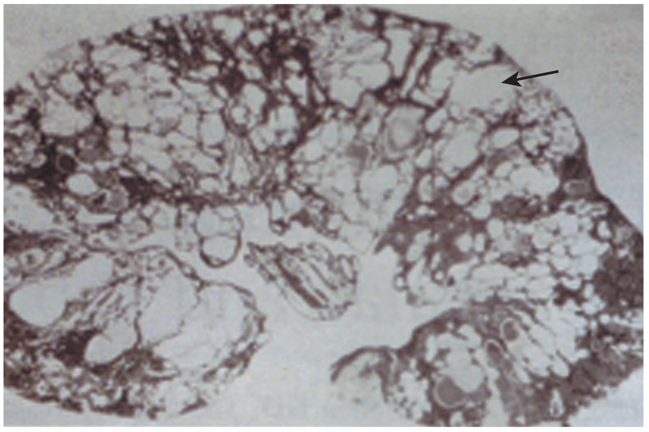

Fig. 72.3 Histologic section of a multicystic dysplastic kidney. The kidney consists in large part of multiple polymorphic cysts *(arrow)*. Normal renal tissue elements cannot be identified.

- Aplasia (agenesis), defined as congenital absence of kidney tissue.
- Simple hypoplasia, defined as renal length less than 2 standard deviations (SDs) below the mean for age, a reduced number of nephrons and normal renal architecture.
- Dysplasia ± cysts, defined as malformation of tissue elements.
- Isolated dilatation of the renal pelvis ± ureters (collecting system).
- Anomalies of position including the ectopic and fused (horseshoe) kidney.

Simple renal hypoplasia is defined by the presence of a small kidney with a decrease in the number of nephrons and normal renal architecture. Viewed through the lens of embryology, normal tissue architecture indicates that patterning of renal tissue elements is normal.

Renal dysplasia is defined as an abnormality of tissue patterning (Fig. 72.2). Renal dysplasia is a polymorphic disorder. At the level of gross anatomy, the dysplastic kidney varies in size from one extreme to the other compared with the mean size for age. However, most dysplastic kidneys are small for age. Dysplastic kidneys also vary in the presence of epithelial cysts, the number of cysts, and cyst size. The multicystic dysplastic kidney (MCDK) is an extreme form of renal dysplasia in which large polymorphic cysts dominate kidney structure (Fig. 72.3). At the level of histopathology, the dysplastic kidney (Fig. 72.2) is characterized by several primary features: (1) abnormal differentiation of mesenchymal and epithelial elements, (2) decreased nephron number, (3) loss of corticomedullary differentiation, and (4) metaplastic transformation of mesenchyme to cartilage and bone. Renal tubular dysgenesis represents a particular form of renal dysplasia and is characterized by the absence or poor development of proximal tubules and is accompanied by thickening of the renal arterial vasculature from the arcuate to the afferent arteries.[8]

An ectopic kidney is classified by its position and its relationship with an independent kidney unit (Fig. 72.4). Simple (nonfused) ectopy refers to a kidney that lies on the correct side of the body but lies in an abnormal position. Kidneys that cross the midline are referred to as crossed renal ectopy. Crossed renal ectopy can occur with and without fusion to

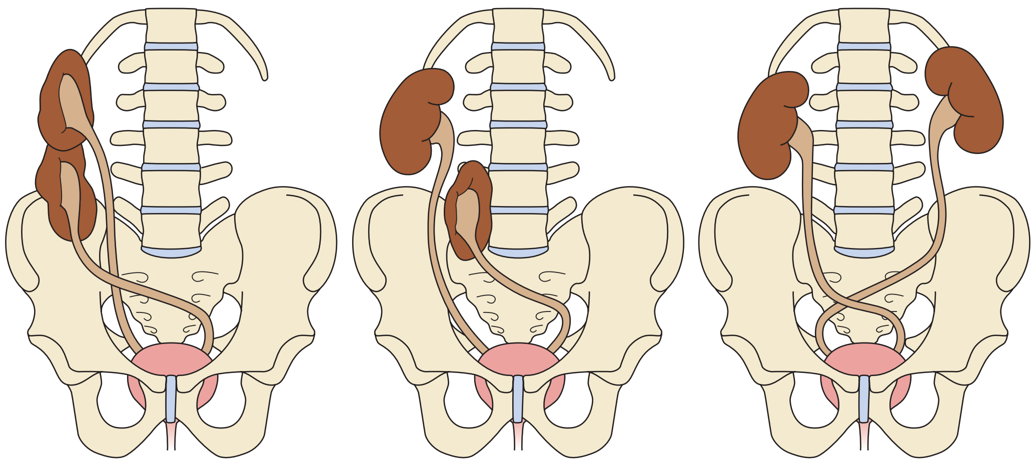

Fig. 72.4 Crossed fused ectopia. Renal ectopia is classified as simple (nonfused), fused, and bilateral. A fused kidney migrates across the midline and fuses to the lower pole of the normally positioned contralateral kidney. A simple crossed (nonfused) kidney migrates across the midline, does not fuse with the normally positioned contralateral kidney, and is usually positioned at the rim of the pelvis. In bilateral ectopia, both kidneys are ectopic and cross the midline with their native ureters, which are inserted normally into the bladder.

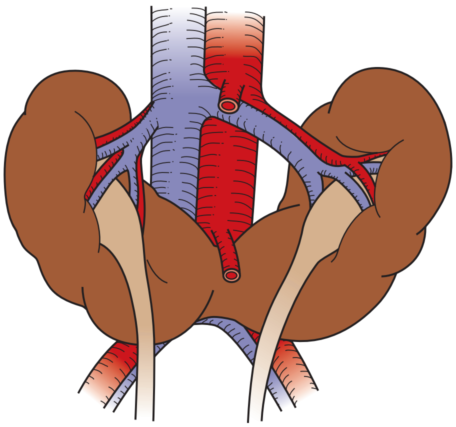

Fig. 72.5 Horseshoe kidney. A horseshoe kidney with fusion of the lower poles. Note that the renal pelvis is positioned anteriorly and the ureter traverses the anterior aspect and the fused lower pole of each kidney.

the contralateral kidney. Ectopic kidneys that do not ascend above the pelvic brim are commonly called pelvic kidneys. Renal fusion occurs when a portion of one kidney is fused to the other. The most common fusion anomaly is the horseshoe kidney, which involves abnormal migration of both kidneys (ectopy), resulting in fusion (Fig. 72.5). This differs from crossed fused renal ectopy, which usually involves abnormal movement of only one kidney across the midline with fusion of the contralateral noncrossing kidney.

EPIDEMIOLOGY OF CONGENITAL ANOMALIES OF THE KIDNEY AND URINARY TRACT

CAKUT is the most frequent malformation detected in utero. Using fetal ultrasound, the incidence of renal and urinary tract malformations is 0.3 to 1.6 per 1000 live-born and stillborn infants.[9] Lower urinary tract abnormalities can be identified in approximately 50% of affected patients, consistent with the common origin of the kidney and ureter from the mesonephric duct. These anomalies include vesicoureteral reflux ([VUR] 25%), ureteropelvic junction obstruction (11%), and ureterovesical junction obstruction (11%).[10] Renal malformations, other than mild antenatal pelviectasis, occur in association with nonrenal malformations in about 30% of cases.[9]

The reported incidence of particular types of renal–urinary tract malformation is no doubt dependent on the ascertainment method. Most incidence rates are not based on population-based studies during pregnancy. Rather, they are based on autopsy series or studies in selected live-born infants. Incidence rates based on methods other than population-based ascertainment may underestimate the true incidence because fetal loss may not be accounted for and CAKUT can be clinically silent in the surviving fetus.

Complete or partial duplication of the renal collecting system is the most common congenital anomaly of the urinary tract.[11] Autopsy studies report an estimated incidence of 0.8% to 5%.[12] A similar rate was reported in a study of 13,705 fetuses with antenatal ultrasounds performed in a tertiary center in Turkey.[13] However, a study that screened 132,686 Taiwanese schoolchildren (6 to 15 years of age) found a lower incidence of 1 in 5000 children.[14] Bilateral renal agenesis occurs in 1:3000 to 10,000 births. Males are affected more often than females. Unilateral renal agenesis has been reported with a prevalence of 1:1000 autopsies. The incidence of unilateral dysplasia is 1 in 3000 to 5000 births (1:3640 for the MCDK) compared with 1 in 10,000 for bilateral dysplasia.[15] The male-to-female ratio for bilateral and unilateral renal dysplasia is 1.32:1 and 1.92:1, respectively.[16] The incidence of renal ectopia is 1 in 1000 autopsies, but the clinical recognition is estimated to be only 1 in 10,000 patients.[17] Males and females are equally affected. Renal ectopia is bilateral in 10% of cases; when unilateral, there is a slight predilection for the left side. The incidence of fusion anomalies is estimated to be about 1 in 600 infants.[18] The most common fusion anomaly is the horseshoe kidney, which occurs with fusion of one pole of each kidney. The reported incidence of horseshoe kidney based upon data from birth defect registries varies from 0.4 to 1.6 to 10,000 live births.[18,19]

PATHOGENESIS OF CONGENITAL ANOMALIES OF THE KIDNEY AND URINARY TRACT

MECHANISMS OF INHERITANCE

The genetics of CAKUT are complex. In the majority of affected patients, congenital renal malformations occur as sporadic events. In approximately 30% of affected individuals, these malformations occur as part of a multiorgan genetic syndrome. Renal–urinary tract malformations and extrarenal malformations can go unrecognized unless a careful phenotypic examination is performed. Over 200 distinct monogenic syndromes feature some type of kidney and urinary tract malformation. Approximately 50 genes have been identified as mutant in human CAKUT, 25 with autosomal dominant inheritance and 15 with autosomal recessive inheritance (Table 72.1).[20-48] Incomplete penetrance with variable expressivity is frequent in affected families. That is, within any particular family in which affected members carry the same mutation, the renal phenotype can vary from agenesis to dysplasia to isolated abnormalities of the collecting system (e.g., hydronephrosis).[20]

The majority of CAKUT occurs without a clear Mendelian pattern of inheritance. In probands with bilateral renal agenesis or bilateral renal dysgenesis and without evidence of a genetic syndrome or a family history, 9% of first-degree relatives were shown by ultrasound to have some type of malformation in the kidney and/or lower urinary tract.[21] A study of CAKUT in asymptomatic first-degree relatives of patients with CAKUT in Turkey revealed a positive family history of CAKUT in 23% and ultrasound evidence of CAKUT also in 23% of individuals.[22] A study of 100 patients with renal hypodysplasia and renal insufficiency demonstrated a gene mutation in 16% of affected individuals.[23] Another study of 204 unrelated patients with CAKUT using target exome sequencing of 330 genes implicated in CAKUT in humans and other species demonstrated pathogenic mutations in 31/204 (17.6%) patients. The 24 pathogenic mutations were identified in seven genes associated with CAKUT: HNF1b,

Table 72.1 Human Gene Mutations Associated With Syndromic Congenital Anomalies of the Kidney and Urinary Tract

Primary Disease	Gene	Kidney Phenotype	References
Alagille syndrome	*JAGGED1* *NOTCH2*	Cystic dysplasia	1170,1171
Apert syndrome	*FGFR2*	Hydronephrosis	1172
Atypical DiGeorge syndrome	*CRKL*	Hypodysplasia, vesicoureteral reflux (VUR), obstructive uropathy	1173
Beckwith–Wiedemann syndrome	*$p57^{KIP2}$*	Medullary dysplasia	1174
Branchio-oto-renal (BOR) syndrome	*EYA1, SIX1, SIX5*	Unilateral or bilateral agenesis/ dysplasia, hypoplasia, collecting system anomalies	1175
Campomelic dysplasia	*SOX9*	Dysplasia, hydronephrosis	1176
Duane radial ray (Okihiro) syndrome	*SALL4*	UNL agenesis, VUR, malrotation, cross-fused ectopia, pelviectasis	1177
Fraser syndrome	*FRAS1* *FREM1* *FREM2*	Agenesis, dysplasia	1178,1179
Isolated renal hypoplasia	*BMP4* *RET*	Hypoplasia, VUR	1180,1181
Hypoparathyroidism, sensorineural deafness, and renal anomalies syndrome	*GATA3*	Dysplasia	1182
Kabuki syndrome	*KDM6A, KMT2D*	VUR, hypodysplasia, ectopia	1183
Kallmann syndrome	*KAL1, FGFR1, PROK2, PROK2R*	Agenesis	1184
Leukemia, acute B cell	*PBX1*	Hypodysplasia, VUR, ectopia, horseshoe kidney	1185
Mammary–ulnar syndrome	*TBX3*	Dysplasia	1186
Okihiro syndrome	*SALL4*	Ectopia, dysplasia	1176
Pallister–Hall syndrome	*GLI3*	Agenesis, dysplasia, hydronephrosis	1187,1188
Renal-coloboma syndrome	*PAX2*	Hypoplasia, vesicoureteral reflux	1189
Renal tubular dysgenesis	*RAS components*	Tubular dysplasia	1190
Renal cysts and diabetes syndrome	*HNF1b*	Dysplasia, hypoplasia	1191
Rubinstein–Taybi syndrome	*CREBBP*	Agenesis, hypoplasia	1192
Simpson–Golabi–Behmel syndrome	*GPC3*	Medullary dysplasia	1193
Smith–Lemli–Opitz syndrome	*7-hydroxy-cholesterol reductase*	Agenesis, dysplasia	1194
Townes–Brock syndrome	*SALL1*	Hypoplasia, dysplasia, VUR	1195
Ulnar–mammary syndrome	*TBX3*	Hypoplasia	1186
Zellweger syndrome	*PEX1*	VUR, cystic dysplasia	1196

PAX2, EYA1, ANOS1, GATA3, CHD7, and KIF14. Sixteen of these 31 patients had extrarenal as well as renal abnormalities.[24] This frequency of pathogenic mutations is considerably higher than that found in a study of 453 patients in which exome sequencing of >200 candidate genes identified pathogenic mutations in 6/453 patients.[25] The difference in results likely is explained by the phenotypes studied. Although Nicolaou et al.[25] studied a wide variety of CAKUT phenotypes including unilateral malformations and lower urinary tract obstruction, Heidet et al. only included bilateral CAKUT and/or familial and/or syndromic forms of CAKUT.[24] Taken together, these results indicate that currently as many as 18% of CAKUT cases can be explained by an established monogenic cause.[20,26] The majority of mutations identified are in *TCF2* (hepatocyte nuclear factor-1β [HNF-1β]; especially in patients with kidney cysts) and PAX2. Some of the mutations are de novo, explaining the sporadic appearance of CAKUT. Careful clinical analysis of patients with *TCF2* and *PAX2* mutations revealed the presence of extrarenal symptoms in only 50%, supporting previous reports that *TCF2* and *PAX2* mutations can be responsible for isolated renal tract anomalies or at least CAKUT malformations with minimal extrarenal features.[27,28] Exome sequencing studies also identify rare heterozygous missense gene variants, some of which are predicted by bioinformatic analyses to be pathogenic.[24,25] Yet, the lack of functional validation for many of these variants underlies a lack of certainty of how such information can be applied to diagnosis and counselling of patients.

Studies suggest that polymorphic variants in genes that control renal development and copy number variants (CNVs) contribute to the pathogenesis of CAKUT. Analysis of a cohort of 168 Caucasian newborns from Montreal suggests that single-nucleotide polymorphisms (SNPs) in *PAX2* are associated with newborn kidney size at the low end of the population spectrum.[29] Analysis of the *RET* gene in the aforementioned Montreal cohort demonstrated that a reduction in newborn kidney volume by 9.7% was significantly associated with an SNP (rs1800860) that encodes an A versus G at codon 1476.[30] Analyses in vitro indicate that the amount of messenger RNA (mRNA) generated from this allele is reduced consistent with decreased *RET* expression in affected individuals. Analysis of genomic CNVs has revealed that 10% to 15% of CAKUT cases are associated with such variants. For example, a study of 522 children with renal hypodysplasia revealed genomic abnormalities in 10.5% of cases.[31] In this and other similar studies, the abnormalities detected are predominantly localized to the renal cyst and diabetes syndrome locus on chromosome 17 or the DiGeorge syndrome region on chromosome 22.

MOLECULAR PATHOGENESIS

The morphologic and genetic events that control kidney development are detailed in Chapter 1. Briefly summarized, formation of the human kidney is initiated at 5 weeks' gestation in the human when the ureteric duct is induced to undergo lateral outgrowth from the wolffian duct and to invade the adjacent metanephric mesenchyme. The ureteric bud then undergoes repetitive branching events, so termed because each event consists of expansion of the advancing ureteric bud branch at its leading tip, division of the ampulla resulting in formation of new branches, and elongation of the newly formed branches. Beginning with the 10th–11th branch generation, the pattern of branching becomes terminally bifid. During branching morphogenesis, 65,000 collecting ducts are formed, as both cortical and medullary collecting ducts, a process that is essential to the function of the mature kidney. During the latter stages of kidney development, tubular segments formed from the first five generations of ureteric bud branching undergo remodeling to form the kidney pelvis and calyces.[32]

Genetic analyses in humans have identified mutant alleles associated with CAKUT. Studies in mice have identified more than 180 genes that are required during renal development.[20] Together, studies in humans and mice have provided complimentary information and critical insights into the likely functions of CAKUT-related genes during human renal morphogenesis. In this section, the functions of genes mutated in human CAKUT and for which a function has been elucidated are reviewed as a framework for understanding the molecular pathogenesis of CAKUT.

URETERIC BUDDING, *ROBO2*, AND *BMP4*

The protooncogene *RET,* a tyrosine kinase receptor, and its ligand, GDNF, regulate ureteric budding and renal branching morphogenesis. RET is expressed on the surface of ureteric cells.[33] GDNF is expressed by metanephric mesenchyme cells.[34] Homozygous deletion of either *Ret* or *Gdnf* in mice causes failure of ureteric outgrowth and renal agenesis. Patients with CAKUT have mutations in the RET/GDNF signaling pathway.[35–37] A study of 122 patients with CAKUT identified heterozygous deleterious sequence variants in *GDNF* or *RET* in 6/122 patients, 5%, whereas another group screened 749 families from all over the world and identified three families with heterozygous mutations in RET.[35] Similar findings have been reported in studies of fetuses with bilateral or unilateral renal agenesis.[36,37]

The site of ureteric bud outgrowth from the wolffian duct is normally invariant and the number of outgrowths is limited to one. Outgrowth of more than one ureteric bud can result in renal malformations including a double collecting system and duplication of the ureter. The position at which the ureteric bud arises from the wolffian duct relative to the metanephric mesenchyme is critical to the nature of the interactions between the ureteric bud and the metanephric mesenchyme. Ectopic positioning of the ureteric bud is associated with renal tissue malformation (dysplasia) due to abnormal ureteric bud–metanephric mesenchyme interactions and is also thought to contribute to the integrity of the ureterovesical junction. Mackie and Stephens postulated[38] that an abnormal position of the ureteral orifice in the bladder is associated with VUR in humans. Consistent with this hypothesis, mutations in *ROBO2* are associated with VUR in humans.[39] ROBO2, a cell surface receptor, is expressed in the nephrogenic mesenchyme.[40] Mice deficient in *Robo2* exhibit ectopic ureteric bud formation, multiple ureters, and hydroureter. Interestingly, the domain of *Gdnf* expression is expanded anteriorly in these mice, suggesting that loss of inhibition of *Gdnf* expression by ROBO2-dependent signaling expands the domain of *Gdnf* expression and results in ectopic ureter budding.[41] Members of the bone morphogenetic protein (BMP) family of secreted proteins also negatively regulate GDNF signaling. *Bmp4* is expressed in stromal cells immediately adjacent to the wolffian duct and the ureteric bud. Mice heterozygous for *Bmp4* exhibit ectopic or duplicated ureteric buds, suggesting that BMP4 suppresses ureteric induction by antagonizing the local effect of GDNF-RET signaling at the normal site of induction on the wolffian duct. Indeed, exogenous BMP4 has been shown to block GDNF-induced ureteric bud outgrowth from the wolffian duct in vitro.[42] Consistent with these observations, mutations in *BMP4* have been identified in humans with CAKUT phenotypes including renal dysgenesis and VUR.[43]

URETERIC BRANCHING, *PAX2*, AND *RET*

Branching of the ureteric bud is initiated immediately following invasion of the metanephric mesenchyme by the ureteric bud. The number of ureteric bud branches elaborated is considered to be a major determinant of final nephron number because each ureteric bud branch tip induces a discrete subset of metanephric mesenchyme cells to undergo nephrogenesis (see Chapter 1: Normal Renal Development). Regulation of ureteric branch number has been informed by complementary studies in humans and mice. These investigations have demonstrated an essential role for *PAX2*, a transcription factor of the paired-box family of homeotic genes that is expressed in the mesonephros and in the metanephros during renal development.

Mutations in *PAX2* cause renal coloboma syndrome (also named papillorenal syndrome), an autosomal dominant disorder characterized by the association of renal hypoplasia, vesicoureteric reflux, and optic nerve coloboma.[44] While the prevalence of the syndrome is unknown, approximately 100 affected families have been reported.[45] A wide range of renal malformations are observed in renal coloboma syndrome and include most frequently oligomeganephronic hypoplasia, renal dysplasia, and vesicoureteric reflux.[46] Ureteropelvic junction obstruction has also been described.[23,27,46] The ocular phenotype is extremely variable. The most common finding is an optic disc pit associated with vascular abnormalities and cilioretinal arteries, with mild visual impairment limited to blind spot enlargement.[47] In other cases, the only ocular anomaly is optic nerve dysplasia with an abnormal vessel pattern and no functional consequence.

The PAX2 gene is located on chromosome 10q24-25. It encodes a transcription factor that belongs to the paired-box family of homeotic genes. The vast majority of the mutations are located in exons 2 and 3 encoding the DNA-binding domain. In 1995, Sanyanusin and colleagues reported heterozygous mutations in two renal coloboma syndrome (RCS) families.[48] Since then, more than 30 mutations have been reported, most of them lying in the second and third exons that encode the paired domain that binds to DNA or in exons 7–9 that encode the transactivation domain.[45,49]

During renal development, *Pax2* is expressed in the wolffian duct, the ureteric bud, and the metanephric mesenchyme. Studies in the 1Neu mouse strain, which is characterized by a *Pax2* mutation, demonstrated decreased ureteric branching in association with decreased nephron number. Decreased ureteric branch number and nephron number are rescued by inhibition of apoptosis in the ureteric lineage.[50,51] Studies in normal term newborns suggest that loss of PAX2 function may also contribute to generating a lower number of nephrons within the range of nephron number (approximately 250,000–1,600,000) observed in humans.[52] Goodyer hypothesized that gene polymorphisms that generate loss of PAX2 function could contribute to mild reductions in nephron number and discovered that a *PAX2* haplotype ($PAX2^{AAA}$) is associated with an approximately 10% decrease in kidney volume in a cohort of newborn infants.[29]

Ureteric branching is also mediated in large part by the GDNF–RET signaling axis. During ureteric branching, RET is expressed on the surface of ureteric tip cells and controls the number and pattern of branches elaborated.[53] The observation that $Ret^{+/-}$ mice exhibit a 22% reduction in nephron number[54] suggested that allelic variants in *RET* associated with decreased RET function contribute to decreased nephron number in humans. Indeed, an *RET* allelic variant which generates a reduced amount of RET mRNA in vitro control with the predominant *RET* allele has been associated with an approximately 10% decrease in total kidney volume, used in this case as a quantitative trait and surrogate measure of nephron number, in a cohort of newborn infants.[30]

CONTROL OF GDNF EXPRESSION IN THE METANEPHRIC MESENCHYME: *SALL1*, *EYA1*, AND *SIX1*

As discussed earlier, GDNF expression by metanephric mesenchyme cells is critical to ureteric branching. In the metanephric mesenchyme, *Sall1, Eya1,* and *Six1* positively control *Gdnf* expression. *Sall1*, a member of the Spalt family of transcriptional factors,[55] is expressed in the metanephric mesenchyme prior to and during ureteric bud invasion. Mutational inactivation of *Sall1* in mice causes renal agenesis or severe dysgenesis and a marked decrease in GDNF expression.[56] Mutations in *SALL1* are associated with Townes–Brock syndrome, an autosomal dominant malformation syndrome characterized by imperforate anus, preaxial polydactyly and/or triphalangeal thumbs, external ear defects, sensorineural hearing loss (SNHL), and, less frequently, kidney, urogenital, and heart malformations.[57,58] *SALL1* mutations have also been identified in patients that lack extrarenal features of Townes–Brock syndrome.[23]

EYA1, a DNA binding transcription factor, is expressed in metanephric mesenchyme cells in the same spatial and temporal pattern as GDNF. EYA1 functions in a molecular complex with SIX1[59] to control expression of *Gdnf*.[60] Both EYA1 and SIX1 are also expressed in developing otic and branchial tissues.[61,62] Mice with EYA1 deficiency demonstrate renal agenesis and failure of GDNF expression.[61] Mutations in *EYA1* and *SIX1* occur in humans with branchio-oto-renal (BOR) syndrome.[59,63] Classic BOR syndrome, an autosomal dominant disorder, is defined by its major features, conductive and/or SNHL, branchial defects, ear pits, and renal anomalies in 95%, 49% to 69%, 83%, and 38% to 67% patients, respectively.[64,65] Renal malformations include unilateral or bilateral renal agenesis, hypodysplasia as well as malformation of the lower urinary tract including VUR, pyeloureteral obstruction, and ureteral duplication. Different renal malformations can be observed in the same family. Many patients have only one or two of these major BOR syndrome features; these findings in the absence of renal findings are termed BO syndrome. Mutations in *EYA1* have been identified, as well, in children with CAKUT who lack any of the extrarenal features of BOR or BO syndromes.[23]

BOR syndrome is transmitted as an autosomal dominant trait with incomplete penetrance and variable expressivity. A mutation in EYA1 has been identified in approximately 40% of patients with BOR syndrome.[65] Mutations are most commonly identified in a highly conserved region called the eyes absent homologous region encoded within exons 9–16. It is estimated that 5% may have a mutation in SIX1.[66] Molecular testing can confirm the diagnosis and provide genetic recurrence risk information to families. However, variability of the phenotype even with the same mutation does not permit accurate prediction of the disease severity. Within the same family, a given mutation may be associated with renal malformation in some individuals, but not in others.

HEDGEHOG SIGNALING, GLI3 REPRESSOR, AND CONGENITAL ANOMALIES OF THE KIDNEY AND URINARY TRACT

Studies in mice and humans have demonstrated a critical role for Hedgehog-dependent signaling during kidney development. Hedgehog ligands function within concentration gradients. Binding of Hedgehog ligand, including Sonic Hedgehog (SHH), to its cognate cell surface receptor stimulates nuclear translocation of full-length GLI transcriptional activators and inhibits proteolytic processing of full length GLI3 to a shorter transcriptional repressor.[67] Loss-of-function mutations in *SHH* have been identified in patients with holoprosencephaly and renal hypoplasia or urogenital malformations.[68] Renal malformations have also been reported in patients exhibiting deletions of chromosome 7q, encompassing the *SHH* gene locus.[69–71] Posttranslational modification of Hedgehog ligand is essential for regulated diffusion of Hedgehog proteins and signaling.[72] A Hedgehog loss-of-function phenotype, including renal agenesis and hypodysplasia, is observed in Smith–Lemli–Opitz syndrome, which is caused by heterozygous mutations in the *DHCR7* gene, encoding sterol delta-7-reductase, that is required in mammalian sterol biosynthesis to convert 7-dehydrocholesterol into cholesterol.[73] Reduced Hedgehog modification with cholesterol may be pathogenic in Smith–Lemli–Opitz syndrome.

Pallister–Hall syndrome is an autosomal dominant multiorgan disorder characterized by multiple renal abnormalities including agenesis or dysplasia, hypoplasia, and hydronephrosis.[74–76] Affected individuals carry frameshift/nonsense and splicing mutations exclusively in the second third of the *GLI3* gene. These mutations are predicted to generate a truncated protein similar in size to GLI3 repressor.[77,78] The pathogenic role of GLI3 repressor was demonstrated in mice with homozygous *Shh* deficiency in which the levels of GLI3 in kidney tissue are elevated. Remarkably, renal dysgenesis in the absence of *Shh* is completely rescued by homozygous inactivation of *Gli3*.[79] Analysis of mice with constitutive expression of GLI3 repressor demonstrated aberrant common nephric duct patterning and ureteric stalk outgrowth associated with ureteric duplication and blind ending ureter.[80] Murine deficiency of *Smo*, which encodes a cell surface protein required for Hedgehog signaling, targeted to the metanephric mesenchyme lineage causes hydronephrosis, dyskinesia of the ureter, and absent expression of cell surface markers that characterized pacemaker cells in the renal pelvis and upper ureter. Homozygous deficiency of GLI3 in these mutant mice rescues these abnormalities.[81]

THE MEDULLA AND GLYPICAN-3

Between the 22nd and 34th week of human fetal gestation, the peripheral (cortical) and central (medullary) domains of the developing kidney are established. Glypican-3 (GPC3), a glycosyl-phosphatidylinositol–linked cell surface heparan sulfate proteoglycan, and the gene that is mutated in Simpson–Golabi–Behmel syndrome, is required for normal patterning of the medulla.[82,83] Medullary dysplasia in mice deficient in *Gpc3* is associated with increased ureteric branching and cell proliferation,[82] and subsequent destruction of medullary collecting ducts due to apoptosis.[83] The defect is thought to be caused by insensitivity to inhibitors of branching morphogenesis including BMPs and enhanced sensitivity to the stimulatory effect of other factors including FGF7. The finding of medullary dysplasia in humans and mice mutant for $p57^{KIP2}$, an inhibitor of cell proliferation, further suggests that regulation of cell proliferation and apoptosis is important in the renal medulla[84] and suggests a further link to human renal medullary dysplasia as observed in some patients with Beckwith–Wiedemann syndrome and *p57*KIP2 mutations.[85]

TCF2, MODY5, AND SPORADIC FORMS OF CONGENITAL ANOMALIES OF THE KIDNEY AND URINARY TRACT

TCF2 encodes HNF-1β, a homeotic DNA-binding transcription factor, which is required during the development of the pancreas, kidneys, liver, and intestine. During mouse kidney development, *TCF2* is expressed in the wolffian duct, ureteric bud, comma and S-shaped bodies, and proximal and distal tubules.[86] Epithelial-specific inactivation of *Tcf2* in the kidney causes renal cystic disease and downregulation of several genes, inactivation of which cause renal cystic disease.[87] In humans, heterozygous mutations in *TCF2* cause maturity-onset diabetes type 5 (MODY5).[86,88] More than 50 different *TCF2* mutations have been reported, most of which are located in the first four exons that encode the DNA-binding domain. In more than one-third of the cases, the gene is entirely deleted.[28,89] Diabetes mellitus is present in approximately 60% of all the cases reported, usually occurs before 25 years of age, and is often associated with pancreatic atrophy.[89–91] In the presence of renal cysts, the term, renal cysts and diabetes syndrome, is used.

While *TCF2* mutations were first identified in MODY5, such mutations are observed more frequently in fetuses with bilateral hyperechoic kidneys. An analysis of 62 newborns or fetuses with antenatally diagnosed bilateral hyperechogenic kidneys revealed that large genomic *TCF2* deletions were the most frequent cause (29%).[92] A complementary retrospective analysis of 377 patients with a *TCF2* mutation demonstrated that isolated hyperechogenic kidney with normal or slightly enlarged size is the most frequent phenotype observed before birth in these patients.[93] After birth, *TCF2* mutations most commonly manifest as bilateral small cortical cysts.[28] However, other manifestations of CAKUT are observed in these patients and include renal hypoplasia and dysplasia, MCDK, renal agenesis, horseshoe kidney, pelviureteric junction obstruction, and clubbing and tiny diverticula of the calyces.[90,93–96] Patients with hyperechoic kidneys as a fetus and bilateral cortical cysts after birth and found to harbor *TCF2* mutations rarely manifest extrarenal anomalies during infancy and childhood.

TUBULAR DYSGENESIS AND MUTATIONS OF RAS SYSTEM ELEMENTS

Renal tubular dysgenesis is a severe perinatal disorder characterized by absence or paucity of differentiated proximal tubules, early severe oligohydramnios, and perinatal death. The latter is usually due to pulmonary hypoplasia and skull ossification defects.[8,97,98] This condition has also been described in clinical conditions associated with renal ischemia including the twin–twin transfusion syndrome, major cardiac malformations, severe liver diseases, and fetal or infantile renal artery stenosis[99] and in fetuses that are exposed in utero to angiotensin-converting enzyme (ACE) inhibitors or angiotensin II (Ang II) receptor antagonists.[100] All of these pathophysiologic states are postulated to lead to chronic hypoperfusion of the fetal kidneys with upregulation of the renin–angiotensin system.[101] Mutations in the genes that encode components of the renin–angiotensin system have been identified in some families.[102] Mutations in the ACE and renin genes have been identified in 65.5% and 20% of cases, respectively. Mutations in angiotensinogen (AGT) and in the angiotensinogen type I receptor occur much less frequently.[103]

CHD1L, CHD7, AND CHARGE SYNDROME

Chromodomain helicase DNA binding protein 1-like protein, CHD1L, is a member of the SNF2 family of helicase-related ATP-hydrolyzing proteins. Like its related family member, CHD7, CHD1L contains a helicase-like region. CHD1L is expressed in early ureteric bud and comma- and S-shaped structures during human kidney development.[104] Heterozygous missense mutations in CHD1L were identified in three patients among 85 with CAKUT.[104] Mutations in CHD7 have also been identified in CHARGE syndrome with 20% of these patients affected with CAKUT phenotypes including horseshoe kidneys, renal agenesis, renal dysplasia, VUR, ureterovesical junction obstruction, posterior urethral valves, and renal cysts.[105,106]

DSTYK AND CONGENITAL ANOMALIES OF THE KIDNEY AND URINARY TRACT

DSTYK is a dual serine–threonine and tyrosine protein kinase that is coexpressed with fibroblast growth factor receptors in the developing mouse and human kidney in both metanephric mesenchyme and ureteric bud cells. Heterozygous mutations in DSTYK have been identified in a small number of individuals (7/311) with CAKUT.[107]

COPY NUMBER VARIANTS, CONGENITAL ANOMALIES OF THE KIDNEY AND URINARY TRACT, AND NEUROPSYCHIATRIC DISORDERS

CNVs are stretches of DNA that are larger than 1 kb in length. Rare CNVs have been implicated in neuropsychiatric and craniofacial syndromes, and in syndromes with CAKUT.[31,108] Sanna-Cherchi et al. examined the frequency of rare CNVs in individuals with CAKUT and identified such variants in 10% of affected individuals compared with 0.2% of population controls.[31] Deletions at the *HNF1* locus (chromosome 17q12) and the locus for DiGeorge syndrome (chromosome 22q11) were most frequently identified, suggesting these are "hot spots" for copy number variation. Interestingly, 90% of the CNVs associated with congenital renal malformations were previously reported to predispose to developmental delay or neuropsychiatric disease, suggesting that there are shared pathways implicated in renal and central nervous system (CNS) development. Similarly, Handrigan et al. demonstrated that CNVs at chromosome 16q24.2 are associated with autism spectrum disorder, intellectual disability, and congenital renal malformations.[108]

THE ENVIRONMENT IN UTERO AND CONGENITAL ANOMALIES OF THE KIDNEY AND URINARY TRACT

There is an increasing body of evidence, derived from human epidemiological studies and animal models, demonstrating an important role for the intrauterine environment and fetal programming in the pathogenesis of renal hypoplasia and predisposition to later kidney disease (reviewed in[109–118]; Table 72.2) Low birth weight or intrauterine growth retardation (IUGR) is generally considered to be due to a suboptimal in utero environment. Here, the fetal kidney is particularly susceptible, leading to reduced nephron number. In humans, IUGR is most often due to uteroplacental insufficiency and maternal undernutrition. Modeling of these disorders in animals causes a significant reduction in nephron endowment.[119] Maternal dietary protein restriction results in decreased nephron number, reduced renal function, and hypertension in a variety of species including rodents and sheep.[120–123] Although the underlying mechanisms are not well defined, there is some evidence suggesting that the maternal diet programs the expression of critical genes required for embryonic kidney development, cell survival,

Table 72.2 Factors Influencing in Utero Environment That Are Associated With Renal Hypoplasia

Fetal Exposure to	Renal Phenotype	References
Uteroplacental insufficiency	Hypoplasia	110
Vitamin A deficiency	Hypoplasia, hydronephrosis/ureter	111
Low protein diet	Hypoplasia	112,113
Hyperglycemia	Agenesis, ectopic/horseshoe, cystic/dysplasia, hypoplasia, hydronephrosis/ureter	114
Cocaine	Agenesis, hypoplasia, hydronephrosis/ureter	115
Alcohol	Agenesis, ectopia/horseshoe, cystic dysplasia, hypoplasia, hydronephrosis/ureter	116,117
Angiotensin-converting enzyme inhibitors and angiotensin II receptor blockers	Renal dysgenesis	118

and renal function.[121,124,125] It is likely that such changes in gene expression are controlled by epigenetic regulation.

Studies in mutant mice have identified a requirement for vitamin A and its retinoic acid receptor signaling effectors in RET expression and ureteric branching.[126] Vitamin A deficiency during pregnancy causes renal hypoplasia and decreased glomerular number and *Ret* expression in the rodent fetus.[127] Interestingly, a single dose of retinoic acid administered at midgestation is able to normalize kidney size and nephron number in rat offspring exposed to maternal protein restriction, raising the possibility of preventative approaches in humans.[128]

Maternal diabetes and in utero exposure to drugs and alcohol are associated with renal hypoplasia in the absence of reduced birth weight. In animal models, offspring of hyperglycemic or diabetic mothers demonstrate a significant nephron deficit.[129] Exposure of human fetuses to ACE inhibitors during the first trimester is associated with an increased risk of renal dysplasia as well as cardiovascular and CNS malformations.[130] Human infants exposed to cocaine in utero have an increased risk of renal tract anomalies.[131] Similarly, infants with fetal alcohol syndrome have a higher incidence of CAKUT.[132] The pathogenic mechanisms underlying fetal alcohol exposure are beginning to be elucidated. Ethyl alcohol administration to pregnant rats at E13.5 and E14.5, stages which correspond to human E5–E7, caused a modest decrease in nephron number but no deleterious effect on fetal kidney weight or maternal weight gain. The expression of *Gdnf* and *Wnt11,* both of which are required during ureteric branching, was reduced, consistent with a decrease in nephrogenesis. Analysis of blood pressure at 6 months of age in pups exposed to ethyl alcohol in utero demonstrated a 10% increase in both males and females compared with age-matched sham-treated controls although GFR and renal vascular resistance differed between sexes.[133]

FUNCTIONAL CONSEQUENCES OF CONGENITAL ANOMALIES OF THE KIDNEY AND URINARY TRACT

A growing body of evidence indicates that the number of functional nephrons formed by 32 to 34 weeks' gestation has important implications for short- and long-term renal function. Infants with simple renal hypoplasia or a moderate to severe degree of hypodysplasia exhibit renal insufficiency. A subtle deficiency in nephron number has been associated with adult-onset hypertension.[134] The association of decreased nephron number with hypertension is consistent with the "Barker hypothesis," which is based on epidemiologic evidence showing a correlation between birth weight and the incidence of cardiovascular diseases and proposes that adult-onset diseases such as hypertension have a fetal origin.[135,136] The human kidney does not exhibit a capacity to accelerate the rate of nephron formation in children born prematurely or to extend the period of nephrogenesis beyond the equivalent of 34 weeks' gestation.[137] Thus the integrity of nephron formation in utero is absolutely critical to postnatal life.

Growth of renal tubules and expansion of glomerular cross-sectional area in utero and after birth are critical to renal functional capacity. The general observation that tubule number, cross-sectional area, and cellular maturation are abnormal in renal dysgenesis is consistent with clinical observations that infants with moderate to severe renal hypoplasia or dysplasia demonstrate a limitation of GFR and tubular function. Here, the developmental maturation of renal structures is discussed in the context of their functions. Illustrative examples are provided for how abnormal differentiation, growth, and maturation in the malformed kidney can limit these functions. Existing knowledge has been generated, for the most part, from the study of maturing preterm and term animals. By contrast, very little data have been derived from the study of animals, such as mutant mice, with renal malformation. Thus interpretation of physiologic abnormalities in humans and experimental animals with renal malformation is largely an extrapolation from developmental studies in experimental animals with normal kidney development.

A major increase in glomerular basement surface area after birth contributes to the maturational increase in GFR during infancy, childhood, and adolescence.[138] Low GFR at birth limits excretion of free water, increasing the susceptibility of newborns to hyponatremia in association with a hypotonic fluid challenge.[139] Maximum urine concentrations achieved by preterm and term infants following fluid restriction are 600 and 800 mOsm/kg, respectively.[140] An adult level of urine concentrating capacity is attained by 6 to 12 months of age.[141] Establishment of cortical and medullary domains during the 22nd to 24th week of gestation is critical to urine concentration.[137] During embryogenesis, the renal cortex grows along a circumferential axis with a tenfold increase in its volume. The renal medulla expands 4.5-fold in thickness along a longitudinal axis, an increase that is mainly due to elongation of the outer medullary collecting ducts.[142] Longitudinal growth of the medulla contributes to lengthening of the loops of Henle such that they reach the inner renal medulla in the mature kidney. Elongation of the loops of Henle is important to the urine concentration mechanism because the magnitude of sodium and urea transport is greatest in longer loops, which generate steeper medullary tonicity gradients. The responsiveness of collecting duct cells to vasopressin is limited in newborns. This is thought to be due to high intrarenal levels of prostaglandins, which antagonize vasopressin.[143]

Maturation of sodium transport in the fetus and infant is dependent on growth and differentiation within the proximal tubule, loop of Henle, and distal tubule. Normal newborn infants are limited in their capacity to respond to sodium restriction by reducing urinary sodium excretion. Interruption of tubule generation, differentiation, and growth, hallmark features of renal dysplasia, contribute to an exaggerated limitation in the capacity to absorb sodium in affected infants and children. The proximal tubule exhibits dramatic growth and maturation during renal development. The epithelium matures from a columnar to a cuboidal epithelium, microvilli are elaborated on the apical and basal domains, and expression of the Na^+,K^+-ATPase and the type 3 sodium-hydrogen exchanger (NHE3) increase.[144–146] At birth, proximal tubule length is heterogeneous between the inner and outer cortex[138]; by 1 month of life proximal tubule length becomes uniform and tubule length and diameter have increased.[144] Maturation of tubule length is associated with the capacity to absorb sodium.[147]

The loop of Henle is also characterized by increased spatial expression of transporters (NKCC2, NHE3, ROMK, Na^+,K^+-ATPase) key to sodium transport.[148] Similarly, expression of

NCCT and epithelial sodium channel (ENaC) is low in the neonatal kidney and increases thereafter.[149]

Urinary potassium secretion is achieved predominantly by secretion of potassium in the cortical collecting duct via apical ROMK K^+ channels. In neonates, K^+ secretion is lower than in children due to the low secretory capacity of the cortical collecting duct. The postnatal increase in K^+ secretion is thought to be due to a developmental increase in the number of ROMK channels[150] as well as the BK potassium channel of the collecting duct.[151] Malformation of tubules in disorders such as renal dysplasia is commonly associated with limited K^+ secretion particularly during infancy.

CLINICAL PRESENTATION OF CONGENITAL ANOMALIES OF THE KIDNEY AND URINARY TRACT

CLINICAL PRESENTATION IN THE FETUS

The majority of renal malformations are diagnosed antenatally, largely because of the widespread use and sensitivity of fetal ultrasound. The fetal kidney can be visualized at 12 to 15 weeks of human gestation. A screening antenatal ultrasound is recommended between 16 and 20 weeks of gestation by which time renal anatomy can be imaged with considerable definition and anomalies can be detected with a sensitivity of approximately 80%.[152] Corticomedullary differentiation is distinct by 25 weeks of gestation and sometimes earlier. The fetal ureters are not normally detected by ultrasound. Visualization of ureters may be indicative of ureteric or bladder obstruction, or VUR. A urine-filled bladder is normally identified at 13 to 15 weeks' gestation.[153] The normal bladder wall is normally thin. If the bladder wall is thick, urethral obstruction such as posterior urethral valves in a male fetus may be present.

Development of the kidney in utero is commonly assessed using fetal renal length standardized for gestational age as a surrogate marker.[154] The volume of amniotic fluid is a surrogate measure of renal function. Fetal urine production begins at 9 weeks of gestation. By 20 weeks' gestation and thereafter, fetal urine is the primary source of amniotic fluid volume.[155] A decrease in amniotic fluid volume, termed oligohydramnios, at or beyond the 20th week of gestation is an excellent indicator of a critical defect in both kidneys, for example bilateral renal dysplasia (or a critical defect in one kidney where a solitary kidney exists), bilateral ureteral obstruction, or obstruction of the bladder outlet. Severe oligohydramnios in the second trimester can result in lung hypoplasia because an adequate amniotic fluid volume is critical for lung development.[156] In its most severe form, oligohydramnios results in Potter syndrome, which consists of a typical facies characterized by pseudoepicanthus, a recessed chin, posteriorly rotated and flattened ears, and flattened nose, as well as decreased fetal movement, clubfoot, hip dislocation and joint contractures, and pulmonary hypoplasia.

The composition of fetal urine is also used as a marker of kidney function. Urine levels of sodium and beta-2-microglobulin decrease with increasing gestational age and urine osmolality increases.[157,158] Impaired resorption occurs in fetuses with bilateral renal dysplasia or severe bilateral obstructive uropathy, resulting in abnormal high urine levels of sodium and beta-2-microglobulin and low urine osmolality.[159] In general, sodium and chloride concentration greater than 90 mEq/L (90 mmol/L) and urinary osmolality less than 210 mOsmol/kg·H_2O (210 mmol/kg·H_2O) are indicative of fetal renal tubular impairment and poor renal prognosis.[160] In addition, urinary beta-2-microglobulin levels greater than 6 mg/L are predictive of severe renal damage with a sensitivity and specificity of 80% and 71%, respectively.[161]

CLINICAL PRESENTATION OF SPECIFIC FORMS OF CONGENITAL ANOMALIES OF THE KIDNEY AND URINARY TRACT

RENAL AGENESIS

A diagnosis of unilateral renal agenesis depends on the certainty that a second kidney does not exist in the pelvis or some other ectopic location. Because absence of one kidney induces compensatory hypertrophy in the existing kidney, the presence of a large kidney on one side supports a diagnosis of unilateral renal agenesis. Unilateral renal agenesis is generally asymptomatic. A solitary kidney is most often detected either during routine antenatal ultrasonography or during the assessment of an accompanying urinary tract abnormality, the occurrence of which has been reported in 33% to 65% of cases.[162] VUR is the most commonly identified urological abnormality, occurring in approximately 37% of patients with unilateral renal agenesis and prompting investigation particularly in newborn infants. Other associated urological anomalies include obstruction of the ureteropelvic junction in 6% to 7%, and ureterovesical junction in 11% to 18% of patients.

RENAL DYSPLASIA

The dysplastic kidney is generally smaller than normal and is characterized with ultrasound by increased echogenicity, loss of corticomedullary differentiation, and cysts of varied size and location. Large cystic elements can contribute to large kidney size. The MCDK is an extreme example of a large dysplastic kidney (Fig. 72.2). Renal dysplasia may be unilateral or bilateral and may be discovered during routine antenatal screening or postnatally when renal ultrasonography is performed in a dysmorphic infant. Bilateral dysplasia is likely to be diagnosed earlier than unilateral dysplasia especially if oligohydramnios is present. Infants with bilateral dysplasia may demonstrate impaired renal function shortly after birth. Associated urinary tract abnormalities include hydronephrosis, a duplicated collecting system, megaureter, ureteral stenosis, and VUR.[160] Clinical presentation may be related to complications such as urinary tract infection (UTI) associated with these disorders.

The MCDK is identified by ultrasound as a large cystic nonreniform mass in the renal fossa and by palpation as a flank mass. The MCDK is nonfunctional and usually unilateral. If bilateral, it is fatal. In unilateral MCDK, associated contralateral abnormalities occur in 25% of cases and can include rotational or positional anomalies, renal hypoplasia, vesicoureteric efflux, and ureteropelvic junction obstruction.[15] While hypertension rarely occurs (0.01%–0.1% of cases), blood pressure should be monitored intermittently during the first few years of life. Wilms tumor and renal cell carcinoma

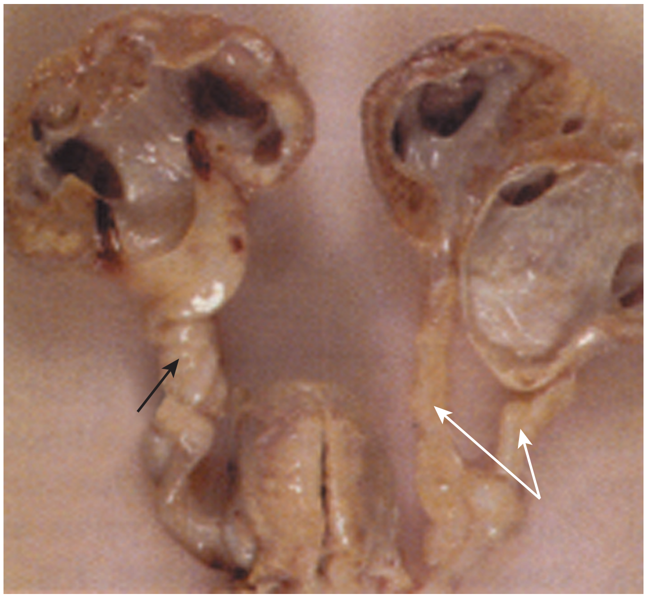

Fig. 72.6 Duplicated collecting system. Duplicated ureters *(white arrows)* are shown on the right. A single dilated ureter *(black arrow)* is shown on the left. Each ureter is dilated due to obstruction at the level of the bladder.

have also been described but the incidence of malignant complications is not significantly different from the general population.[163] The natural history of an MCDK is gradual reduction in size such that the kidney eventually cannot be detected using noninvasive imaging. At 2 years of age, an involution in size by ultrasound occurs in up to 60% of the affected kidneys.

DOUBLE COLLECTING SYSTEM

Complete or partial duplication of the renal collecting system is the most common congenital anomaly of the urinary tract (Fig. 72.6).[11] A double collecting system is thought to result from duplication of the ureteric bud with the superior bud associated with the upper renal pole and the inferior bud with the lower renal pole. In complete duplication, the kidney has two separate pelvicalyceal systems and two ureters. The ureter from the lower collecting system usually enters the bladder in the trigone, whereas the ureter from the upper collecting system can have a normal insertion in the trigone or be inserted ectopically in the bladder or elsewhere. In boys, insertion can occur in the posterior urethra, ejaculatory ducts, or epididymis and in girls into the vagina or uterus. Ectopic insertion of the ureter can result in obstruction or VUR. Depending upon the location of the ectopic insertion, incontinence also may be present. Partial duplication is more common than complete duplication. In these cases, the kidney has two separate pelvicalyceal systems with either a single ureter or two ureters that unite prior to insertion into the bladder.

RENAL ECTOPY

Renal ectopy (Fig. 72.5) results from disruption of the normal embryologic migration of the kidneys. Rapid caudal growth during embryogenesis results in migration of the developing kidney from the pelvis to the retroperitoneal renal fossa. With ascension comes a 90° rotation from a horizontal to a vertical position with the renal hilum finally directed medially. Migration and rotation are complete by 8 weeks of gestation. Simple congenital ectopy refers to a low-lying kidney that failed to ascend normally. It most commonly lies over the pelvic brim or in the pelvis and is termed a pelvic kidney. Less commonly, the kidney may lie on the contralateral side of the body, a state that is termed *crossed ectopy without fusion*. Affected individuals generally have no symptoms and the condition is identified during an imaging study performed for some other indication; however, some patients develop symptoms due to complications, such as infection, renal calculi, and urinary obstruction.[154] The ectopic kidney is often characterized by decreased function. In a case study of 82 cases of unilateral renal ectopy, decreased renal function was detected by ^{99}mTc-dimercaptosuccinic acid (DMSA) renal scan in 74 patients.[154] A high incidence of other urological abnormalities has been associated with renal ectopy. The most frequent of these is VUR, which occurs in 20% of crossed renal ectopy, 30% of simple renal ectopy, and 70% of bilateral simple renal ectopy.[154] Other associated urological abnormalities include contralateral renal dysplasia (4%), cryptorchidism (5%), and hypospadias (5%).[17]

RENAL FUSION

Renal fusion occurs when a portion of one kidney is fused to the other (Fig. 72.5). The most common fusion anomaly is the horseshoe kidney, which involves abnormal migration of both kidneys (ectopy). The horseshoe kidney differs from crossed fused renal ectopy, which usually involves abnormal movement of only one kidney across the midline with fusion of the contralateral noncrossing kidney. In more than 90% of horseshoe kidneys, fusion occurs at the lower poles; as a result, two separate excretory renal units and ureters are maintained. The isthmus (fused portion) may lie over the midline (symmetric horseshoe kidney) or lateral to the midline (asymmetric horseshoe kidney). Depending on the degree of fusion, the isthmus can be composed of renal parenchyma or a fibrous band. Fusion is thought to occur before the kidneys ascend between the fourth to ninth week of gestation from the pelvis to their normal dorsolumbar position. If large portions of the renal parenchyma fuse, the fusion anomaly loses its horseshoe appearance and appears as a flattened disc or lump kidney. Early fusion also causes abnormal rotation of the developing kidneys. As a result, the axis of each kidney is shifted so that the renal pelvis lies anteriorly and the ureters either traverse over the isthmus of the horseshoe kidney or the anterior surface of the fused kidney. Fusion anomalies seldom ascend to the dorsolumbar position of normal kidneys and are typically found in the pelvis or at the lower lumbar vertebral level (L4 or L5). The blood supply of the fused kidney is variable and may come from the iliac arteries, aorta, and at times, the hypogastric and middle sacral arteries.

The majority of patients with renal fusion are asymptomatic. Some, however, develop urinary tract obstruction which presents with loin pain, hematuria, and may be associated with UTIs due to urinary stasis or vesicoureteric reflux. Renal calculi may occur in up to 20% of cases.[164] Other associated urological anomalies include ureteral duplication, ectopic ureter, and retrocaval ureter. Investigations can include static imaging (renal ultrasound) and functional imaging such as a ^{99}mTc-DMSA scan and a voiding cystourethrogram (VCUG). Renal calculi are reported to occur in 20% of cases. Obstruction resulting in urinary stasis and complicating UTI have been thought to be the major contributing factors to stone formation. Patients with a horseshoe kidney appear to have an increased risk for Wilms tumor. This was illustrated in a retrospective review of 8617 patients from the National Wilms Tumor Study, between 1969 to 1998, that identified 41 patients with Wilms tumor in a horseshoe kidney.[165]

In crossed fused ectopy, the ectopic kidney and ureter cross the midline to fuse with the contralateral kidney but the ureter of the ectopic kidney maintains its normal insertion into the bladder. In most cases, the ectopic kidney is positioned inferiorly to the contralateral kidney. The contralateral kidney can either retain its normal dorsolumbar position or is positioned lower in the pelvis or lower lumbar vertebral level (L4 or L5). Most patients with crossed fused ectopy are asymptomatic and are detected coincidentally, often by antenatal ultrasonography. As is true in patients with horseshoe kidney, most patients have an excellent prognosis without need for intervention. In some cases, complications can occur, including obstructive uropathy due to extrinsic ureteric compression by aberrant blood vessels or ureteropelvic junction obstruction, renal calculi, urinary infection, and VUR.[166]

CLINICAL MANAGEMENT OF CONGENITAL ANOMALIES OF THE KIDNEY AND URINARY TRACT

Because CAKUT play a causative role in 30% to 50% of cases of end-stage renal disease (ESRD) in children,[167] it is important to diagnose and initiate therapy to minimize renal damage, prevent or delay the onset of ESRD, and provide supportive care to avoid complications of ESRD.

OVERALL APPROACH TO MANAGEMENT OF CONGENITAL ANOMALIES OF THE KIDNEY AND URINARY TRACT IN UTERO AND IN THE IMMEDIATE POSTNATAL PERIOD

Counseling of families during pregnancy is a key element in the management of CAKUT. Coordinated consultation among professionals in the disciplines of obstetrics, pediatric nephrology, pediatric urology, and neonatology is critical. Consistent and clear clinical information regarding diagnosis and prognosis should be provided during pregnancy and after birth. The level of certainty regarding the severity of the diagnosis and prognosis has a major impact on decision-making during pregnancy and in the immediate postnatal period. Intervention in utero has been designed to (1) reduce renal damage arising from urinary tract obstruction, and (2) rescue pulmonary development in the face of urinary tract obstruction and oligohydramnios. To date, little evidence exists that relief of urinary tract obstruction in utero prevents the development of associated renal dysplasia or renal scarring. By contrast, insertion of a bladder–amniotic cavity shunt in the fetus with obstruction below the bladder neck can rescue oligohydramnios and pulmonary hypoplasia.[166,168] Diagnostic and therapeutic management after birth should be anticipated via the coordinated actions of obstetricians, neonatologists, pediatric nephrologists, and pediatric urologists and should include an immediate assessment in the postnatal period of the need for specialized imaging, assessment of renal function, and management of nutrition and electrolytes.

After delivery, a detailed history and careful physical examination should be performed in all infants with an antenatally detected renal malformation. The examination should include the respiratory system to assess the presence of pulmonary insufficiency; the abdomen to detect the presence of a mass that could represent an enlarged kidney due to obstructive uropathy or MCDK or a palpable enlarged bladder, which could suggest posterior urethral valves; the ears, because outer ear abnormalities are associated with an increased risk of CAKUT; and the umbilicus, because a single umbilical artery is associated with an increased risk of CAKUT.

In newborns with bilateral renal malformation, a solitary malformed kidney, or a history of oligohydramnios, an abdominal ultrasound is recommended within the first 24 hours of life because an intervention such as decompression of the bladder with a transurethral catheter may be required. Newborn infants with unilateral involvement do not need immediate attention. In these infants, a renal ultrasound is generally performed after 48 hours of age and within the first week of life. Ultrasound examination before 48 hours of age may not detect collecting system dilatation because a newborn is relatively volume contracted during this period.[169] The serum creatinine estimates the extent of renal impairment and should be used when there is bilateral renal disease or an affected solitary kidney. The serum creatinine concentration at birth is similar to that in the mother (usually ≤1.0 mg/dL [88 μmol/L]). Thus serum creatinine should be measured after the first 24 hours of life. It declines to normal values [serum creatinine 0.3 to 0.5 mg/dL (27–44 μmol/L]) within approximately 1 week in term infants and 2 to 3 weeks in preterm infants.

MANAGEMENT OF SPECIFIC TYPES OF CONGENITAL ANOMALIES OF THE KIDNEY AND URINARY TRACT

RENAL DYSPLASIA

Renal dysplasia is frequently associated with collecting system anomalies, particularly VUR. Thus a VCUG should be considered in all patients with renal dysplasia, particularly after detection of hydronephrosis by ultrasound and/or occurrence of a UTI. The presence of VUR or some other collecting system abnormality in the normal contralateral kidney places children with unilateral renal dysplasia at increased risk of long-term sequelae of renal scarring from recurrent UTI. A DMSA radionuclide scan can provide further information on the differential function of each kidney, which may be useful in management decisions regarding surgical interventions. The prognosis of renal dysplasia depends on whether there is unilateral versus bilateral disease. In general, the long-term outcome of unilateral renal dysplasia is excellent, particularly if there is a normal contralateral kidney. Serial ultrasonography can assess compensatory renal growth of a normal contralateral kidney and any further change in the size of the abnormal kidney.

The natural history of an MCDK is gradual reduction in size such that the kidney eventually cannot be detected using noninvasive imaging. At 2 years of age, an involution in size by ultrasound occurs in up to 60% of the affected kidneys. Renal ultrasound is generally recommended at an interval of 3 months for the first year of life and then every 6 months up to involution of the mass, or at least up to 5 years.[170] Compensatory hypertrophy of the contralateral kidney is expected and should be monitored by renal ultrasound. Hypertension is unusual; blood pressure should be monitored intermittently during the first few years of life. Medical therapy is usually effective in treating hypertension in the small number of affected patients, but nephrectomy may be curative in resistant cases. Wilms tumor and renal cell carcinoma have also been described but the incidence of malignant complications is not significantly different from the general population.[163]

RENAL ECTOPY AND FUSION

Detection of an ectopic kidney provides a basis to determine renal function and any associated urinary tract anomalies. Reduced renal function in the ectopic kidney can be determined by radionuclide scan. An abdominal and pelvic ultrasound is indicated to determine the presence of collecting system abnormality. A VCUG should be undertaken, particularly if there is hydronephrosis and in the aftermath of a UTI to detect possible VUR or urinary tract obstruction. If VUR is detected, then prophylactic antibiotics should be considered, especially in patients who have a history of UTIs.

Further evaluation is based upon the results of a renal ultrasound, VCUG, and serum creatinine. No further evaluation is required in the patient with a normal appearing contralateral kidney and no evidence of hydronephrosis in the ectopic kidney. If the serum creatinine is elevated or if the contralateral kidney appears abnormal, a DMSA renal scan should be performed to assess differential renal function. A diuretic renogram with (^{99}mTc)-mercaptotriglycylglycine (MAG-3) or technetium ^{99}mTc-diethylenetriamine pentaacetic acid (DTPA) should be performed to detect obstruction if there is severe hydronephrosis and the VCUG is normal. If the hydronephrosis is mild or moderate and the VCUG is normal, then follow-up ultrasonography should be performed 3 to 6 months later. If there is progressive hydronephrosis, then an MAG-3 or DTPA diuretic renogram should be performed to detect obstruction.

LONG-TERM OUTCOMES OF RENAL MALFORMATION

Clinical outcomes in CAKUT vary widely from no symptoms whatsoever to CKD resulting in a need for renal replacement during a period ranging from the newborn period to the 4th and 5th decades of life. Risk factors for mortality during infancy and early childhood include coexistence of renal and nonrenal disease, prematurity, low birth weight, oligohydramnios, and severe forms of renal–urinary tract malformation (agenesis, hypodysplasia).[171] In a case series of 822 children with prenatally detected CAKUT that were followed for a median time of 43 months, Quirino et al. reported a mortality of 1.5% and morbidities including UTI, hypertension, and CKD in 29%, 2.7%, and 6% of surviving children, respectively.[172] A faster rate of decline of renal function in patients with CAKUT and CKD has been associated with a urine albumin-to-creatinine ratio greater than 200 mg/mmol compared with less than 50 mg/mmol (eGFR –6.5 mL/min/1.73 m^2/year vs. –1.5 mL/min/1.73 m^2/year), and with more than two (vs. <2) febrile UTIs (eGFR –3.5 mL/min/1.73 m^2/year vs. –2 mL/min/1.73 m^2/year). A greater decline in eGFR occurs during puberty (eGFR –4 mL/min/1.73 m^2/year vs. –1.9 mL/min/1.73 m^2/year).[173] A study examining the risk for dialysis in patients with CAKUT demonstrated a significantly higher risk for patients with a solitary kidney compared with nondisease controls.[174] These results raise the possibility that the prognosis for a solitary apparently normal kidney may not be as "normal" as previously thought. Finally, a study of CAKUT patients receiving some form of replacement therapy and registered within the European Dialysis and Transplant Association Registry showed that some of these patients only require renal replacement in the third, fourth, or fifth decade of life. The finding that the mean age at which patients with CAKUT require dialysis and/or transplantation is 31 years indicates that children with CAKUT are at risk of developing a requirement for dialysis and/or transplantation as adults.[7]

GLOMERULAR DISEASES

NEPHRITIC SYNDROME AND RELATED DISEASES

DIFFERENTIAL DIAGNOSIS OF GLOMERULAR HEMATURIA

A urine sediment that consists of >5 red blood cells/high-power field together with acanthocytes, red cell casts, or mixed red and white cell casts is characteristic of glomerular hematuria and is called a nephritic urine. Hematuria, impaired kidney function, and hypertension define a nephritic syndrome.[175] The differential diagnosis of glomerular hematuria is extensive and includes, but is not limited to, structural defects of the glomerular basement membrane (GBM); (e.g., Alport syndrome [AS]); thin basement membrane nephropathy (TBMN), inflammatory/autoimmune processes affecting the GBM (e.g., Goodpasture syndrome; anti-GBM antibody disease), systemic diseases affecting the glomerular capillaries (e.g., thrombotic microangiopathy [TMA]), systemic lupus erythematosus (SLE), antineutrophil cytoplasmic antibody (ANCA) vasculitis, antiphospholipid syndrome (APS), immunoglobulin A (IgA) nephropathy (IgAN), Henoch–Schönlein purpura (HSP), membranous nephropathy (MN), immune complex membranoproliferative glomerulonephritis (IC MPGN)/C3 glomerulopathy (C3G), and postinfectious GN (PIGN), and—to a lesser extent—nephrotic syndrome (NS). Identifying the specific diagnosis underlying glomerular hematuria can be challenging and requires in addition to the obligatory microscopic urinalysis (urine sediment; see earlier discussion) a detailed review of the personal and family medical history, a thorough clinical examination with focus on extrarenal disease manifestations, and a complete laboratory workup (including blood and urine) for the parameters specific for the aforementioned differential diagnoses.

Glomerular hematuria is a common first symptom of glomerulopathy and can mark the beginning of a disease process subsequently characterized by (glomerular) proteinuria and a progressive decline in kidney function, leading to CKD, and eventually resulting in ESRD. Presently, a total of six genes are known to cause familial microscopic hematuria (FMH): CFHR5, MYH9, and FN1 (not discussed here), as well as COL4A3, COL4A4, and COL4A5.[176]

ALPORT SYNDROME AND THIN BASEMENT MEMBRANE NEPHROPATHY

PATHOGENESIS

AS and TBMN define a spectrum of hereditary diseases affecting collagen type IV, the major component of the GBM.[177,178] Collagen IV is made of α chains (monomers), three of which combine to form a triple helical heterotrimer, or protomer, with the following structural domains: (1) a 7S triple helical domain at the N terminus, (2) a central triple helical collagenous domain, and (3) a noncollagenous (NCl) trimer at the C terminus. The collagen IV network exists as a combination of protomers, which is stabilized via disulfide cross-links between triple helical collagenous domains of the collagen IV chains.[179]

The triple helical collagen IV protomers are composed of defined combinations of six distinct monomeric chains (α1–6). In the embryonic kidney, the GBM is composed of α1 and α2 chains in the combination of α1.α1.α2, whereas the mature kidney contains the α3, α4, and α5 chains in the combination of α3.α4.α5.[179] In contrast to the embryonic α1.α1.α2 protomer, the mature α3.α4.α5 protomer forms a higher number of disulfide bonds, thus enhancing its mechanical stability toward the hydrostatic (blood) pressure the glomerular capillary loop is constantly exposed to. Mutations in the COL4A3, COL4A4, or COL4A5 genes result in the formation of defective collagen IV protomers forming a GBM with (partial) persistence of the embryonic α1.α1.α2 chains and with decreased mechanical stability, which in turn is the cause of progressive deterioration of the GBM structure and loss of function in AS and TBMN.[177–179]

Additional factors contributing to the decreased stability and increased permeability of the GBM as well as progressive glomerulopathy in AS include compensatory overproduction of atypical laminin isoforms (laminin α1 and α5),[180,181] the production of transforming growth factor β1 by podocytes,[181,182] and matrix metalloproteinase (MMP)-mediated proteolytic removal of the embryonic collagen IV protomers (α1.α1.α2).[181,183,184] Recently, crosstalk between the podocytes and abnormal collagen IV of the GBM via the discoidin domain receptor (DDR) 1 and α2β1 integrin has been identified as critical for AS pathogenesis.[185,186] DDR1 is a tyrosine kinase transmembrane receptor with collagens I–V as its ligand.[187] DDR1 regulates extracellular matrix remodeling by controlling the adhesion and proliferation of renal mesangial cells.[188] It has been postulated that the podocyte detects altered collagen protomers via DDR1 receptors, resulting in the upregulation of various cytokines and growth factors, ultimately leading to progressive glomerulopathy via inflammation and fibrosis.[186] An additional possible pathogenic mechanism involves α2β1 integrin, present on the basolateral surface of podocytes. α2 integrin binds to a specific sequence of the collagen IV protomer.[189] α2β1 integrin can be upregulated in signaling to downstream mechanosensor proteins in the presence of abnormal collagen chains, thus resulting in overproduction of extracellular matrix by podocytes.[190,191]

CLINICAL MANIFESTATIONS

First described in 1927 by A.C. Alport,[192,193] the triad of familial nephritis, deafness, and ocular changes was subsequently named AS.[193] The clinical manifestation of AS at disease onset in childhood is typically mild with hematuria only. Untreated, AS progresses to proteinuria, CKD, and eventually ESRD in adulthood.

AS is caused by mutations in the collagen IV α3, α4, and α5 genes (COL4A3, COL4A4, and COL4A5).[178,179] COL4A3 and COL4A4 are encoded on chromosome 2 (2q35-37); COL4A5 is encoded on the X chromosome (Xq26-48). About 65% of AS cases are inherited in an X-linked fashion via COL4A5, about 15% in an autosomal recessive fashion via mutations in COL4A3 and COL4A4, and about 20% in an autosomal dominant fashion with mutations in COL4A3 and COL4A4.[179] While male AS patients with an X-linked COL4A5 mutation are always severely affected with progression to ESRD in 50% typically during the 2nd decade of life,[194] female carriers present with a wide phenotypic spectrum ranging from microscopic hematuria to ESRD.[178] Of note, the clinical presentation is similar between males with hemizygous COL4A5 mutations and individuals with homozygous or compound heterozygous COL4A3 and COL4A4 mutations.[178] AS can vary in both severity and organ involvement (mosaicism) in females due to random inactivation of one of the two alleles of each X chromosome.[195]

Disease severity of AS is determined by the type of mutation with a strong genotype–phenotype correlation (point or missense mutations versus deletions, frame shift, or truncating mutations) and the type of inheritance. In a large cohort of X-linked AS patients, missense mutations were the most common, occurring in 51% of cases, and conferred the most favorable prognosis with a median time to ESRD of 37 years, as compared with patients with deletions, who reach ESRD by age 22 (95% confidence interval, 16–23 years).[196]

AS disease manifestation is also determined by the organ distribution pattern of the collagen IV α chains involved in the formation of the mature collagen IV protomers. Collagen IV α3.α4.α5 is not only found in the GBM, but also in the basement membrane of the renal tubules, the inner ear, and the posterior lens capsule.[197] Collagen IV α5.α5.α6 is found in the basement membrane of the eye and in visceral smooth muscle cells of the gastrointestinal (esophagus), the respiratory (trachea), and the female reproductive tract.[178,198] Thus AS is also characterized by cochlear and ocular involvement. Progressive SNHL is usually present by late childhood or early adolescence.[197,199] Ocular findings include corneal opacities, anterior lenticonus, fleck retinopathy, and temporal retinal thinning. Rarer findings include posterior dysmorphous corneal dystrophy, giant macular hole, and maculopathy—all causing loss of visual acuity.[197,200] In addition, in very rare cases AS can manifest leiomyomatosis affecting the visceral smooth muscle cells of the esophagus, the trachea, or the female genital tract.[179,201]

THIN BASEMENT MEMBRANE NEPHROPATHY

TBMN, previously termed benign FMH, is the most common cause of persistent glomerular hematuria in adults and children, and has a prevalence of approximately 1%. TBMN is characterized by a benign clinical course with minimal or no proteinuria and normal renal function. The family history is typically positive for microscopic hematuria and ultrastructural examination of a renal biopsy specimen via electron microscopy (EM) reveals uniform thinning of the GBM to approximately half of its normal thickness.[202,203]

Approximately 40% to 50% of TBMN patients carry heterozygous mutations in COL4A3 or COL4A4. This genetic finding and the clinical observation that TBMN patients occasionally present with a more severe disease course characterized by marked proteinuria and progressive glomerulopathy have given rise to the concept of a collagen IV disease spectrum with AS at its severe end and with TBMN at its mild end. Thus TBMN patients can be considered as heterozygous carriers of autosomal-recessive AS.

ESTABLISHING THE DIAGNOSIS OF ALPORT SYNDROME AND THIN BASEMENT MEMBRANE NEPHROPATHY

Diagnostic criteria for AS include the combination of hematuria and a positive family history of hematuria, chronic renal failure, or both, high-frequency SNHL, pathognomonic ocular anomalies (see earlier discussion), or characteristic ultrastructural changes in glomerular kidney or skin basement membranes.[204,205] A renal biopsy is diagnostic. In AS, ultrastructural examination of a renal biopsy specimen via EM reveals GBM thinning associated with early stages of disease, and thickening and multilamellation associated with late stages of the disease (Fig. 72.7).[184,206] By contrast, in TBMN, ultrastructural examination reveals uniform thinning of the GBM to approximately half of its normal thickness without thickening and multilamellation (Fig. 72.7).[202,203] Immunofluorescence (IF) staining identifies the absence of the mutant collagen IV α5 chain in the most frequent form of AS, autosomal dominant AS, affecting approximately 85% of cases. Of note, while X-mosaicism might obscure the correct diagnosis, an IF-based diagnosis can in principle also be established via skin biopsy.[207]

TREATMENT

While originally aiming to establish the blood pressure target for optimal renal protection in children via blockade of the renin–angiotensin–aldosterone system (RAAS), the ESCAPE Study Group identified that both blood pressure control and a decrease in proteinuria in CKD were significant independent predictors of delayed progression of renal disease.[208] Groundbreaking studies in COL4A3 knockout mice[209] established the benefit of ACE inhibitor and angiotensin receptor blocker (ARB) treatment in this animal model of progressive AS. While untreated knockout mice died from ESRD after approximately 70 days (71 ± 6 days), RAAS blockade resulted in a significant delay in disease progression with an increase in life span by 100% (150 ± 21 days) using ACE inhibitors,[210] and by 40% (98 ± 16 days) in ARB-treated mice.[211]

Current guidelines for the treatment of AS suggest the initiation of ACE inhibitors and/or ARBs at the onset of overt proteinuria to prolong renal survival and delay the need for RRT. ACE inhibitors should be first line, with ARB or aldosterone inhibition used as second-line agents.[212] RAAS blockade has been studied in children with AS.[194,213] The comparison of the ARB losartan with placebo or amlodipine over 12 weeks in 30 children resulted in a significant reduction in proteinuria.[214] In a retrospective evaluation of 283 AS patients followed for 20 years in the European Alport Registry, three treatment subgroups were identified: (1) patients in whom ACE inhibitor treatment was commenced with onset of microalbuminuria, (2) with proteinuria of greater than 0.3 g/day, or (3) with CKD III–IV. ESRD with onset of dialysis treatment was delayed by 3 years in the CKD group, and by 18 years in the proteinuria group. Of note, in 15 sibling pairs, in whom treatment initiation occurred later in the older than in the younger sibling, the median age of ESRD was 27 years in the older, but 40 years in the younger sibling. Proteinuria was reduced in all patients, but treatment benefit was more sustained if treatment was commenced prior to advanced stages of CKD.[194]

Increasing evidence points toward aldosterone inhibition to have a blood pressure–independent, antiproteinuric effect

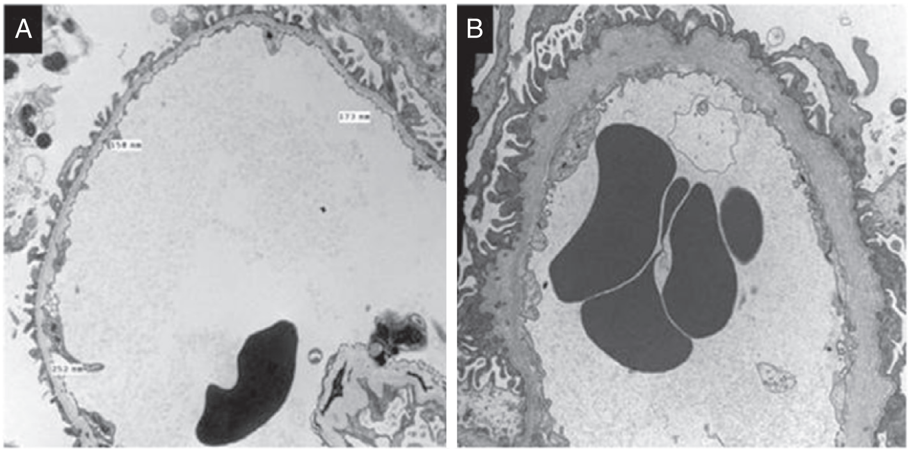

Fig. 72.7 Electron microscopy findings in thin basement membrane disease (TBMD) and Alport syndrome (AS): (A) TBMD shows diffuse thinning of the glomerular basement membrane (GBM) lamina densa compared with that of age-matched controls (best established in the same laboratory). (B) AS shows an irregular GBM with thinning and thickening, the latter becoming more prominent over time, and with splitting/fragmentation of the lamina densa resulting in a multilaminated/basket-weave appearance often containing stubs of cellular processes with electron lucent areas and small electron dense particles. A and B: electron microscopy ×10,000.

in adults.[214–219] Aldosterone inhibition was also found to be beneficial in children: in five children with persistent proteinuria despite ACE inhibitors or ACE inhibitor/ARB combination treatment, significant improvement in proteinuria was achieved and maintained for at least 18 months via aldosterone inhibition.[220] Of note, spironolactone may offer renoprotection in pediatric CKD patients with aldosterone breakthrough.[221]

The reported experience of calcineurin inhibitors (CNIs) such as cyclosporine in AS patients involves three case series involving a total of 32 patients.[222–224] Although a significant and sustained improvement in proteinuria was found, this was associated with a significant reduction in GFR, calcineurin toxicity on renal biopsy, and hypertension. Nephrotoxicity and hypertension likely preclude the use of cyclosporine in AS, particularly in children. Whether tacrolimus could offer a more favorable profile may warrant further study.

Numerous novel treatments for AS are currently under investigation, some of which are still in the phase of preclinical development, while others are in clinical trials. These treatments include chemokine receptor 1 (CCR 1) antagonists,[225] BMP 7,[226] MMP inhibitors,[227] DDR1 and integrin α2β1 antagonists,[186,191] and bone marrow transplant/stem cell therapy.[228–231]

NEPHROTIC SYNDROME

NS is one of the most common types of kidney diseases occurring in children. Children with NS typically present with massive proteinuria, edema, and hypoalbuminemia, which is almost always accompanied by marked hyperlipidemia. While NS typically occurs as a primary disorder in children, it can also be associated with a variety of systemic illnesses. The clinical manifestations of NS are increasingly understood to result from structural and/or functional abnormalities in the glomerular filtration barrier. These abnormalities most often involve the podocyte, the primary target cell of injury in NS, but in some cases may result from abnormalities in the extracellular GBM. In most children, the specific cause of disease remains unknown. However, expanding genetic studies are steadily revealing greater numbers of genetic mutations that result in the aforesaid observed abnormalities.

CLINICAL CLASSIFICATION AND DEFINITIONS

The diagnosis of NS requires the presence of edema, massive proteinuria (>40 mg/m^2/h, or a urine protein-to-creatinine ratio > 2.0), hypoalbuminemia (<2.5 g/dL), and hyperlipidemia.[232–236]

Clinical remission is defined by a marked reduction in proteinuria (to <4 mg/m^2/h, or urine albumin dipstick of 0 to trace for 3 consecutive days) in association with resolution of edema.[235,237]

Clinical relapse is defined by a recurrence of severe proteinuria (>40 mg/m^2/h, or urine albumin dipstick ≥ 2+ on 3 successive days), typically with a recurrence of edema.[232,233,235]

Patients who enter clinical remission in response to daily glucocorticoid treatment alone are referred to as having steroid-responsive or steroid-sensitive NS (SSNS).

A majority of children who develop NS will experience at least one relapse of their disease. While many will have only infrequent relapses (≤3 relapses per year), a significant proportion (~33%) of children who develop NS will experience multiple relapses. If a child has four or more relapses in any 12-month period, or two relapses in the first 6 months after diagnosis, he or she should be labeled as having frequent relapsing NS (FRNS).[238]

Some children respond to initial glucocorticoid treatment by entering clinical remission, but develop a relapse either while still receiving glucocorticoids, or within 2 weeks of discontinuation of treatment following a steroid taper. Such children may require continued low-dose treatment with glucocorticoids to prevent this rapid development of relapse, and thus should be labeled as having steroid-dependent NS (SDNS).[238]

Patients with either FRNS or SDNS are at slightly increased risk of developing complications of NS, as well as disease progression to CKD or ESRD (SDNS > FRNS). However, these risks are generally felt to be intermediate between those of children with SSNS and those with steroid-resistant NS (SRNS), who are at a far higher risk. In addition, children with FRNS, SDNS, and most importantly SRNS often require prolonged or more frequent glucocorticoid treatment which increases their risks for steroid-induced side effects compared with children with SSNS.

Patients with NS who fail to enter clinical remission after 4 to 8 weeks of daily glucocorticoid treatment are referred to as having SRNS.[234,235] It should be noted, however, that there are significant discrepancies in the literature regarding the definition for SRNS. While some authors define SRNS as a failure to enter clinical remission after 4 weeks of daily prednisone at a dose of 60 mg/m^2/day, others define it as failure to enter remission after 4 weeks of daily prednisone at a dose of 60 mg/m^2/day followed by 4 weeks of alternate-day prednisone at a dose of 40 mg/m^2/dose[239] or 4 weeks of prednisone at a dose of 60 mg/m^2/day followed by three intravenous pulses of methylprednisolone at a dose of 1000 mg/1.73 m^2/dose.[239] While several groups have to date developed evidence- and/or expert-opinion-based treatment protocols to define SRNS, these ongoing discrepancies in the definition of SRNS make direct comparisons of the efficacy of therapies for NS challenging. Indeed, development of a worldwide initial treatment protocol for new-onset NS in children could greatly enhance the nephrology community's ability to collaborate more effectively on the development and testing of novel future treatments for NS. The important implication of being diagnosed with SRNS, however, is that children who present with or develop SRNS are at significantly higher risk for the development of disease complications, as well as for progression to CKD or ESRD.[240]

EPIDEMIOLOGY

The estimated annual incidence of NS ranges from 2 to 7 new cases per 100,000 children,[234,241–244] while the estimated prevalence is approximately 16 cases per 100,000 children, or 1 in 6000 children.[198,208] Gender differences are present in NS, but appear to be age dependent. Among younger children, boys are almost twice as likely to develop NS as girls. However, by adolescence and into adulthood the proportion of males and females developing NS becomes essentially equal.[244] Ethnic differences in NS are also well-known. While idiopathic NS is six times more common among Asian versus Caucasian children in the United Kingdom,[245] idiopathic NS is comparatively less common among African children, in whom SRNS due to structural glomerular lesions are more common.[246] By contrast, NS in the United States appears to occur relatively proportionately among children of various ethnic backgrounds.[247] Race, however, does appear to have an important correlation with the histologic lesions associated with NS. In a study performed in Kansas City, focal segmental glomerulosclerosis (FSGS) was found in 44% of African-American children, but in only 17% of Caucasian children.[243] In addition, another study reported that 47% of African-American children had FSGS, whereas only 11% of Hispanic children and 18% of Caucasian children with NS had FSGS.[247] Thus African-American children appear to be at dramatically increased risk for developing FSGS compared with other races.

Idiopathic NS is also known to occur in families. It has also been reported to occur in identical twins.[248] A very large European cohort of 1877 children reported that 3.3% of children had affected family members, typically siblings,[249] and that disease often developed in siblings at the same ages, and with similar clinical outcomes and renal biopsy findings.[250]

NS most commonly presents at 2 years of age, and almost 80% of cases of NS present in children younger than 6 years of age.[234,244] Notably, the age at presentation appears to also be predictive of the underlying histologic lesion associated with NS. The International Study of Kidney Diseases in Children (ISKDC) reported that approximately 80% of children who developed NS before 6 years of age had minimal change NS (MCNS), whereas only 50% of those with FSGS, and 2.6% of children with MPGN presented before 6 years of age.[251] In this landmark study, the median age at presentation of MCNS was 3 years versus 6 years for FSGS versus 10 years for MPGN.[251] Together, these findings suggest that the likelihood of NS being associated with MCNS decreases with increasing age at onset, whereas the likelihood for having the far less-favorable diagnoses of FSGS or MPGN increases.[251,252]

Failure to enter clinical remission of NS in response to daily glucocorticoid treatment (i.e., SRNS) has very important implications for the risk of developing progressive CKD later in life. The Southwest Pediatric Nephrology Study Group found that within 5 years of being diagnosed with SRNS, 21% of children with FSGS developed ESRD and another 23% developed CKD.[253] Thus among children diagnosed with FSGS, the risk of developing CKD or ESRD within 5 years approaches 50%.

Finally, while the overall prevalence of NS has remained relatively stable over the last 30 years, there appears to have been a dramatic increase in the incidence of FSGS and a corresponding decrease in the incidence of MCNS.[243,247] This increased incidence of FSGS must be interpreted cautiously, however, because renal biopsies are typically only performed in children with atypical presentations, older ages at presentation, or those who develop SRNS. Thus it remains possible that this apparent increase in the incidence of FSGS simply indicates that a larger percentage of children with SRNS undergo renal biopsies than in the past.

PATHOGENESIS

The molecular pathogenesis of most forms of NS is still not clear. However, significant evolving information over the last 10 to 20 years has clarified that the primary target cell of injury during NS is the podocyte. All forms of this disease involve structural changes in podocytes, including cell swelling, as well as retraction and effacement (spreading) of the podocyte's distal foot processes, leading to the formation of a diffuse cytoplasmic sheet overlying the GBM. Additional podocyte structural changes include the formation of vacuoles, development of occluding junctions with displacement of the normal slit diaphragms that extend between podocyte foot processes, and in some areas detachment of podocytes from the underlying GBM. These podocyte structural changes, along with detachment from the GBM, have been demonstrated to result in the proteinuria that characterizes NS.[243,247,251] Table 72.3 shows the variety of podocyte structural abnormalities seen in essentially all forms of NS, as well as the variety of potential causes (some with potential overlap) that have been identified to date for the various forms of this disease.

Extensive investigations over several decades have failed to identify a single unifying cause for idiopathic NS. However, many secondary causes for disease have now been identified, which include various infections, allergies, immunizations, drugs, malignancies, and autoimmune disorders. Of note, Table 72.3 shows that there is potential overlap among some of these identified causes, which highlights that we still have only partial understanding of the molecular cause(s) of idiopathic NS. New secondary causes for idiopathic NS continue to be identified each year, particularly with regard to genetic causes. Indeed, more than 60 genes have now been identified for which mutations of various types have been found to lead to either idiopathic NS or FSGS (discussed later). Several recent publications document the growing number of gene mutations found to be associated with NS.[254–260] In general, however, the majority of mutations identified to date have been found to affect podocyte structure or function, with smaller numbers affecting the GBM or immune pathways. While the number of gene mutations associated with NS is expected to continue to grow in the future, the vast majority of patients who present with NS, particularly children, still have no identified secondary cause for their idiopathic NS. Over time it is hoped and anticipated that the list of identified secondary causes for idiopathic NS will continue to expand as we learn more about the molecular pathogenesis of this disease. For interested readers, a nice review has recently been published that provides a more comprehensive discussion of idiopathic NS.[256]

Of particular note is that almost one-third of patients who present with NS have some history of allergies, asthma, or atopy.[235,261–264] Consistent with this, allergies, immunizations, and autoimmune disorders (Table 72.3) are all known causes of NS. Together, these findings suggest a strong, but indirect, correlation between idiopathic syndrome and dysregulation of the immune system. Moreover, the fact that almost all treatments known to induce clinical remission of NS are immunosuppressants further strengthens this argument. Indeed, it has been suggested that idiopathic NS may be either an immune disorder to which the kidney

Table 72.3 Abnormalities Underlying Nephrotic Syndrome

Podocyte Structural Abnormalities Seen in Nephrotic Syndrome	Potential Causes for Idiopathic Nephrotic Syndrome Identified to Date	Potential Causes for Focal Segmental Glomerulosclerosis Identified to Date
Cell swelling Retraction (effacement) of distal foot processes Formation of vacuoles Development of occluding junctions between podocytes Apical displacement of normal junctions (i.e., slit diaphragms) between podocytes Detachment of podocytes from underlying glomerular basement membrane (GBM)	Soluble plasma factor(s) Genetic predisposition (i.e., mutations) Dysregulation of podocyte structure Dysregulation of podocyte function Dysregulation of immune system function (T cells, B cells [more recent]) Malignancies (i.e., Hodgkin disease, non-Hodgkin lymphoma, leukemia, etc.) Drugs (i.e., nonsteroidal antiinflammatory drugs, D-penicillamine, lithium, tyrosine kinase inhibitors, captopril) Infections (hepatitis B, hepatitis C, HIV, parasitic, mycoplasma, etc.) Allergies (food, pollen, dust, bee stings, cat fur, etc.) Immunizations Autoimmune disorders (systemic lupus erythematosus, celiac disease, allogeneic stem cell transplantation, myasthenia gravis, etc.) Systemic diseases (i.e., diabetes mellitus, Henoch–Schönlein purpura, sarcoidosis, etc.)	Soluble plasma factor(s) (especially in recurrent focal segmental glomerulosclerosis) Genetic predisposition (i.e., mutations) Immunologic podocyte injury (i.e., human immunodeficiency virus [HIV] nephropathy, cytomegalovirus, other viruses) Toxic podocyte injury (lithium, bisphosphonate, heroin, other medications?, cytokines) Inflammatory podocyte injury (APOL1 variant, cytokines) Hemodynamic podocyte injury (morbid obesity, low birth weight, reflux nephropathy) Structural GBM abnormalities (inherited, acquired) APOL1 focal segmental glomerulosclerosis

(i.e., the podocyte) is especially susceptible, or a kidney disorder to which the systemic immune system is especially susceptible.

The pathogenesis of FSGS has been the subject of intensive study for several decades. During this time it has become clear that FSGS is not a single disease, but a pattern of glomerular injury that can represent the final outcome of a variety of mechanistically distinct pathologic processes. As shown in Table 72.3, FSGS may result from circulating plasma factors, which are most clearly seen in recurrent FSGS after renal transplantation, or due to a variety of other forms of glomerular and/or podocyte injury. These include inflammatory, immunologic, structural, or hemodynamic injury to podocytes, as well as the GBM and glomerular parietal epithelial cells. Several recent advances in our understanding of the pathophysiologic, genetic, and histologic features of FSGS have led to a more mechanistic and clinically useful proposed classification of this multifaceted disease. While the historical classification of FSGS separated this condition into primary FSGS (i.e., idiopathic) versus secondary FSGS (i.e., due to known causes),[265] integration of more robust clinical, histologic, and genetic knowledge of this disease has led to the recent proposal that FSGS be categorized into six forms: (1) primary FSGS, (2) adaptive FSGS, (3) APOL1 FSGS, (4) genetic FSGS, (5) virus-associated FSGS, and (6) medication-/toxin-associated FSGS. Among these, the most common causes are primary FSGS, adaptive FSGS, and APOL1 FSGS, which all appear to occur with similar prevalence in the United States, with lower prevalence of genetic FSGS, virus-associated FSGS, and medication-/toxin-associated FSGS.[255] In this schema, primary FSGS is thought to be due to suspected circulating plasma factor(s), whereas adaptive FSGS is thought to be due to either conditions with elevated total kidney GFR such as cyanotic congenital heart disease,[266] androgen abuse,[267] sleep apnea[268] or conditions with reduced functional nephron mass such as renal dysplasia, reflux nephropathy, repeated episodes of acute kidney injury (AKI), or premature or small-for-gestational age birth.[269]

Note that while the precise relationships among the diverse potential causes of FSGS remain uncertain, the podocyte has emerged as the final common target cell of injury for all of these causes. Moreover, any of the aforementioned forms of FSGS, and likely any chronic glomerular or interstitial disease that leads to reduced nephron mass, can potentially result in podocyte injury and glomerular changes consistent with adaptive FSGS. In this setting, progressive loss of injured or dead podocytes into the urine (i.e., podocyturia) often leads to depletion of podocytes in the glomerulus (i.e., podocytopenia), which then induces podocytes to hypertrophy to try to compensate for these losses. These (mal)adaptive changes are now thought to result in the development of podocyte sheer forces which further exacerbate podocyte injury,[270] and ultimately lead to increases in glomerular pressures and flows in the remaining intact glomeruli that, in turn, induce further glomerular injury and disease progression.[255]

CLINICAL PRESENTATION

Most children who develop idiopathic NS present with acute or subacute onset of periorbital or generalized edema. These findings combined with the presence of 3–4+ proteinuria on a dipstick urinalysis (~300–2000 mg/dL [17.7–118 mmol/L]) typically establish a preliminary diagnosis of NS. The extent of proteinuria can be better quantified by obtaining a spot urine protein-to-creatinine ratio (typically >2.5 mg/mg), and the diagnosis can be confirmed by documenting the presence of hypoalbuminemia (typically <2.5 g/dL [<25 g/L]) and hyperlipidemia.

Table 72.4 Primary Causes of Childhood Nephrotic Syndrome

Minimal change nephrotic syndrome (MCNS)
Focal segmental glomerulosclerosis (FSGS)
Mesangial proliferative glomerulonephritis
Membranoproliferative glomerulonephritis (MPGN)
Membranous nephropathy (MN)

Childhood NS is most often due to primary glomerulopathies. These include MCNS, FSGS, mesangial proliferative glomerulonephritis, MPGN, and MN (Table 72.4). However, a careful history may reveal potential secondary causes for this disease, including infections, systemic diseases, drugs, malignancies, autoimmune disorders, and allergies, as detailed in Table 72.3.

Physical examination can also often provide suggestive clues about the diagnosis. For example, while periorbital and peripheral edema, and ascites are common in most children with NS, this appears not to be similar for the presence of hypertension. In a very large international study of children with NS, the ISKDC, it was determined that hypertension was far less common in children found to have MCNS (~21%) than in children found to have either of the less favorable diagnoses of FSGS or MPGN, where the incidence was approximately 50%.[251]

The laboratory evaluation of children presenting with NS always begins with confirmation of the presence of hypoalbuminemia and hyperlipidemia, as noted earlier, and determination of the renal function. The presence of AKI is not uncommon, and is often attributed to intravascular volume depletion and renal hypoperfusion. While this may be only prerenal AKI, in more severe cases AKI may be found to be due to acute tubular necrosis, likely reflecting more severe renal hypoperfusion resulting in renal tubular injury. In the ISKDC study in 1978, 32% of children with MCNS had elevated serum creatinine, while 41% of those with FSGS and 50% of those with MPGN had elevated creatinine.[251] Of note, a recent large multicenter study of 370 children admitted to the hospital with NS performed through the Midwest Pediatric Nephrology Consortium identified a tripling of the incidence of AKI among this cohort.[271] While the cause of this dramatic increase in AKI incidence is not yet known, it is suspected that it may be related to the significant increase in the use of CNIs as alternative treatments for NS in the last decade, as these agents are well-known to have vasoconstrictive effects on the glomerular vasculature.

Microscopic hematuria is also not uncommon in children with NS. Microscopic hematuria was found in 23% of children with MCNS versus 48% of children with FSGS versus 59% of children with MPGN in the ISKDC study.[251] By contrast, the presence of macroscopic hematuria is unusual, and should prompt the suspicion of a possible renal venous thrombosis.

In addition to the aforementioned laboratory studies, serologic evaluation to screen for potential secondary causes of NS is often performed in children presenting at an older age, or who have a history or signs or symptoms of systemic illness. The finding of low serum complement levels can identify children with MPGN, SLE, or PIGN, and positive serum antinuclear antibody and antidouble-stranded DNA studies can confirm the diagnosis of SLE. Screening for viral infections can identify patients with disease due to hepatitis B (typically associated with MN), hepatitis C (typically associated with MPGN), or human immunodeficiency virus (HIV; typically associated with collapsing variant of FSGS). In addition, a complete blood count found to be abnormal could potentially identify patients with leukemia, lymphoma, or SLE.

Because immunosuppressive treatments are the mainstay of therapy for childhood NS, it is recommended to place a purified protein derivative test in all children prior to initiating treatment, to screen for possible previously undiagnosed tuberculosis. In addition, consideration should be given to obtaining varicella IgG titers to screen for immunity to this virus, which can be potentially life-threatening in immunocompromised hosts. Such knowledge can be very helpful to ensure optimal management of a nonimmune child with NS under immunosuppressive treatment with varicella zoster Ig.

Finally, while a renal ultrasound is not routinely indicated in children with NS, development of gross hematuria represents a clear indication to obtain an ultrasound with Doppler evaluation of the renal vasculature to exclude possible development of a renal venous thrombosis.

CONSIDERATION OF RENAL BIOPSY

The role of the renal biopsy in the diagnosis and management of children with NS remains somewhat controversial. In the 1970s the ISKDC study included renal biopsies for all children with NS before treatment initiation.[278] Since that time the clinical indications have been debated, with some pediatric nephrologists reserving biopsies for older children in whom diagnoses other than MCNS are more likely to be found. Among others, renal biopsies are reserved for those children in whom there is a clinical or serologic suspicion that the child may have a glomerulonephritis. Renal biopsies are also often performed by some, but not all, nephrologists to determine baseline and follow-up levels of glomerular and interstitial scarring when a child is being transitioned to an alternative therapy with significant potential nephrotoxicity, such as the CNIs cyclosporine and tacrolimus. Despite these variations in practice patterns, however, almost all pediatric nephrologists agree that a renal biopsy should be performed in a child who presents with or subsequently develops SRNS.

HISTOLOGICAL CLASSIFICATION

From a clinical perspective, most children who present with idiopathic NS under the age of 10 to 12 years receive empiric initial treatment with oral glucocorticoids (discussed later). Only if they present with or subsequently develop clinical steroid resistance (or steroid dependence in some cases) do such patients routinely undergo a renal biopsy. While renal biopsy findings at the level of EM almost always reveal numerous abnormalities involving the podocytes (Table 72.3), at the light microscopic level many patients have no or only minimal other histologic abnormalities. Dating back to the early days of renal biopsies when only light microscopy (LM) was available, such patients are still often referred to as having "MCNS" or "minimal change disease."

The primary glomerulopathies that comprise childhood NS include MCNS, FSGS, mesangial proliferative glomerulonephritis, MPGN, and MN (Table 72.4). The detailed histologic features of each of these diseases have been addressed elsewhere in the text, and the reader is referred to these sections for more detailed images and descriptions of these features.

The secondary glomerulopathies associated with childhood NS include systemic diseases such as SLE, HSP, diabetes mellitus, sickle cell disease, and sarcoidosis. In addition, other secondary causes of disease include infections such as HIV, hepatitis B, hepatitis C, malignancies such as leukemia and lymphoma, and drugs such as nonsteroidal antiinflammatory drugs (NSAIDs), ACE inhibitors (captopril), D-penicillamine, and gold. Finally, morbid obesity, pregnancy (preeclampsia), and allergies to food, bee stings, and even immunizations have all also been associated with the development of NS. The detailed histologic features of most of these diseases have also been presented elsewhere in the text and the reader is referred to these sections for more detailed images and descriptions of these histologic features.

TREATMENT OF NEPHROTIC SYNDROME

The primary treatment of NS in children has remained daily oral glucocorticoids for more than 60 years. Unlike adults, in whom prolonged steroid treatment has been associated with increases in remissions of NS,[272,273] the vast majority of children respond to a 4- to 8-week course of oral glucocorticoids by entering into complete clinical remission.

Interestingly, while almost all pediatric nephrologists will initiate glucocorticoids upon diagnosis of NS, it is known that spontaneous remissions occur in approximately 5% of patients, often within the first 1 to 2 weeks.[274,275] Such findings suggest that delaying slightly the initiation of treatment in some cases could be reasonable.

Initial treatment of NS in children who present at the age of 1 year or more includes daily high-dose oral glucocorticoids (i.e., most commonly prednisone at 2 mg/kg/day with maximal dose of 60 mg/day) for 4 to 8 weeks. Most children (~75%) enter complete remission within 2 weeks, and the vast majority of children (~95%) enter remission within 4 weeks.[235,276] Continued treatment for an additional 4 weeks typically identifies only a few additional patients able to enter a complete remission, with the remainder either failing to respond or responding with only a partial resolution of their proteinuria. Such patients are then labeled as having SRNS. Children with SRNS are at a dramatically higher risk than those with SSNS for the development of progressive renal disease (see the "Outcomes" section), and almost always undergo a renal biopsy to confirm a histologic diagnosis. Regardless of the histologic diagnosis, however, treatment decisions in children are guided primarily by either the success (i.e., SSNS) or the failure (i.e., SRNS) of oral glucocorticoids to induce a complete clinical remission.

STEROID-SENSITIVE NEPHROTIC SYNDROME

Over the last few decades, multiple protocols have been reported for the initial treatment of children presenting with NS. The initial widely accepted approach was reported by the ISKDC and suggested treatment with prednisone in divided doses for 4 weeks at 60 mg/m^2/day (or 2 mg/kg/day up to a maximal dose of 80 mg/day), followed by a taper to 40 mg/m^2/day (or 1.5 mg/kg/day up to a maximal dose of 60 mg/day) for an additional 4 weeks.[235] Since then there have been numerous subsequent reports of variations on this regimen.[277–281] Recently, a comprehensive meta-analysis has systematically compared many of these approaches.[281] Despite all of these publications over the years, however, there remains a fairly wide distribution of initial treatment practices among pediatric nephrologists in the United States,[282] as well as among pediatric nephrologists in many other countries around the world. The unfortunate result of this is that the worldwide pediatric nephrology community has yet to come to consensus on one single definition of a glucocorticoid treatment dose and duration that clearly defines when a child should be labeled as having SRNS. As noted earlier, because the vast majority of nephrologists base their treatment decisions on the distinction between SSNS and SRNS, such variations in this definition impede the ability to directly compare the effectiveness of new treatments for this disease developed primarily for children with SRNS.

Once complete remission has been achieved with one of the aforementioned protocols, many pediatric nephrologists elect to taper the prednisone to alternate days with weekly dosage reductions to discontinue the medication over about 6 weeks, although significant practice variation regarding prednisone taper also remains.[282]

STEROID-DEPENDENT AND FREQUENT RELAPSING NEPHROTIC SYNDROME

Once diagnosed with NS, the majority of children (50%–75%) will experience one or more relapses of disease. While some relapses can be transient and remit spontaneously as noted earlier,[283] unfortunately approximately 50% of children with initial SSNS ultimately develop either frequent relapses (FRNS; see definition earlier) or steroid dependence (SDNS; see definition earlier).[277,280] Such relapses require additional courses of oral prednisone, and thus greater overall glucocorticoid exposure and risk for side effects, compared with children with fewer relapses.[278]

Several additional factors also increase the risk for relapses. Children who present with NS under the age of 5 years, as well as those who are boys, are both at greater risk for frequent relapses.[284] Perhaps the greatest risk for relapses, however, occurs during intercurrent infections. Indeed, upper respiratory infections are the single most common cause for relapses of NS in children, and approximately 70% of relapses have been reported to be preceded by an upper respiratory infection.[285] Despite this strong correlation, however, a large multicenter study of more than 200 children performed by the ISKDC failed to identify any other specific factors predictive of frequent relapses, including time to initial remission of disease, time between first remission and first relapse, MCNS histologic subtypes, clinical findings, or laboratory findings.[286] One exception to this was the identification of a strong correlation between the number of relapses in the initial 6 months of disease and the subsequent clinical course, with more early relapses predicting a continued frequent relapsing course of disease. Together, these findings suggest that earlier age at presentation, male gender, and early frequent relapses all are associated with an increased risk for more frequent future relapses.

Treatment of relapses typically involves reinitiation of oral prednisone at the standard dose of 2 mg/kg/day (maximal dose of 80 mg/day) divided twice daily until the urine protein is trace or negative for 3 consecutive days by urine dipstick. Preferably these urine samples are obtained on first-morning voids, so as to minimize the risk for confounding by possible orthostatic proteinuria, which occurs more commonly in older children. Once in remission, prednisone is typically tapered to 1.5 mg/kg on alternate mornings for 4 weeks (ISKDC protocol)[235] or gradually tapered further (~0.25 mg/kg/dose) every week until discontinuation in approximately 6 weeks. For children with FRNS, this taper can be slowed significantly to be weaned and discontinued over 5 to 6 months to try to reduce the frequency of relapses. Those children unable to remain in complete remission during this prednisone wean, or who relapse within 2 weeks of discontinuation of prednisone, should be labeled as having SDNS. Such patients can be treated generally as noted earlier, but the alternate-day prednisone tapered only to the lowest dose that has historically kept the patient in remission, and continued for 6–18 months, followed by gradual further tapering of the dose every 2 weeks until discontinuation.

For children in whom the aforementioned measures fail to adequately control the disease, or in whom significant signs and symptoms of steroid toxicity develop, there are several alternative treatments that are used. While these agents as a group can often reduce the frequency of relapses, many of them have significant toxicities of their own.

Alkylating agents, which include cyclophosphamide and chlorambucil, are historical treatments that, while still used, are used increasingly infrequently. Typical courses of oral cyclophosphamide include administration of 2 mg/kg/day for 8 to 12 weeks, while chlorambucil is typically prescribed at 0.2 mg/kg/day for 8 to 12 weeks. Both treatments have been reported to reduce the frequency of relapses in approximately 75% of children with FRNS, whereas their efficacy among children with SDNS is only 30% to 35%.[287,288] Importantly, these agents have been associated with significant toxicities, including leukopenia, hair loss, thrombocytopenia, seizures (chlorambucil only), hemorrhagic cystitis (cyclophosphamide only), severe infections, malignancies, and fatalities.[289] For these reasons, other less toxic alternatives have gained in popularity compared with these treatments in recent years.

Levamisole is another alternative treatment used to treat many children with FRNS and SDNS. Similar to most other treatments for NS, its mechanism of action in NS is not known. However, unlike all other reported treatments for NS, levamisole is known to act not as an immunosuppressant, but as an immunostimulant. Despite this, it has been widely reported to prolong remissions and reduce the overall exposure to glucocorticoids in children with both FRNS and SDNS.[290–294] Typical dosing includes administration of 2.5 mg/kg PO on alternate days, and a typical treatment duration ranges from 4 months to a year. This regimen has been reported to be generally effective and safe, although reported side effects include flulike symptoms, neutropenia, agranulocytosis, vomiting, seizures, rash, and hyperactivity. Unlike alkylating agents, the beneficial effects of levamisole (as well as most other alternative agents) do not extend beyond the period of treatment.

Mycophenolate mofetil (MMF) and its closely related compound mizoribine are additional alternative agents that have gained in popularity to treat FRNS and SDNS in recent years. Although their mechanism of action in NS remains unclear, these compounds are known to inhibit lymphocyte DNA synthesis and cellular proliferation via inhibition of inosine monophosphate dehydrogenase, a key enzyme in purine biosynthesis.[295] These agents also interfere with adhesion of activated lymphocytes to endothelial cells via inhibition of glycosylation of selected adhesion molecules.[296] From a clinical perspective, these effects are selective only for T lymphocytes and B lymphocytes, because all other cells in the body have a “salvage pathway” that enables purines to still be synthesized in the presence of these agents.[297] These drugs have been widely used to reduce the frequency of relapses in children with both FRNS and SDNS,[298–301] enabling significant reductions in overall glucocorticoid exposure and thus the incidence and severity of their side effects. A typical treatment dose is approximately 600 mg/m^2/dose given twice daily (maximum dose ~1000 mg BID). While most children do not have significant side effects that limit treatment, more prevalent side effects include nausea, vomiting, diarrhea, bone marrow suppression, anemia, and gastrointestinal bleeding. Notably absent for this list, however, is the induction of renal injury (see later). Given their general effectiveness, modest side-effect profiles, and lack of renal toxicity, MMF and mizoribine have become very popular alternative treatments for children with FRNS and SDNS in recent years.

CNIs include cyclosporine and tacrolimus, and have become far more widely used alternative agents in the last decade. While their exact mechanism of action in NS is not fully understood, both agents are known to inhibit T-cell activation via inhibition of calcineurin-induced interleukin-2 gene expression, an early step in T-cell activation,[302] and cyclosporine has additionally been reported to act directly on podocytes to stabilize their actin cytoskeleton.[303] Use of these agents has been widely reported to reduce the frequency of relapses,[304,305] but unlike the alkylating agents noted earlier which have beneficial effects well beyond their course of treatment, many children relapse when CNIs are tapered or discontinued. CNIs can also have several side effects. These include hypertension, mild AKI, hyperkalemia, hypertrichosis (mostly cyclosporine), gingival overgrowth (mostly cyclosporine), and most importantly an increased risk for calcineurin nephrotoxicity which manifests as irreversible interstitial fibrosis. This typically occurs in a striped pattern that is reflective of the interstitial vascularization, and its incidence and severity appear to correlate generally with the duration and dose of drug exposure. Despite this, these agents have gained much favor in recent years as perhaps the preferred alternative treatments for FRNS and SDNS.

STEROID-RESISTANT NEPHROTIC SYNDROME

Despite optimal management, approximately 10% to 20% of children with NS will initially present with or subsequently develop SRNS. Notably, prior to labeling a child as having SRNS it is essential to exclude several potentially treatable

Table 72.5 Current, Emerging, and Potential Future Treatment Options for Children With Steroid-Resistant Nephrotic Syndrome

Immunosuppressive	Immunostimulatory	Nonimmunosuppressive
Alkylating agents[a] (cyclophosphamide, chlorambucil)	Levamisole[a]	Angiotensin-converting enzyme inhibitors
Pulse intravenous methylprednisolone[a]		Angiotensin receptor blockers (ARBs)
Plasmapheresis/immunoadsorption[a]		Vitamin E (antioxidant)[a]
Mycophenolate mofetil (and mizoribine)		
Calcineurin inhibitors (cyclosporine, tacrolimus)		
Anti-CD20 monoclonal antibodies (rituximab, ofatumumab[b])		
Low-density lipoprotein apheresis[b] (LIPOSORBER)		
ACTHar[b]		
CTLA-4-Ig[b] (Abatacept)		
ARB + endothelin type A receptor antagonist[b] (Sparsentan)		
p38 mitogen-activated protein kinase inhibitor[c] (Losmapimod)		
Peroxisome proliferator-activated receptor-γ agonist[c] (Pioglitazone)		
Anti-interleukin-13 monoclonal antibody[c] (Lebrikizumab)		

[a]Less commonly used treatment.
[b]Emerging treatment.
[c]Potential future treatment.
ACTH, Adrenocorticotropic hormone.

causes for treatment failure, including poor medication adherence, poor medication absorption, possible underlying infection, and possible malignancy. Once diagnosed with SRNS, a child's risk for both the development of extrarenal complications of NS and for the development of ESRD and/or CKD increases greatly. For this reason, both the alternative treatments described earlier, as well as additional alternative treatments, are often employed in an attempt to induce a complete remission of proteinuria. Although the results of the renal biopsy are helpful in guiding the clinical management of patients with SRNS, it is generally accepted that the best way to reduce an SRNS patient's risk of developing progressive CKD or ESRD is to utilize these alternative treatments, either singly or in combination, to induce a complete remission. Induction of only a partial clinical remission, however, is controversial as it provides an unproven and unclear reduction in risk for future CKD or ESRD. Despite this, most clinicians agree that achieving lower levels of proteinuria, however induced, are associated with generally better clinical outcomes and fewer disease complications.

Treatment of SRNS in children includes the use of a wider array of alternative treatments than for FRNS and SDNS. Importantly, almost all treatments used for FRNS and SDNS are notably less effective in the setting of SRNS, and it is thus not uncommon for multiple alternative treatments to be added to one another or even initiated simultaneously in an attempt to induce a complete clinical remission of NS. Unfortunately, all of the available alternative treatments are only partially effective, and many are associated with significant toxicities of their own.

Although significant advances have been made in expanding the array of alternative treatments available to treat SRNS in the last several years, most published data reporting their usage, efficacy, and toxicity remain derived from anecdotal reports and uncontrolled clinical trials, with relatively few randomized controlled trials completed that have been able to confirm the safety and efficacy of these newer treatments. This lack of high-quality clinical trial data, combined with the lack of a standardized definition for the diagnosis of SRNS, makes it difficult to directly compare the relative effectiveness of the various alternative treatments currently available, as well as to compare emerging alternative treatments with the existing ones. Despite these limitations, Table 72.5 lists the current, emerging, and several potential future treatment options for children with SRNS.

The overall management of children with SRNS can be divided into four major categories: (1) supportive therapy, (2) immunosuppressive therapy, (3) immunostimulatory therapy, and (4) nonimmunosuppressive therapy. Supportive therapy for SRNS includes the provision of general medical care, management of edema, and management of dyslipidemia (Table 72.6). General medical care includes the identification and treatment of suspected infection, which may present as pneumonia, cellulitis, peritonitis, or sepsis.[271] It also includes the identification and treatment of suspected thrombosis, which may present with gross hematuria with or without acute oliguria (renal venous thrombosis), respiratory distress (pulmonary thrombosis), asymmetric swelling of an extremity (deep venous thrombosis), or neurologic symptoms (cerebral venous thrombosis).[306–308] Additional important aspects of general care include the maintenance or restoration of the patient's intravascular volume, maintenance of adequate protein intake,[274] and alteration of immunizations while the patient remains immunosuppressed.[309–311,311a,311b]

Management of the chronic edema associated with SRNS includes use of moderate restriction of fluid intake, moderate restriction of salt intake, and the judicious use of diuretics. Severe anasarca can be reasonably effectively treated with 25% albumin at a dose 1 g/kg infused over 4 to 6 hours, followed by IV furosemide at a dose of 1 to 2 mg/kg to induce diuresis. This can be repeated every 8 to 12 hours as clinically indicated, assuming care is taken to ensure that mobilization of interstitial fluid into the intravascular space is not too rapid, which may result in pulmonary edema and possible congestive heart failure.[312] Additional approaches to control severe edema include elevation of the extremities,

Table 72.6 Major Components of Supportive Therapy for Children With Steroid-Resistant Nephrotic Syndrome

General Medical Care	Management of Edema	Management of Dyslipidemia
Identification and treatment of suspected infection • Pneumonia • Cellulitis • Peritonitis • Sepsis Identification and treatment of suspected thrombosis • Gross hematuria +/-acute oliguria (renal venous thrombosis) • Respiratory distress (pulmonary thrombosis) • Asymmetric extremity swelling (deep venous thrombosis) • Neurologic symptoms (cerebral venous thrombosis) Maintenance or restoration of intravascular volume Maintenance of adequate protein intake (130%–140% of recommended dietary allowances) Alteration of immunizations while immunosuppressed	Avoidance of excessive fluid intake Elevation of extremities Moderate dietary salt restriction Judicious use of diuretics for severe edema Head-out water immersion	Avoidance of high fat intake Regular exercise (>30 min/day of moderately intense activity) Consideration of hydroxymethylglutaryl-coenzyme A reductase inhibitors (statins) Consideration of low-density lipoprotein apheresis

as well as the use of head-out water immersion, which although not commonly considered, has been reported to induce strong diuretic and natriuretic effects in patients with NS.[313]

Management of the chronic dyslipidemia that almost always accompanies SRNS includes the avoidance of high fat intake, introduction of a regular daily exercise regimen, pharmacologic therapy when the aforementioned approaches fail, and more recently the use of low-density lipoprotein (LDL) apheresis. The dyslipidemia in NS typically includes marked elevations in LDL, very-low-density lipoprotein, cholesterol, lipoprotein A, and often triglyceride levels, with variable abnormalities in high-density lipoprotein levels.[314] This chronic dyslipidemia has been clearly associated with increased risks for atherosclerosis, myocardial infarction, cardiac death, cerebral infarctions, and accelerated progression of CKD, although primarily among adults.[315–319] Conservative treatment of the dyslipidemia in SRNS is commonly ineffective, and pharmacologic treatment is thus sometimes required to control severe chronic lipid abnormalities, although none of the treatments have been approved by the Food and Drug Administration for use in children with NS to date. The most common treatment includes the use of hydroxymethylglutaryl-coenzyme A inhibitors (HMG-CoA inhibitors; i.e., statins), for which there are a few uncontrolled trials in children reporting moderate efficacy without significant toxicity.[320–322] Among adults with NS, statin usage has been generally well tolerated, with only mild elevations in liver enzymes and creatinine kinase, and rarely diarrhea, reported.[323]

Among the pharmacologic treatments for children with SRNS, the immunosuppressive alternative treatments are the most widely employed. The most widely used currently available and emerging alternative treatments for SRNS will be described briefly in the following section, highlighting their known mechanisms of action, and their popularity of usage among pediatric nephrologists. In addition, a few potential future treatments for SRNS will be briefly described.

Several alternative treatments for SRNS have already been described in detail earlier for the management of FRNS and SDNS. These treatments include cytotoxic agents, levamisole, and MMF and mizoribine. While they are sometimes still used to treat children with SRNS, their reported efficacy among children with SRNS is far lower and they are thus gradually falling out of favor among pediatric nephrologists. One notable exception is that MMF continues to be widely used in attempt to maintain remission among children with SRNS in whom complete remission has been induced with another alternative treatment.

Another alternative treatment that has gradually fallen out of favor for SRNS is the use of high-dose pulse IV methylprednisolone. Like most drugs used to treat NS, its mechanism of action in NS is not fully known. However, multiple reports have documented that in addition to their known antiinflammatory effects, glucocorticoids also act directly on podocytes to protect against and enhance recovery from NS-related cellular injury, suggesting that at least part of the beneficial effects of glucocorticoids result from direct action on the primary target cell of injury during NS.[324,325] While courses of this drug have demonstrated relatively good efficacy (38%–82% complete remission rates) among children with SRNS for many years, treatment has been associated with toxicity in 17% to 25% of children, and due to concerns about the long-term side effects of steroids, has long been controversial among pediatric nephrologists.[326–331]

Another less widely used treatment for SRNS is plasmapheresis. While it is widely used for the treatment of recurrence of NS following renal transplantation in treatment of a presumed circulating factor which induces increased glomerular permeability,[299,332–336] use of plasmapheresis in children with SRNS before renal transplantation has never become widely accepted.[337] Immunoadsorption, a related extracorporeal treatment to directly bind potential circulating factors on a column, has also been reported to have some efficacy, but has not gained as much popularity in the United States as it has in Europe.[337–339]

The CNIs cyclosporine A and tacrolimus have become among the most widely used treatments for children with SRNS over the last 10 to 15 years. Numerous reports have documented their ability to induce complete remissions in significant percentages of children with SRNS (33%–59%), as well as partial remissions in many of the remaining patients.[284,305,340–343] Their mechanism of action in NS remains uncertain, but these drugs are currently thought to exert their beneficial clinical effects via their immunosuppressive actions, as well as via direct action on podocytes to stabilize the podocyte actin cytoskeleton.[303] Despite these beneficial effects, CNIs have been associated with several potentially significant side effects that can limit their clinical usefulness. These include hypertension, mild AKI, hyperkalemia, gingival overgrowth (primarily cyclosporine), hypertrichosis (primarily cyclosporine), headaches, nausea, and vomiting. The most concerning side effect, however, is the development of irreversible interstitial fibrosis (calcineurin nephrotoxicity). This side effect appears to be generally associated with the cumulative exposure to CNIs, and thus increases with longer time of treatment and higher dosages.[284,344–348] This is especially problematic for children with SRNS, because many of whom respond well to CNIs will experience future relapses, and not infrequently become dependent on calcineurin treatment to maintain remission, thus increasing their risk for future nephrotoxicity.[349] Because SRNS is a potentially reversible disease, prolonged usage of CNIs which may induce irreversible side effects has raised important concerns about their long-term usage in children with SRNS. In this light, many pediatric nephrologists choose to perform a renal biopsy prior to initiation of a CNI to determine the baseline extent of interstitial fibrosis, as well as repeat biopsies at periodic intervals (usually about every 2 years or so) to screen for potential development or progression of interstitial fibrosis as long as a child requires continued calcineurin treatment due to his or her inability to be transitioned to another alternative treatment.

More recently, the use of anti-CD20 monoclonal antibodies (mAbs) has become increasingly popular as an alternative treatment for SRNS (and sometimes for FRNS or SDNS). Rituximab is a chimeric (i.e., combined mouse + human) mAb directed at the CD20 antigen on B lymphocytes, while its newer closely related biologic agent atumumab is a fully humanized mAb directed at this same antigenic target.[350] While originally developed to treat lymphomas, these biologics are highly effective in essentially eliminating circulating B lymphocytes, and it is thought that their mechanism of action in NS results from reductions in B lymphocyte numbers and/or function. An additional potential mechanism of action directly on podocytes has also been reported, but remains somewhat controversial.[351] Several reports have now demonstrated that anti-CD20 mAb therapy is effective in inducing remission in children with SRNS, as well as a landmark randomized controlled trial confirming rituximab's effectiveness in prolonging remission in children with FRNS.[352] Unlike most other alternative treatments for SRNS, rituximab is administered as one to four infusions over 1 to 4 weeks (usually 375 mg/m^2 over several hours) and has a prolonged duration of effect that lasts for approximately 5 to 9 months, when the B lymphocytes destroyed by this therapy gradually recover toward normal levels. The efficacy of this treatment has been reported in mostly small case series and uncontrolled trials, where it has been found generally to induce complete remission in approximately 25% of children with SRNS, and partial remission in another approximately 25% of children. Despite premedication, side effects of anti-CD20 mAb therapy are relatively common, occurring in 20% to 30% of cases, and include rash, flushing, and rarely acute respiratory distress. In most cases, slowing or stopping the infusion temporarily has allowed completion of most infusions, but in rare cases treatment has had to be discontinued. Because of its efficacy, long-term clinical benefits, and elimination of noncompliance due to drug infusion, anti-CD20 mAb treatment for SRNS (as well as for FRNS and SDNS) has gained wide popularity in recent years. Owing to its high price, however, its use remains restricted by private and/or public payers in some settings, and it is still unavailable in several areas of the world.

Sparsentan is an emerging alternative treatment for SRNS. It is an orally active dual-acting ARB combined with a highly selective endothelin type A receptor antagonist. A recent report of its use in a phase 2 trial in children and adults with primary FSGS is currently underway to determine how effective this treatment may be in reducing proteinuria in patients with primary FSGS.[353]

Another emerging alternative treatment for SRNS is LDL apheresis, which is not a drug but a device. This is an extracorporeal therapy that was originally developed to treat familial hypercholesterolemia, but its usage has been expanded for several years in Japan to also include the treatment of hyperlipidemia in NS. Strikingly, it has been reported to not only reduce the hyperlipidemia of NS very effectively, but also to induce complete remission in approximately 25% of adults and children diagnosed with SRNS, and to induce partial remission in another 25% of patients.[354] This efficacy is comparable with that of rituximab, and has thus raised several new questions about whether the dyslipidemia seen in NS is simply a consequence of disease, as has been historically thought, or whether it might in fact play a pathophysiologic role in the development or the progression of disease in patients with SRNS who remain clinically nephrotic.[314]

One potential future therapy for the treatment of SRNS (and possibly SSNS) is the Peroxisome proliferator-activated receptor-γ agonist pioglitazone. This drug is a member of a family of insulin-sensitizing compounds developed to treat type 2 diabetes in adults, but which has been found in recent reports in animals and humans to have potential in the treatment of glomerular disease.[355–358] Moreover, pioglitazone has recently been reported to directly protect podocytes against NS-related injury, and to both reduce proteinuria directly and enhance glucocorticoid-induced proteinuria reduction in experimental NS in rats.[324–359] Of greater clinical interest, introduction of pioglitazone to a child with multidrug-resistant NS resulted in a significant reduction in proteinuria (80%, but not complete remission) in the face of a simultaneous 64% reduction in overall immunosuppression.[360] Importantly, no toxicity related to pioglitazone treatment was reported in the first 2 years of treatment.

OUTCOME

The long-term outcome for childhood NS is excellent overall. The vast majority of children who develop NS will have SSNS, among which approximately one-third of children will have no relapses, one-third will have infrequent relapses (i.e., <4 relapses per year), and another third will have frequent relapses (≥4 relapses per year, or FRNS). Nonetheless, the most significant predictor of a child's likelihood of outgrowing NS without the development of CKD is entry into complete clinical remission in response to glucocorticoids. By extension, with the continuing expansion of alternative treatments able to induce complete remission among children with SRNS, we now infer that induction of complete remission by any of these treatments confers similar reduction in risk for the development of CKD or ESRD. By contrast, while the risk for the development of CKD or ESRD among children with SSNS is <5%, this risk rises dramatically for children who present with or subsequently develop SRNS, where the risk of CKD and/or ESRD approaches 50% within 5 years.[240] Lastly, mortality in NS is now <5%, and is most commonly due to infection.[233,237] Importantly, this represents a dramatic drop from the 67% mortality estimated to exist 60 to 70 years ago, prior to the development of antibiotics and introduction of glucocorticoids. Despite these major advances, deaths do still occur among children with NS, most commonly among those with SRNS who remain in relapse and are often exposed to prolonged immunosuppression.

In summary, NS is among the most common forms of kidney disease in children. While significant progress had been made in recent years in developing more effective treatments for this disease, the molecular pathophysiology of the most common form of this disease, SSNS, remains unknown, as do the mechanisms of action of many of the newer treatments that have improved the clinical outcomes of children with SRNS. Additional research to better define the molecular pathophysiology of the disease is needed to enable the development of more targeted and effective and less toxic treatments for this very common kidney disease in children.

COMPLEMENT-ASSOCIATED KIDNEY DISEASES

Complement-associated kidney diseases comprise a spectrum of conditions, which are caused by genetic mutations and/or the presence of autoantibodies disrupting regulation of the innate immune complement system.[360–363] An increasing number of diseases and conditions are linked to complement regulatory defects that result in chronic complement overactivation. Complement-associated diseases manifest in many tissues, but predominantly affect the kidneys.[364–366] Two of the most prominent examples for complement-associated kidney diseases are MPGN/C3G and atypical hemolytic uremic syndrome (aHUS).[365] In aHUS, mutations of complement regulators and inhibitory anticomplement factor H (CFH) autoantibodies are found in approximately 50% of cases resulting in complement dysregulation on endothelial surfaces.[367] By contrast, in C3G, which encompasses both C3 glomerulonephritis (C3GN) and dense deposit disease (DDD), complement dysregulation is caused by the presence of autoantibodies and/or mutations resulting in complement dysregulation in plasma (fluid phase) with subsequent deposition of C3 degradation products along and/or within the GBM.[368] The firm link of aHUS and C3G pathogenesis to complement [alternative pathway (AP)] dysregulation allows for new, specific treatment strategies directed at reestablishing complement control, and raises hopes for better patient outcomes.[266,369]

COMPLEMENT SYSTEM

The classical pathway (CP) of complement is initiated by Igs or ICs and thus is especially involved in autoimmune diseases and the Ig/IC-mediated form of MPGN. The lectin pathway (LP) is activated by repetitive carbohydrate structures found, for example, on the surface of bacteria. By difference, the AP of complement is constitutively active via a process called "tick over" and requires tight regulation. All three activation pathways converge in the activation of the central complement component C3 to C3b. Together with complement factor B (CFB)—activated by complement factor D—and properdin (CFP), C3b forms the AP C3 convertase (C3bBb), which rapidly amplifies C3 activation. The resulting activation products are involved in inflammatory responses (C3a; anaphylatoxin) or deposit on bacterial surfaces (C3b; opsonization), rendering them a target for phagocytosis. C3b also initiates C5 activation via the formation of the C5 convertase (C3bBbC3b), thus contributing to inflammation (C5a; anaphylatoxin) and initiation of the terminal pathway (TP) of complement (C5b) with the formation of the membrane attack complex (MAC) C5b-9 (Fig. 72.8).[360–363,370]

Complement regulation is provided by soluble and membrane-anchored proteins, which limit complement activation in a timely and spatial fashion both in fluid phase and on cell surfaces. The most important complement regulator, fluid-phase CFH (FH), regulates complement via (1) limiting C3b binding to surfaces and formation of the C3 convertase, C3bBb (competition), (2) inactivating/cleaving C3b (cofactor activity), and (3) accelerating the natural decay of the C3 convertase, C3bBb (decay-acceleration activity; Fig. 72.8).

While the C-terminal region of CFH is responsible for surface recognition and binding to C3b, its N-terminal region provides cofactor and decay-accelerating function.[371] Of note, aHUS—a disease characterized by uncontrolled complement activation on endothelial cells—is associated with mutations in the C-terminal region of CFH, while in C3G—a disease associated with enhanced C3 conversion in fluid phase—mutations are found in the N-terminal region (Figs. 72.8 and 72.9).[368,372]

Besides CFH there are five proteins sharing sequence and structural homology with CFH – the CFH-related proteins (CFHR) 1–5. CFHR1 regulates complement on the level of the C5 convertase, C3bBbC3b.[373] CFHR1, CFHR2, and CFHR5 form homodimers and heterodimers via shared short consensus repeats (SCRs) 1 and 2, which compete with CFH for C3b binding, thus physiologically modulating its function.[374,375] Mutations within the CFHR locus (1q32) result in copy number variations (deletions and duplications) and the formation of fusion proteins, which more avidly than nonmutant proteins form CFHR homodimers and heterodimers. These atypical complexes compete with CFH for C3b binding in a manner exceeding physiological modulation, thus preventing CFH from executing its physiological complement regulatory functions. This "CFH deregulation" has recently been identified as disease causing in C3G patients.[374,375]

In addition, cellular surfaces are protected via complement receptor 1 (CR1) and membrane cofactor protein (MCP; CD46), both acting as cofactors to CFI, via decay-accelerating factor (DAF; CD55) and via CD59, which prevents assembly of the MAC (C5b-9; Fig. 72.8).[360–363,370]

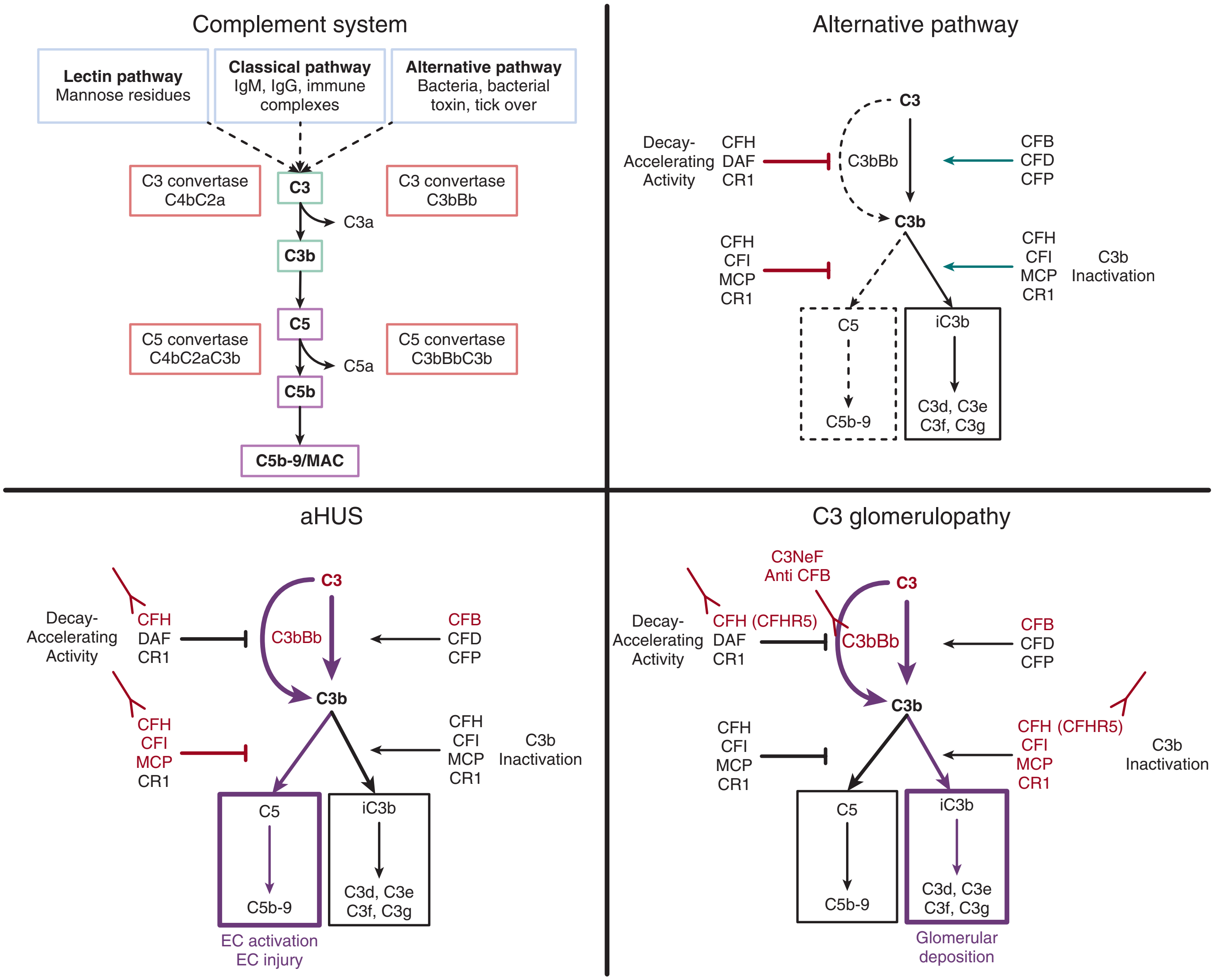

Fig. 72.8 Complement and its link to disease. Complement activation via the lectin, classical (immune complexes), or alternative pathway (constitutively active) resulting in C3, followed by C5 activation, terminal pathway activation and formation of the MAC (Cb-9; for details, see text). Alternative pathway: a small amount of C3 is constantly activated and turned into C3b. If not inhibited, C3b leads via an amplification loop to the activation of C5 followed by the induction of the terminal pathway and the formation of the MAC, C5b-9. Physiologically, C3b is immediately inactivated via plasma or endothelial cell membrane–anchored regulators (such as CFH, CFI, MCP, CR1) and cleaved to become C3d-g. aHUS: aHUS is caused by unrestricted activation of the complement alternative pathway on vascular endothelial cells. Loss-of-function mutations in genes encoding complement regulators (such as CFH, CFI, MCP), gain-of-function mutations of complement activators (such as C3, CFB), or CFH autoantibodies result in a shift of the balance between C3 activation and inactivation toward C3 activation with subsequent induction of the terminal pathway. C3 glomerulopathy: C3 glomerulopathy is caused by an overactive C3 amplification loop, which can be caused via gain-of-function mutations in components of the C3 convertase complex, C3bBb (such as C3 and CFB), loss-of-function mutations in C3 amplification regulators (such as CFH, CFHRs, CFI, MCP), autoantibodies inhibiting complement regulators (such as CFH), or stabilizing complement activators (such as CFB, C3NeF). Accordingly, excessive amounts of C3b metabolites (C3d-g) are generated, which deposit in the glomerular filter and cause a local inflammatory response. *aHUS,* Atypical hemolytic uremic syndrome; *C3NeF,* C3 nephritic factor; *CFB,* complement factor B; *CFD,* complement factor D; *CFH,* complement factor H; *CFI,* complement factor I; *CFP,* properdin; *CR1,* complement receptor 1; *DAF,* decay-accelerating factor; *Ig,* immunoglobulin; *MAC,* membrane attack complex; *MCP,* membrane cofactor protein.

C3 inactivation results in the formation of iC3b, C3dg, and C3d, degradation products of C3b, which can be interpreted as indirect measure of complement activation, but are also of biological relevance, as they interact with the adaptive immune system and deposit within and along the GBM where they initiate the histomorphological changes typical for IC-MPGN/C3G (Fig. 72.10).[376,377]

IMMUNE COMPLEX MEMBRANOPROLIFERATIVE GLOMERULONEPHRITIS AND C3 GLOMERULOPATHY

MPGN is characterized by the histopathological pattern of mesangial and endocapillary proliferation and thickening

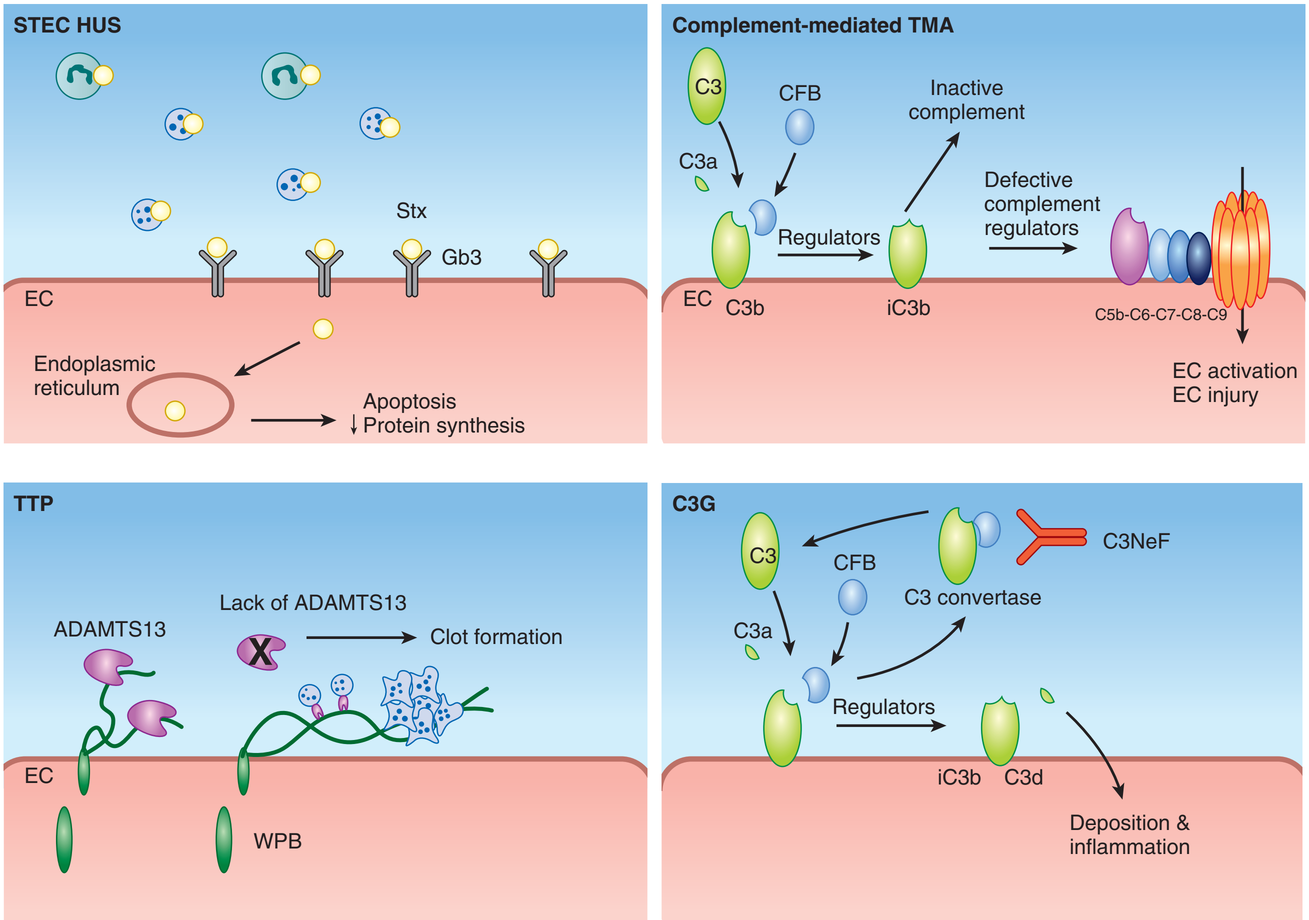

Fig. 72.9 Mechanism of complement-mediated disease. STEC–HUS is caused by bacterial exotoxins produced, for example, by *Escherichia coli* O157:H7. Toxin binds to the endothelial expressed Gb3 receptor, is internalized, transferred to the ER, and results in apoptosis or necrosis interfering with protein translation. Complement-mediated TMA: Complement-mediated TMA is caused by unrestricted activation of the complement alternative pathway on vascular endothelial cells. In primary TMA, congenital or acquired defects in the activation or regulation of the complement alternative pathway cause unrestricted activation of the terminal pathway. In secondary TMA, complement overactivation in secondary TMA can be caused, for example, by inflammation, drug toxicity, shear stress. TTP: TTP is caused by the genetic or acquired functional defect in the serine protease ADAMTS13, which physiologically cleaves vWF multimers upon release from the vascular endothelium. With impaired ADAMTS13 function, vWF multimers persist and provide a nidus for the formation of platelet rich clots. C3G: C3G is caused by an overactive C3 amplification loop caused by mutations or autoantibodies affecting the physiological decay of the C3 convertase complex, C3bBb. Excessive amounts of activated C3 (C3b), and subsequently of C3b metabolites (C3d-g), are generated, which deposit in the glomerular filter and cause a local inflammatory response. *ER,* Endoplasmic reticulum; *STEC–HUS,* Shiga toxin *Escherichia coli*–hemolytic uremic syndrome; *TMA,* thrombotic microangiopathy; *TTP,* thrombotic thrombocytopenic purpura; *vWF,* von Willebrand factor.

of the capillary with the formation of double contours. This results from the deposition of Ig/ICs and/or complement proteins in the mesangium and/or along the capillary wall of the glomerulus.[378,379]

MPGN has recently been reclassified into (1) a primary complement-mediated C3G with predominant C3 deposition, and (2) a secondary Ig/IC-mediated glomerulonephritis with predominant IgG/IgM deposition on immunohistochemistry (IC-MPGN).[379,380] This new classification system encompasses all glomerular lesions with predominant C3 staining (≥2 orders of magnitude stronger than any other immune reactant on a scale ranging from 0 to 3),[380] and, in addition, enables the inclusion of cases with C3 deposition without a membranoproliferative pattern.[380] C3G is considered a primary complement disease, where complement deposition is a result of defective control of the AP of complement.[381,382] By contrast, IC-MPGN may be idiopathic or secondary to a spectrum of underlying diseases or conditions including infections, autoimmune diseases, malignancies, or monoclonal gammopathy (Table 72.7), and is characterized by Ig/IC deposition in the kidney with subsequent activation of the CP.[383–386] The diagnosis of C3G is based on LM and EM.[380] C3GN and DDD can present on LM as MPGN, diffuse proliferative/mesangioproliferative glomerulonephritis, and necrotizing or crescentic glomerulonephritis.[380,387] C3GN and DDD can be differentiated by EM: C3GN and DDD are characterized by discrete C3 deposits in the mesangium and along the capillary walls, DDD by C3 deposits in the mesangium and within the GBM.[380]

PATHOGENESIS

During recent years major progress has been made in the understanding of IC-MPGN and C3G pathogenesis via the investigation of disease causing genetic and autoimmune

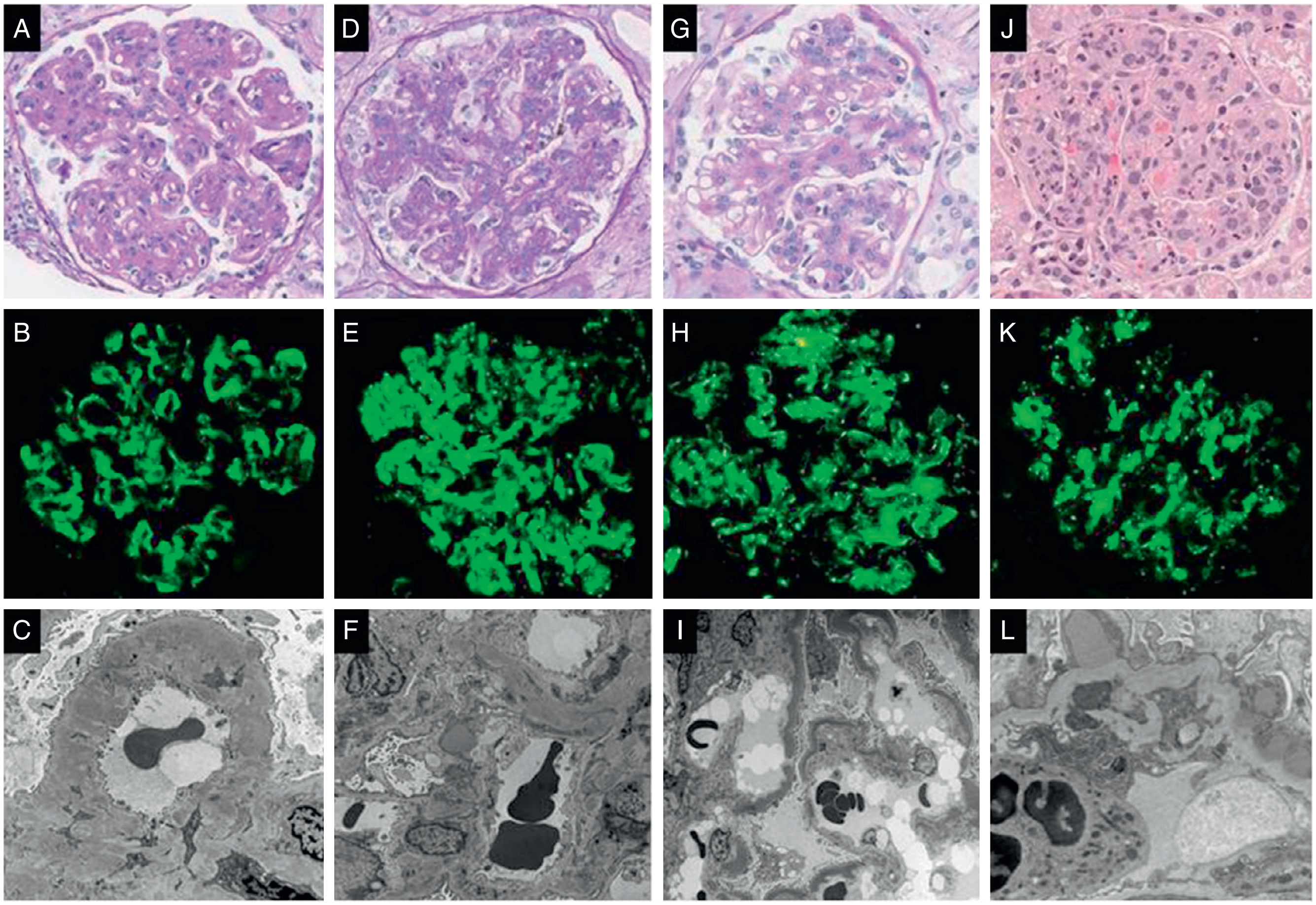

Fig. 72.10 Histopathologic characteristics in membranoproliferative glomerulonephritis (MPGN) – immune complex-mediated (IC-MPGN) and C3 glomerulopathy (C3G). Also shown in this figure and frequently in the differential diagnosis are dense deposit disease (DDD) and postinfectious glomerulonephritis (PIGN). Panels highlight the microscopic features of (A–C) MPGN, (D–F) C3G, (G–I) DDD, and (J–L) PIGN by light microscopy *(top row)*, immunofluorescence microscopy *(middle row)*, and electron microscopy *(bottom row).* MPGN has typical features of glomerular lobular accentuation, mesangial expansion by matrix and cells, endocapillary hypercellularity, and capillary wall double contouring (A); coarse granular peripheral capillary and variable mesangial staining for C3 (B), invariably with one or more immunoglobulins (Ig), usually IgG and/or IgM; and mesangial and subendothelial deposits with cellular interpositioning and new basement membrane formation (C). C3G often shows a membranoproliferative (D) or mesangial proliferative pattern; dominant C3 staining (with little or no staining for immunoglobulin) in the mesangium and capillary walls (E); and a combination of mesangial, subendothelial, and subepithelial deposits, and often including subepithelial hump-type deposits (F). DDD can show an MPGN pattern (G) just as commonly as a mesangial proliferative, proliferative/exudative, or crescentic pattern; dominant glomerular C3 staining, often in a pseudolinear capillary pattern with mesangial coarse granules or rings (H); and pathognomonic electron dense transformation of the basement membrane (I), often also with mesangial deposits. PIGN shows a proliferative glomerulonephritis with prominent neutrophilic infiltration (J); mesangial and often spotty granular capillary staining for C3 (K), often with IgG; and subepithelial hump-type deposits with scattered mesangial and subendothelial deposits (L). A, D, and G: Periodic acid–Schiff ×400; J: hematoxylin and eosin ×400; B, E, H, K: C3 ×400; C and L: electron microscopy ×10,000; F: electron microscopy ×6000; I: electron microscopy ×4000.

defects affecting activation and regulation of the complement system—in particular its AP—identified in patients both in vitro and in vivo, and via testing of new complement modulating drugs in animal models recapitulating such defects.[369,388,389]

Principles of C3G pathogenesis can be derived from spontaneously occurring and genetically engineered animal models.[368] A lesion consistent with MPGN II/DDD developed in pigs with complete CFH deficiency due to a spontaneously occurring truncating CFH mutation. Notably, administration of CFH (via plasma or as purified protein) delayed disease progression.[390–394] A similar phenotype developed also in genetically engineered mice ($cfh^{-/-}$) with complete CFH deficiency.[395] In both animal models, C3 accumulates along the GBM in the setting of systemic hypocomplementemia, and immune EM identified complement components (i.e., C3, C5, C5b-9) within these deposits.[390,395] An essential role for the AP in disease pathogenesis was demonstrated by the observation that the C3G phenotype in the $cfh^{-/-}$ mice could be rescued if mice were also genetically deficient in the AP activation protein, factor B.[395] Utilizing mice lacking complement factor I (CFI) ($cfh^{-/-}$; $cfi^{-/-}$), it was demonstrated that renal C3 deposition is complement factor I (FI) dependent as mice did not develop renal C3 deposition unless they were treated with FI containing serum.[396]

In humans, the phenotype associated with complete CFH deficiency was not limited to C3G but included also the TMA aHUS. Besides the differences in the complement systems of mice versus man, this observation indicates that AP defects can give rise to a spectrum of renal pathologies. aHUS-associated FH mutations typically occur within the CFH

Table 72.7 Causes of Secondary Membranoproliferative Glomerulonephritis

Infectious diseases: bacterial/viral/protozoal	• Hepatitis B, C, Epstein–Barr virus, human immunodeficiency virus • Endocarditis/visceral abscesses • Infected ventriculoatrial shunts/empyema • Malaria, schistosomiasis, mycoplasma • Tuberculosis, leprosy • Epstein-Barr virus infection • Brucellosis
Systemic immune diseases	• Cryoglobulinemia • Systemic lupus erythematosus • Sjögren syndrome • Rheumatoid arthritis • Hereditary deficiencies of complement components • X-linked agammaglobulinemia
Neoplasms/dysproteinemias	• Plasma cell dyscrasia • Fibrillary and immunotactoid glomerulonephritis • Light-chain deposition disease • Heavy-chain deposition disease • Light- and heavy-chain deposition disease • Leukemias and lymphomas (with cryoglobulinemia) • Waldenstrom macroglobulinemia • Carcinomas, Wilms tumor, malignant melanoma
Chronic liver disease	• Chronic active hepatitis (B and C) • Cirrhosis • Alpha-1-antitrypsin deficiency
Miscellaneous	• Thrombotic microangiopathy • Sickle cell disease • Partial lipodystrophy • Transplant glomerulopathy • Niemann–Pick disease (type C)

surface recognition domains SCR18–20 (CFH C terminus). In CFH null mice with transgenic expression of a mutant FH protein functionally mimicking human aHUS-associated CFH mutations (CFHΔ16-20; $cfh^{-/-}$.FHΔ16-20), systemic hypocomplementemia was ameliorated as the mutant protein still had an intact complement regulatory domain (CFH N terminus). These animals spontaneously developed aHUS (not C3G), and the aHUS phenotype was dependent on C5 activation[397,398] as opposed to the C3G phenotype observed in the $cfh^{-/-}$; $cfi^{-/-}$ mice substituted with CFI containing serum (see earlier text), which was C3 dependent.[399] Mice lacking CFH expression by hepatocytes (hepatocyte $cfh^{-/-}$) showed significantly reduced but not absent peripheral CFH levels. At baseline, these mice did not develop C3 deposition along the GBM, but mesangial C3 accumulation consistent with C3G. However, after induction of complement activation via nephrotoxic serum they developed C5-dependent TMA.[400]

Taken together, these studies indicate that the aHUS phenotype depends on defective endothelial surface complement regulation combined with an (partially) intact complement system allowing for the presence and activation of sufficient amounts of C5 to induce the TP-mediated endothelial damage, whereas the C3G phenotype depends on

Table 72.8 Complement Activation Markers in Serum and Tissue Biopsy

Measurement	Interpretation
Markers in Serum	
Increased C3 split products (C3d)	Sign for C3b inactivation
Decreased Bb	Sign for alternative pathway activation
Increased sC5b-9	Sign for terminal pathway activation
Markers in Tissue Biopsy	
Increased C3(c)	Sign for complement deposition
Increased C3 split products (C3d)	Sign for C3b inactivation
Increased sC5b-9	Sign for terminal pathway activation

fluid phase complement (AP) dysregulation with the generation of C3-sufficient amounts of C3b split products to deposit along and within the GBM (Fig. 72.8).

In patients with IC-MPGN/C3G, a central role for complement in disease pathogenesis has long been recognized, as decreased C3 levels are a hallmark feature of the clinical presentation. Mutations or autoantibodies result in the loss of control of AP C3 convertase via one or more of the following mechanisms:

- Autoantibodies to components of the AP C3 convertase stabilizing this complex, thus prolonging its natural decay.
- Mutations in or autoantibodies to CFH resulting in the absence or loss of function of CFH.
- Mutations in C3 or CFB resulting in their gain of function with an exceedingly stable AP C3 convertase.

All these scenarios result in an enhanced rate of C3 activation with subsequent enhanced levels of MAC/C5b-9 as found in C3G patients.[372]

In further support of a complement-mediated pathogenesis, the presence of AP and TP proteins was demonstrated in the glomerulus of patients with C3G via a combination of laser microdissection and mass spectrometry.[381,382]

The association of low C3 and MPGN was first reported in 1965, leading to the hypothesis of the presence of a C3 cleavage factor in GN patients.[401] Subsequently, an antibody binding and stabilizing the AP C3 convertase (C3bBb), thus enhancing C3 activation and C3b generation, was detected and named C3 nephritic factor (C3NeF).[402] C3NeF was detected in 86% of DDD and 46% of C3GN patients, respectively, and is associated with decreased C3 levels.[372] C3NeF levels show significant intraindividual and interindividual variability and do not necessarily correlate with disease activity or treatment status.[403]

Detection of C3NeF is challenging, and current assays involve measurement of IgG binding to the C3 convertase or the detection of the ability to stabilize the AP C3 convertase.[404,405] C3NeF was also reported in SLE,[406–408] PIGN,[382] meningococcal meningitis,[409] as well as in healthy individuals.[404] The finding of C3NeF does not exclude the coexistence of complement mutations, and despite C3NeF positivity, comprehensive complement diagnostic workup (Table 72.8) is required.[410]

Recently, additional nephritic factors—autoantibodies to C4[411] and C5[412]—have been identified in rare cases of C3G patients. Other autoantibodies also detected in C3G patients include anti-CFB and anti-C3b antibodies.[413,414] Nephritic factors bind to and stabilize their respective convertase complexes, thus prolonging the natural decay and rendering them inert for CFH-mediated decay acceleration. By contrast, anti-CFH autoantibodies—frequent in aHUS—can rarely also be detected in C3G patients: while aHUS-associated anti-CFH autoantibodies recognize the C terminus of CFH,[415] C3G-associated anti-CFH autoantibodies such as the CFH mini autoantibody (i.e., antibody fragment-like λ-light chain dimer) block the interaction of CFH and C3b, thus abolishing the cofactor activity of CFH in cleaving C3b.[416,417]

To date, complement mutations have been reported in the following complement genes: CFH, CFHR5, CFI, MCP/CD46, C3, and CFB (Fig. 72.8).[372,374,418–431]. In a series of 134 patients with MPGN I, C3G, and DDD, mutations in *CFH, CFI,* and *MCP/CD46* were detected in 16.6%, 17.2%, and 19.6% patients, respectively (Fig. 72.8).[372]

More recently, genetic variants including CNVs with deletions (internal), duplications, and the formation of hybrid genes within the gene locus encoding for the CFHRs (1q32) have been associated with C3G.[432,433] CFHR5 nephropathy was defined as a specific subtype of C3G arising from an internal duplication within the *CFHR5* gene.[422] CFHR5 nephropathy presents in childhood with persistent microscopic hematuria, synpharyngitic gross hematuria, and a strong family history of ESRD. The disease has been identified and reported primarily in the Cypriot population, but also in two patients of non-Cypriotic background.[422,434–436]

Several more mutations and hybrid variants within the *CFHR* gene locus have been identified and linked to C3G (Fig. 72.8). The function of CFHR CNV and hybrid gene C3G was unclear until it was recognized that CFHR1, CFHR2, and CFHR5 via (almost) identical N-terminal SCR1 and SCR2 were naturally forming homodimers and heterodimers and as such competing with CFH for surface and C3b binding. Under physiological conditions, this competition provides a physiological balance and allows for titration of CFH function. However, in the presence of *CFHR* mutations with the resulting CNV and hybrid gene/protein formation, nonphysiological multimers with enhanced affinity for the shared binding partners form. This scenario results in the loss of CFH complement regulatory function and is termed "CFH deregulation."[401,402]

Risk haplotypes were identified in CFH, C3, and MCP/CD46, with the CFH Y402H haplotype more frequently reported in DDD and the MCP/CD46 652A4G polymorphism in C3G and MPGN I.[372,418] The presence of two or more complement haplotypes significantly increased the risk of disease manifestation.[372,418] Of note, in a recent study in 140 IC-MPGN and C3G patients, two important findings were made: (1) different from previous claims, mutations in genes encoding AP proteins were found in both subgroups regardless of their histopathological pattern (17% and 18%, respectively); (2) these mutations alone did not increase the risk of disease manifestation, unless mutations combined with specific susceptibility variants (i.e., MCP/CD46 c.-366G>A in IC-MPGN; CFH V62 and thrombomodulin [THBD]/CD141 A473V in C3G).[437]

Taken together, these findings support the central role of complement defects for IC-MPGN/C3G pathogenesis and highlight the complexity of the underlying genetic and/or autoimmune defects. They also identify the crucial role of complete genetic and biochemical analyses in the advanced management of IC-MPGN/C3G patients.

CLINICAL MANIFESTATIONS

MPGN and C3G (C3GN and DDD) are rare diseases with an annual incidence of 1 to 2 per million (both pediatric and adult).[438,439] Once thought to be primarily pediatric diseases, a recent study noted that the mean age at diagnosis for MPGN and C3G was 20.7 ± 16.8 years, 18.9 ± 17.7 years, and 30.3 ± 19.3 years, respectively.[372] Within pediatric patients, DDD presents at a younger age with 70% of patients diagnosed prior to age 15.[440] There is a slight bias with males predominating in MPGN and C3G.[372,438]

The clinical manifestation of MPGN and C3G typically includes hematuria, proteinuria (up to nephrotic range), hypertension, and impaired kidney function with significant overlap of symptoms between disease subtypes.[441,442] NS at disease onset has been reported in 40% to 65% of patients with MPGN, 30% to 40% with DDD, and 25% to 40% with C3GN, respectively. Regardless of disease subtype, microscopic hematuria has been reported in 40% to 75%, gross hematuria in 10% to 20%, and nonnephrotic proteinuria in 30% to 40% of patients. Acute glomerulonephritis is common and is accompanied by hypertension in 30% to 60%, and impaired kidney function in 20% to 50% of patients.[372,378,438,443] Although rare, rapidly progressive glomerulonephritis (RPGN) has also been reported in patients with MPGN and C3G.

Hypocomplementemia is long considered a key feature in MPGN,[442] and differential findings in serum C3 and C4 levels were used to distinguish between MPGN I (low C3 and low C4) and other types of MPGN (low C3 and normal C4).[444] Applying the current classification, IC-MPGN as IC-mediated disease is commonly associated with depressed C3 and C4 levels, whereas C3GN and DDD as primary complement-driven diseases are associated with depressed C3 but normal C4 levels.[372,436] There were, however, important caveats noted in these studies as patients did not always follow the expected pattern of C3/C4 levels, and a significant subgroup of patients presented even with normal C3 levels.[372,438] It appears that while assessment of C3 and C4 levels is suggestive of IC-MPGN or C3G (C3GN and DDD), suppressed C3 or C4 levels are not sufficient to establish these diagnoses.[372,438] Of note, patients with CFHR5 nephropathy are also reported to have normal serum C3 levels.[445]

In addition to decreased C3 levels, C3G patients—both C3GN and DDD—were found to also present with decreased CFB levels, increased C3b breakdown products, and elevated soluble C5b-9 levels.[446]

Extrarenal manifestations of C3G are uncommon and include acquired partial lipodystrophy (aPL)[447] and ocular C3 deposits[424] similar to soft drusen seen in age-related macular degeneration (AMD). aPL is presumably caused by complement dysregulation on adipocytes and results in the breakdown of the fat tissue of the upper parts (i.e., head and trunk) of the body. aPL typically precedes the renal manifestation of MPGN and may be associated with decreased C3 levels and the presence of C3NeF.[447,448] Soft drusen consist of C3 split products and CFH,[449,450] may be associated with polymorphisms/mutations in complement genes,[391] and may

Table 72.9 Spectrum of Thombotic Microangiopathies

Thrombotic Microangiopathy (TMA)	Role of Complement	Diagnostic Test
Thrombotic thrombocytopenic purpura	Elevated C5b-9	Biochemistry
Hemolytic uremic syndrome		
STEC–HUS/eHUS	Transient activation possible	Biochemistry; if persistent C activation, consider genetics
Atypical HUS	Continuous (mutation and abs.)	Biochemistry and genetics and antibody screen
Other hereditary forms of TMA	N/A	Biochemistry; if persistent C activation, consider genetics
DGKE		
INF2		
TMA postsolid organ transplantation	Transient or continuous (mutation and abs.)	Biochemistry and genetics and antibody screen
De novo TMA		
TMA recurrence		
TMA posthematopoietic stem cell transplantation/bone marrow transplantation	Transient or continuous (mutation and abs.)	Biochemistry and genetics and antibody screen
TMA associated with pregnancy	Transient or continuous (mutation and abs.)	Biochemistry and genetics and antibody screen
TMA associated with glomerulonephritis	Continuous C activation possible (mutation and abs.)	Biochemistry and genetics and antibody screen
TMA associated with drugs	Transient C activation possible	Biochemistry; if persistent C activation, consider genetics
TMA associated with metabolic disease	C activation possible	Biochemistry; if persistent C activation, consider genetics
TMA associated with infections (e.g., *Streptococcus pneumoniae*, human immunodeficiency virus, influenza)	Transient C activation possible	Biochemistry; if persistent C activation, consider genetics
TMA associated with malignant hypertension	Transient C activation possible	Biochemistry; if persistent C activation, consider genetics

Persistent: after initial cause has resolved; Biochemistry: Complement activation tests (see Table 72.3).
eHUS, Escherichia coli-induced hemolytic uremic syndrome; *STEC,* Shiga toxin *Escherichia coli*.

also be associated with the risk of developing AMD.[451] The long-term risk for visual problems is about 10%.[452]

DIAGNOSIS

MPGN and C3G most commonly present as acute nephritis or NS. More subtle presentations are also seen, and there is considerable clinical overlap with the presentations of other forms of renal disease. Complement analysis (e.g., C3 and C4 levels, CFB levels, C3b breakdown products, soluble C5b-9) may be useful; however, normal results do not exclude the diagnosis.[372] The measurement of nephritic factors or other complement autoantibodies (e.g., anti-CFH, CFB, C3b) may also be supportive; however, these assays are neither standardized nor widely available and lack diagnostic specificity. Genetic complement testing may be valuable and is—in particular in the background of the increasing number of identified disease-causing mutations in the CFHR locus—highly recommended. Ultimately, the diagnosis is finalized by renal biopsy (Table 72.9).

The differential diagnosis of a setting of nephritic/NS with hypocomplementemia (in particular, low C3 and C4 levels) is narrow and mainly includes IC-MPGN/C3G and SLE. Extrarenal disease manifestations such as arthralgias; rashes; neurologic, hepatic, lung, or cardiac involvement combined with laboratory evidence of immune dysregulation including antibody-mediated hemolysis, leukopenia, coagulopathy, myositis, or hepatitis support a diagnosis of SLE as opposed to MPGN. Although exceedingly rare in children, cryoglobulinemia may also present with multiorgan involvement and systemic findings. IC-mediated (secondary) forms of MPGN (IC-MPGN) may present with decreased C3 and C4 levels.[453] The combination of decreased C3 and normal C4 levels points toward PIGN and shunt nephritis. The history of a recent episode of pharyngitis, possibly accompanied by elevated antistreptolysin (ASO) titer or anti-DNase titers, in combination with decreased C3 levels, defines the signature of PIGN. The differential diagnosis of C3G may be considered only if the clinical course is atypical, that is, the C3 level does not return to normal within 8 to 10 weeks of presentation.[382] In such cases—in particular, in the background of a family history of unexplained ESRD—a kidney biopsy should be considered.[382] As MPGN and C3G may present with normal complement levels, they should also be considered in any differential diagnosis for glomerular disease even if the patient's C3 and C4 levels are normal. CFHR5 nephropathy should be considered even in patients outside Cyprus for patients with persistent microscopic hematuria, and synpharyngitic gross hematuria with or without a family history of ESRD.

TREATMENT

To date, no treatment standards exist for patients with IC-MPGN or C3G, and current treatment recommendations are mainly based on small size single-center studies and expert opinions.

ACE inhibitors or Ang II receptor antagonists (ARBs) are used in many patients with IC-MPGN/C3G due to their antiproteinuric and antihypertensive effect, and their use was associated with better renal survival.[372] Treatment of arterial hypertension and CKD/ESRD follows established standards.

Prednisone is considered the first-line agent in patients with IC-MPGN who present with (nephrotic range) proteinuria with or without renal failure. In children a beneficial effect of long-term low-dose prednisone treatment was shown for proteinuria and long-term renal outcome even though only a subgroup of patients will (partially) respond. A typical treatment regimen consists of a prednisone dose of 60 mg/m^2 (maximal 60 mg/day) × 4 weeks (induction) followed by 40 mg/m^2 on alternate-day for 6 to 12 months (maintenance). Of note, no beneficial effect was shown in patients with DDD.

MMF was administered to IC-MPGN and C3G patients alone or in combination with prednisone and was successful in achieving sustained improvement of proteinuria and renal function after 1 year.[454] In patients with primary GN, including MPGN, a partial or complete remission was achieved in 70% after 1 year.[455] In another study, in pediatric patients with MPGN I treated with MMF and prednisone for a mean of 40 months, 56% achieved complete or partial remission. However, treatment failed in all patients with decreased C3 levels.[456] In addition, no treatment benefit was found in patients with DDD. In a recent analysis, renal outcome in 22 IC-MPGN/C3G patients treated with a combination of steroids and MMF was compared with 18 patients treated with different immunosuppressive regimens including steroids alone or in combination with cyclophosphamide. While in principle immunosuppressive (as opposed to no) treatment was beneficial, renal outcome in patients commenced on the combination of steroids and MMF was significantly better than in patients commenced on any other single immunosuppressive regimen.[457]

CNIs (cyclosporine A and tacrolimus) were found to have questionable benefit in IC-MPGN/C3G patients. CNIs were used in steroid-resistant MPGN patients. One study demonstrated efficacy of the combination of low-dose steroids and CNI in patients with refractory MPGN with respect to reduction of proteinuria and decline in renal function in 94% after 2 years.[458] In five adult steroid and cyclophosphamide-resistant MPGN patients, the combination of steroids and tacrolimus resulted in complete or at least partial remission in all patients after 24 weeks or longer.[459,460] In two children with MPGN and suboptimal response to long-term prednisone, a rapid and complete remission was achieved with tacrolimus.[461] In another two patients with DDD low-dose prednisone and cyclosporine A were able to induce remission.[462,463] In another study no treatment benefit of CNI was found in DDD patients.[452]

In analogy to other autoimmune diseases, the pathogenetic role of C3NeF and other complement antibodies (see earlier discussion) has prompted the use of B-cell depleting agents such as rituximab in IC-MPGN/C3G. Several case reports of MPGN (in particular, IC-MPGN) patients describe partial or complete remission after administration of rituximab (in 50% combined with steroids).[464–468] By contrast, two DDD patients did not show any improvement in proteinuria or renal function, but were rescued by eculizumab therapy (discussed later).[469,470]

With the emerging role for the complement AP in IC-MPGN/C3G pathogenesis, complement-targeted treatment strategies are increasingly considered for such patients. Treatment strategies include plasma infusion (PI) or exchange (PLEX) as described for patients with MPGN I and DDD at first disease manifestation and recurrence in native and graft kidneys. Partial or complete remission is reported in about 80% of published cases.[466,471–483]

In one patient with IC-MPGN, despite transient improvement of renal function and proteinuria, clinical condition worsened including seizures, respiratory distress, and sustained anuria. Because of a CFHR1 deletion (without CFH antibodies), low C3 and elevated sC5b-9, indicating AP activation, and TMA-related symptoms, the patient was switched from PLEX to eculizumab. This resulted in dramatic clinical improvement including proteinuria and renal function.[473]

Three case reports, including two siblings with DDD caused by a CFH mutation (CFH SCR4), and one patient with MPGN I with an MCP/CD46 mutation were treated for more than 3 years with plasma. Fresh frozen plasma as maintenance therapy, individualized according to the patient needs (e.g., intercurrent infection), was able to prevent disease progression and stabilize kidney function.[423,427,473] However, PI might not be enough as induction therapy.[484]

Recently, the complement blocker eculizumab has become available. Eculizumab is a humanized monoclonal anti-C5 antibody, which prevents C5 activation and TP induction with the formation of the terminal complement complex (C5b-9).[485] While so far approved only for the use in paroxysmal nocturnal hemoglobinuria (PNH), aHUS, and myasthenia gravis, eculizumab has also been used in the treatment of IC-MPGN/C3G, and recently reports of its successful use for this indication were published either in context of investigator-initiated clinical trials or as (single) case reports.[470,473,486,487] IC-MPGN and C3G (C3GN and DDD) patients were treated for disease manifestation due to genetic mutations or the presence of autoantibodies such as C3NeF in native or graft kidneys at first presentation or disease recurrence. Treatment response spans from “no” to “partial” or “full” response.[486] Although a favorable treatment response seems to be associated with elevated sC5b-9 levels and shorter disease duration,[488] criteria clearly identifying individuals with the potential to respond to eculizumab treatment are still lacking.

Long-term outcome data for MPGN patients—in particular, applying the new classification—are limited. Disease progression in MPGN patients occurs regardless of the underlying MPGN subtype in 40% to 50% within 10 years of diagnosis.[441,442] Renal function (eGFR), nephrotic range proteinuria, hypertension, and age at presentation are negatively correlated with long-term renal outcome.[372,438,441,442] In addition, the presence of glomerular crescents or DDD by biopsy is an independent predictor of disease progression.[438]

Disease recurrence of MPGN and DDD in renal grafts has been well described with recurrence rates ranging from 10% to 100%. However, the impact of disease recurrence on allograft survival remains controversial. Applying the new classification, MPGN recurrence rates for renal grafts are reported as 43%, 55%, and 60% for MPGN, DDD, and C3GN, respectively. The impact on allograft survival was not reported.[372] In another study with 13 transplants all of the six DDD and four of the seven C3GN patients had disease recurrence with graft failure due to recurrence in three of the DDD, and three of the four C3GN patients with recurrence.[438] Overall, allograft survival after 5 years was 69%.

POSTINFECTIOUS (POSTSTREPTOCOCCAL) GLOMERULONEPHRITIS

PATHOGENESIS

The new MPGN classification (discussed earlier) also includes acute postinfectious (PIGN) or poststreptococcal glomerulonephritis as part of the MPGN disease spectrum.[380] PIGN is characterized by strong mesangial C3 with codominant IgG staining (IF), and the formation of mesangial, subendothelial, and subepithelial humps (EM).[380] Of note, some patients present with C3 deposition only without IgG deposition and/or humps.

PIGN can be diagnosed clinically without a renal biopsy.[489] It usually follows a bacterial infection—typically pharyngitis or pyoderma—with β-hemolytic group A *Streptococcus pneumoniae* with a latency of 7 to 14 days and is usually associated with an increased antistreptolysin (ASO) titer. The bacterial infection results in the formation of circulating ICs, which deposit in the kidneys and cause an inflammatory response with leukocyte recruitment and complement activation via both the CP and AP.[490,491] While numerous antigens have been considered as being responsible for streptococcal nephritogenicity,[492] currently two antigens, the nephritis-associated plasmin receptor (NAPlr) and a cationic cysteine proteinase known as streptococcal pyrogenic exotoxin B, are favored. Of note, both of these fractions are capable of activating the AP.[491]

As infections—such as streptococcal infections—are also common triggers for the manifestation of IC-MPGN/C3G, and the PIGN typical humps also occur in biopsies of patients with IC-MPGN/C3G, the differential diagnosis can be difficult and mostly relies on clinical observations.[490,491]

CLINICAL MANIFESTATIONS

PIGN is the most common cause of acute glomerulonephritis in childhood with a median age at presentation of 6 to 8 years. Occurrence prior to age 2 is rare. The male-to-female ratio of PIGN is 2:1, but is not different with respect to the presence or absence of a preceding pharyngitis or pyoderma.[490] Three phases of disease manifestation can be distinguished: (1) the latent phase, that is, the phase from bacterial infection to disease manifestation (1–2 weeks); (2) the acute phase (about 2 weeks); and (3) the recovery phase, that is, the phase of resolution of edema, normalization of blood pressure, resolution of hematuria and proteinuria, and normalization of C3 levels (4–6 weeks).[490]

The clinical manifestation of PIGN is characterized by a nephritic syndrome with gross hematuria, proteinuria, edema hypertension, and low C3 levels.[490,491] PIGN usually follows a benign course and has an excellent prognosis in children, which significantly worsens in elderly patients or in individuals with preexisting renal disease.[490,491] In rare cases, PIGN can also be associated with AKI,[492] ESRD,[493,494] and death.[490]

A subgroup of patients clinically diagnosed as PIGN follows an atypical course showing ongoing C3 consumption with low C3 levels, persistent hematuria and/or proteinuria beyond the expected period of recovery, and progressive decline in renal function.[495] Motivated by the clinical similarity between IC-MPGN/C3G and atypical PIGN, and the ongoing signs of complement activation with decreased C3 levels in PIGN patients with incomplete recovery, PIGN patients were screened for complement defects, and AP mutations and autoantibodies (i.e., C3NeF) were detected in patients with "atypical" PIGN.[382,496,497]

In a recent study of 33 pediatric patients clinically diagnosed as PIGN, reassessment of the renal biopsies resulted in the reclassification of 24% of patients as IC-MPGN/C3G according to the new MPGN classification.[495] Review of the clinical course of the reclassified patients revealed that 72% of patients with typical PIGN recovered completely as compared with only 25% of IC-MPGN/C3G patients. PIGN patients had a more severe disease course at baseline but showed a more favorable long-term outcome. Of note, C3 levels of PIGN patients were significantly lower at baseline than C3 levels of IC-MPGN/C3G patients—a difference that disappeared at later follow-up with a trend toward normalization in both groups.[495] While these findings are in keeping with a disease spectrum spanning from IC-MPGN/C3G to PIGN, the study lacks information on possibly underlying complement defects. Prospective trials including full complement workup in patients with typical and atypical PIGN will be required to fully proof this new concept.

TREATMENT

Typical PIGN takes a benign course with excellent prognosis, and thus does not require specific treatment. Supportive measures mainly include management of edema, and blood pressure control. Still controversial are a potential treatment benefit of antibiotics to treat the underlying bacterial infection, and of immunosuppressants, which are typically used in cases with RPGN and the finding of crescents on renal biopsy.[490,491]

MEMBRANOUS NEPHROPATHY

MN, which is also sometimes referred to as membranous glomerulonephritis, is a relatively rare cause of glomerular disease among children. It may present with proteinuria, hematuria, or in more severe cases with AKI and/or NS. In contrast to the low frequency in children, MN is the most common primary cause of NS among adults, where it represents one-fourth to one-third of all adults presenting with NS.[498–502] MN can occur either as a primary disease (i.e., idiopathic MN) or as a disease secondary to a variety of causes such as infections, medications, or autoimmune diseases.

EPIDEMIOLOGY

The incidence of MN is strikingly different between children and adults. Among adults it occurs at a frequency of about 1.2/100,000 population per year, whereas among children it occurs at only about 0.05–0.1/100,000 population per year, or approximately 10 to 20 times less commonly than in adults. Moreover, while MN represents 25% to 35% of new cases of NS among adults, it represents less than 5% of new cases of NS among children.[251,439,503–505] However, within the pediatric age group the incidence of MN is sixfold to 20-fold higher in children who present in adolescence (i.e., 13–18 years) than in children who present under the age of 12 years.[506–508]

Based on data from the US Renal Data System (USRDS), MN has a similar incidence among African-American versus Caucasian children, and occurs approximately equally among girls and boys.[509]

PATHOGENESIS

MN can be either a primary disease, termed idiopathic MN, or secondary to a wide variety of causes. The many causes of MN in children are shown in Table 72.10. This table highlights that MN is essentially a pathologic finding that likely represents a variety of disease states.[503,510]

Idiopathic MN is characterized by IC formation associated with the GBM (Fig. 72.11). Four stages of disease have been described, which begins with subepithelial deposits on a normal GBM (stage 1), followed by the deposits extending projections into the GBM (stage 2), then incorporation of the deposits into the GBM (stage 3), and ultimately remodeling of the GBM associated with absent deposits (stage 4).[511] A more detailed discussion of the histology of MN is described elsewhere. Unfortunately, despite the variety of histologic features known to occur in MN, these features have not been able to be used to accurately predict clinical outcomes in children.[510,512,513]

Recently, the molecular cause for many cases of idiopathic MN has been identified. Investigators discovered that almost 70% of patients presenting with idiopathic MN (but not secondary MN) had antibodies directed against phospholipase A2 receptor (PLA2R), a transmembrane glycoprotein receptor on the surface of podocytes.[514] Indeed, these same antibodies could be eluted from renal biopsy samples of patients with idiopathic MN (but not secondary MN).[514] By contrast, PLA2R staining was reported in only 45% of children with MN, suggesting that the sensitivity of PLA2R staining of renal biopsies in children is lower than in adults with idiopathic MN.[515] Importantly, anti-PLA2R antibody titers have been demonstrated to be an excellent biomarker of disease activity in patients with MN, with levels being increased during relapses, absent during remissions, and rituximab-induced depletion of anti-PLA2R titers correlating with clinical responses.[516,517]

CLINICAL MANIFESTATIONS

The clinical presentation of children with MN can range widely, from asymptomatic proteinuria to SRNS. Unfortunately, due to the very low incidence of disease in children, few data are available about the relative distribution across this clinical spectrum among children.

Table 72.10 Causes of Membranous Nephropathy in Children

	Secondary				
Primary	**Infections**	**Medications**	**Autoimmune Diseases**	**Neoplasms**	**Miscellaneous**
Idiopathic membranous nephropathy	Hepatitis B	Captopril	Systemic lupus erythematosus	Neuroblastoma	Sickle cell disease
	Hepatitis C	Nonsteroidal antiinflammatory drugs	Sjögren syndrome	Angiomatoid fibrous histiocytoma	De novo, postrenal transplantation
	Streptococcal infections	Penicillamine	Sarcoidosis	Ovarian tumor	Mercury exposure
	Syphilis	Infliximab	Autoimmune hepatitis		
	Tuberculosis	Tiopronin	Primary biliary cirrhosis		
	Epstein–Barr virus	Etanercept			
	Cytomegalovirus	Gold			
	Malaria				

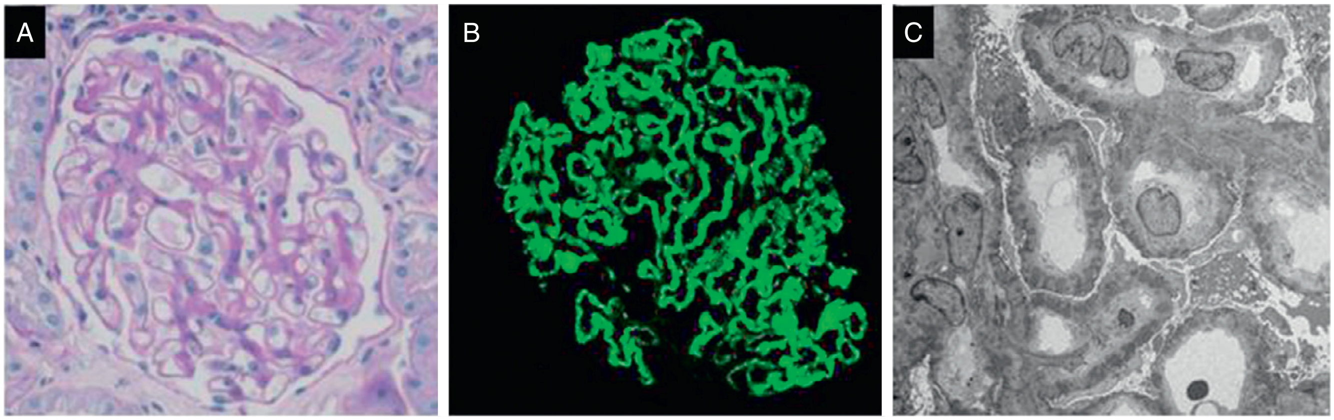

Fig. 72.11 Histopathologic findings in membranous nephropathy (MN) on light, immunofluorescence, and electron microscopy. MN manifests as global/homogeneous glomerular capillary wall thickening, the degree of which relates to the stage of deposits, (A) from mild when early to thick when late; (B) granular capillary staining for IgG that follows the contours of the basement membrane, and almost always with C3; and (C) diffuse subepithelial deposits close to the podocyte in early stages and progressively incorporating into the basement membrane over time. A: Periodic acid–Schiff ×400; B: immunoglobulin G ×400; C: electron microscopy ×5000.

TREATMENT

As in adults, the decision about whom and how to treat children with MN centers around determination of the relative risk versus benefit of any specific therapy. In general, because of few published pediatric studies in this area, such decisions for children with MN are usually derived from data published in adult populations. Notably, however, children typically have very different comorbid factors than adults and inherently different life expectancy potential, and these factors must be considered in making treatment decisions. As such, based primarily on adult literature, children with asymptomatic proteinuria are often treated conservatively with only close observation for potential development of more ominous risk factors of disease progression. Reported risk factors of progressive CKD include severe proteinuria (>8 g/day), baseline renal insufficiency, and the rate of decline in renal function over the initial 6 to 12 months of observation.[518] Children presenting with subnephrotic range proteinuria (urine protein-to-creatinine ratio <2.0 mg/mg), normal renal function, and minimal sclerosis on renal biopsy are often treated with ACE inhibitors and/or ARBs.[498] This approach can generally be expected to achieve only approximately 30% reduction in proteinuria, however, and thus would be insufficient for patients with more severe proteinuria.[519–521]

For children who present with NS, the risk–benefit calculation is notably altered. Given the 30% incidence of spontaneous remission reported among adults with MN, an initial 6-month period focused on antiproteinuric and antihypertensive treatment without immunosuppression has been recommended in adults without life-threatening symptoms or worsening renal function.[522] However, in children such patients may very well have already had steroid treatment started as presumed initial management of childhood NS, which complicates the decision about the use of immunosuppression early in the disease course. Although no controlled trials have examined steroid treatment of MN in children, a report from 1966 found that steroid treatment of children with MN (80% with NS) resulted in a 50% remission rate for those with NS.[523] Importantly, children diagnosed with MN on renal biopsy have almost always been selected for having steroid-resistant NS, because the standard of care for childhood SRNS is a course of oral steroids, and no biopsy is usually performed among children with steroid-sensitive NS.

While monotherapy with either steroids or alkylating agents has been reported to be generally ineffective in adults, combination treatment with these agents has been reported to have reasonably good efficacy among both adults and children.[510,524–528] In addition, combination treatment with steroids and CNIs has also been reported to have similar or better clinical efficacy in both adults and a few pediatric studies.[466,497–501,529–533] By contrast, reported studies of the use of MMF to treat MN in adults appear not to support its use, although a small case series of MMF use in children with MN reported some benefit in proteinuria reduction.[534–537] A more recently introduced treatment for MN is adrenocorticotropic hormone, which seems in adults to have relatively similar efficacy to that of steroids combined with alkylating agents, although no data are available for children.[538–540] Another more recent treatment approach has been the use of rituximab, and the results to date suggest that its efficacy may surpass earlier treatment approaches, particularly when combined with the recent use of anti-PLA2R titers to guide therapy.[516–543]

Table 72.11 Current Treatment Options for Children With Membranous Nephropathy

Immunosuppressive	Nonimmunosuppressive
Oral glucocorticoids (usually in combination with other agents)	Angiotensin-converting enzyme inhibitors
Alkylating agents (cyclophosphamide, chlorambucil)	Angiotensin receptor blockers
Calcineurin inhibitors (cyclosporine, tacrolimus)	
Rituximab	
Adrenocorticotropic hormone	
Mycophenolate mofetil[a] (and mizoribine)	
Pulse intravenous methylprednisolone[a]	

[a]Less commonly used treatment.

In summary, while there are very few data to guide the treatment of MN specifically among children, therapy combining steroids with alkylating agents or CNIs appears to be reasonably effective, and the more recent introductions of adrenocorticotropic hormone and rituximab appear to offer the potential for somewhat improved outcomes, potentially with fewer side effects. Table 72.11 shows a summary of the current treatment options for children with MN. While conservative treatment with ACE inhibitor or ARB therapy may suffice for children with modest proteinuria, immunosuppressive treatment appears necessary to induce remission in the majority of patients. The choice of the use of immunosuppression is complicated in children by the fact that most have already been treated with oral steroids prior to the renal biopsy establishing the diagnosis of MN. Unfortunately, due to the very low incidence of MN in children, there are no randomized trials to help guide optimal therapy.

OUTCOMES

There are few data documenting long-term outcomes in children with MN. However, reports from the North American Paediatric Renal Trials and Collaborative Studies (NAPRTCS) group and the USRDS suggest that the incidence of MN among children with CKD to be 0.5%, and that MN represents approximately 0.5% of all pediatric patients with ESRD.[544,545]

THROMBOTIC MICROANGIOPATHY

TMA defines a spectrum of diseases, which manifest clinically with microangiopathic hemolytic anemia (MAHA), nonimmune thrombocytopenia, and organ dysfunction. Central to the development of TMA is the activation and injury of the microvascular endothelium, which result in platelet and neutrophil recruitment, thrombus formation, inflammation, thromboembolic occlusion of the dependent microvasculature, and subsequent organ failure (Fig. 72.12).[366,367,546]

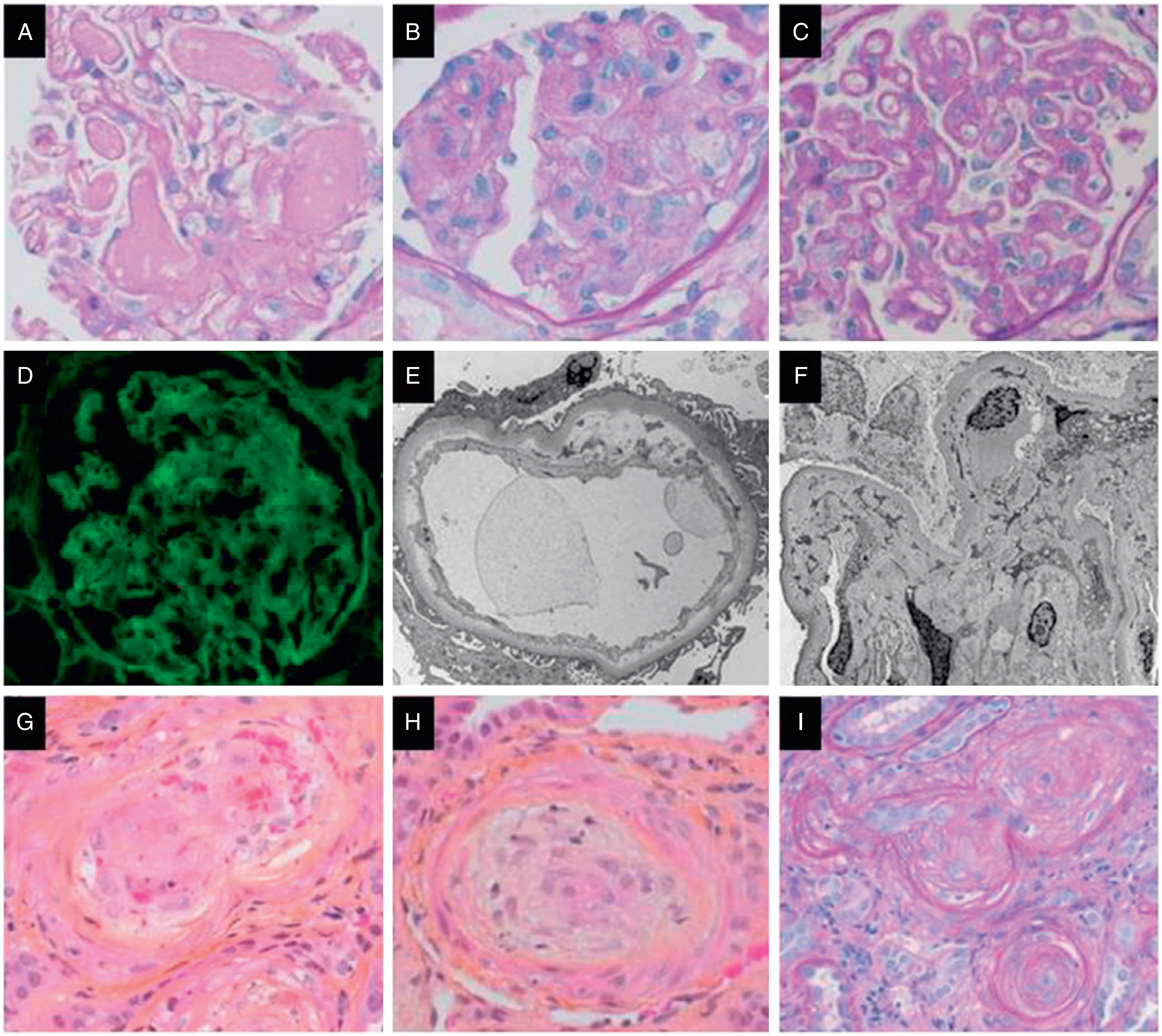

Fig. 72.12 Histopathologic findings in thrombotic microangiopathy (TMA) within glomeruli and arterioles/small arteries in acute and chronic stages. Panels highlight the microscopic features of TMA within glomeruli by light microscopy *(top row),* immunofluorescence and electron microscopy *(middle row)* as well as light microscopic features of TMA within arterioles/small arteries *(bottom row).* Glomerular changes include (in the acute stage) (A) thrombi and/or (B) mesangiolysis and endothelial cell swelling; and (in the chronic stage) (C) capillary wall double contouring; (D) immunofluorescence microscopy shows absent staining for immunoglobulin (Ig; although some trapping can occur in the chronic stage); (E) electron microscopy shows (in the acute stage) subendothelial lucent widening, including with particulate or flocculent material, and sometimes with fibrin tactoids; and (in the chronic stage) (F) cellular interpositioning with new basement membrane formation and subendothelial expansion. Arterioles/small arteries show (in the acute stage) (G) thrombi and/or intimal fibrin and/or red cell fragments, and/or (H) intimal edema; and (in the chronic stage) (I) onion skin-type change from myointimal proliferation.

HUS defines a systemic TMA mainly targeting the kidneys but potentially also involving other organ systems such as the brain, heart, lungs, gastrointestinal tract, and skin.[366,367,546–548] HUS was initially described in 1955 by Gasser as clinical triad of hemolytic anemia, thrombocytopenia, and acute renal failure.[549] The cause of this "typical" form of HUS was unraveled in 1983 when Karmali identified Shiga toxin (ST)/Vero toxin produced by enterohemorrhagic *Escherichia coli* (EHEC) as cause of the disease (STEC or VETEC, respectively).[550–552]

Historically, HUS was defined as pathological entity strictly distinct from the clinically and pathologically almost indistinguishable thrombotic thrombocytopenic purpura (TTP).[553] TTP was first described in 1923 by Moschcowitz in a patient with acute febrile illness, anemia, petechiae, paralysis, and coma with hyaline thrombi in terminal arterioles and capillaries throughout most organs including the kidneys.[554] Subsequently, TTP was defined by the clinical pentad of fever, purpura/hemorrhage with thrombocytopenia, hemolytic anemia, neurological manifestations, and variable degrees of renal dysfunction.[555] TTP is caused by deficiency of ADAMTS13 (a disintegrin and metalloproteinase with a thrombospondin type 1 motif, member 13), a protease that cleaves von Willebrand factor (vWF) multimers upon release from endothelial cells.[555–557]

While histopathologically indistinguishable, a rare subgroup of patients presenting with familial manifestation, significantly higher rates of renal and extrarenal morbidity, and high risk of posttransplant disease recurrence was identified defining a new disease entity called aHUS.[558–560] aHUS is caused by dysregulation of the complement AP on the level of the vascular endothelium. The subsequent endothelial injury induces a sequence of inflammation and coagulation with thromboembolic events, which is shared with HUS.[561] The use of the term aHUS was later used to refer to all non-STEC-caused cases of HUS including (but not exclusively applied to) patients with recurrent disease associated with a more severe clinical course and poorer outcomes.

Today, a wide range of conditions occurring in context of an underlying disease or condition including infections, hypertension, pregnancy, transplantation, medications, and metabolic conditions are considered secondary TMA (Table 72.9).[548,562] While previously not appreciated, secondary and/or causal involvement of the complement system via underlying genetic or autoimmune susceptibility factors are now increasingly recognized to occur in patients with secondary TMA.[563,564] Thus the historical nomenclature of TMA distinguishing between HUS, aHUS, TTP, and nonrelated secondary TMAs is problematic as it emphasizes distinct differences in the individual disease pathogenesis, which today is conflicting with the increasing notion of shared pathogenetic mechanisms. These insights are critical as they have the potential to identify new therapeutic targets for these diseases which to date often poor prognosis.[369,388]

DIAGNOSIS

The diagnosis of TMA is defined by clinical criteria including MAHA, thrombocytopenia, and renal and extrarenal disease manifestation.[563–565]

The initial assessment of a TMA patient must on one hand consider the three most likely differential diagnoses—STEC–HUS, TTP, and aHUS—and on the other hand consider the expanding spectrum of secondary TMA (Fig. 72.9; Table 72.9). Of note, while not defining, complement defects in form of mutations or anti-CFH autoantibodies can also be present in secondary TMA, or complement activation can occur transiently in the absence specific complement defects.[563–565] Although some forms of secondary TMA present undoubtedly (e.g., pregnancy-associated TMA, transplant-associated [TA] TMA, Streptococcal pneumoniae [SP] TMA) other forms will have to be actively excluded (e.g., cblC defect–associated TMA). Table 72.8 provides an overview of meaningful clinical tests relevant to the identification of diagnoses/conditions possibly underlying secondary TMA (Table 72.9).[563–565]

STEC–HUS typically follows a specific clinical sequence: diarrhea developing 2 to 12 days after ingestion of the disease causing bacteria (e.g., *E. coli* O157:H7) is initially nonbloody but becomes bloody in 90% of cases after 1 to 3 days. Microbiological identification of *E. coli* in the stool culture, detection of antilipopolysaccharide (LPS) antibodies specific for disease-causing bacteria, and ST detection in blood via polymerase chain reaction (PCR) confirm the diagnosis.[366,551,552]

TTP is typically characterized by an ADAMTS13 activity level of less than 10%. Detection of anti-ADAMTS13 autoantibodies (inhibitors) or identification of an ADAMTS13 mutation confirms the diagnosis of TTP.[553,555]

By contrast, aHUS is in the first place a diagnosis of exclusion of secondary TMA, STEC–HUS, and TTP. Global complement assays such as CH50 and AP50 allow to screen for complement and AP activation, respectively. Commercially available assays such as the Wieslab enzyme-linked immunosorbent assay (ELISA) allow for the individual assessment of the complement CP, LP, and AP.[377,566,567] The utility of in vitro/ex vivo hemolytic assays for monitoring treatment efficacy in individual patients has recently been demonstrated.[568] Similarly, the prospective eculizumab trial demonstrated full complement blockade (i.e., CH50 <20%) with the recommended treatment regimen.[569]

Additional biomarkers such as serum C3, CFB, and sC5b-9 levels have been used as surrogate markers of complement activation.[570,571] While low C3 and CFB levels indicating AP activation and complement consumption and high sC5b-9 levels indicating TP activation have been reported in aHUS patients,[572,573] C3 and sC5b-9 levels can remain normal during the acute phase of aHUS, revealing a limitation in their utility as markers to monitor disease activity and treatment efficacy.[571,574]

Diagnostic workup of patients with suspected aHUS must include genetic analysis and screening for anti-CFH autoantibodies. While in the past, genetic workup of aHUS patients consisted in Sanger sequencing of panels of target genes including the complement genes currently associated with aHUS (see earlier discussion), today genetic testing is increasingly offered via whole-exome sequencing (WES) or whole-genome sequencing (WGS), thus allowing not only for comprehensive coverage of all known complement genes but also for opening genetic testing up to include so far unknown, potentially disease-causing genes, which is of particular importance for patients without identified complement mutation. Anti-CFH autoantibodies are tested via ELISA against a positive control with international standardization efforts currently underway (Table 72.8).

As complement-mediated injury of the vascular endothelium is central to aHUS pathogenesis, biomarkers of TP activation (C5a and sC5b-9), vascular inflammation (soluble tumor necrosis factor receptor-1 [sTNFR1]), endothelial activation (soluble vascular cell adhesion molecule-1 [sVCAM1]) and damage (THBD [CD141]), coagulation (prothrombin fragment 1+2 [F1+2], D-dimer), and renal injury (clusterin, tissue inhibitor of metalloproteinases-1 [TIMP-1], liver fatty acid binding protein-1 [L-FABP-1], β2-microglobulin, cystatin-C) were assessed and found to be elevated in patients with aHUS regardless of the underlying complement defect.[575] Of note, these markers were suppressed to varying degrees in response to eculizumab therapy.[575]

Recently, an ex vivo assay determining complement activation on endothelial cells was developed to monitor disease activity and potential treatment efficacy.[571] When incubating nonactivated endothelial cells with serum of aHUS patients with active disease, endothelial C3 and C5b-9 deposition were enhanced, whereas complement deposition returned to baseline when serum of aHUS patients in remission was used. Preactivation of endothelial cells with adenosine 5′-diphosphate (ADP), however, unraveled subclinical complement activation in aHUS patients in clinical remission, as incubation of endothelial cells with their serum resulted in enhanced C3 and C5b-9 deposition. This assay became clinically useful when serum of eculizumab-treated aHUS patients was tested: serum of one patient with clinical breakthrough despite eculizumab therapy demonstrated enhanced endothelial complement deposition, which subsided upon dose increase; serum of another patient in stable remission did not result in enhanced endothelial complement deposition despite increased dosing intervals.[571]

INFECTIOUS FORMS OF THROMBOTIC MICROANGIOPATHY

STEC–HUS

PATHOGENESIS

HUS is the most common cause for childhood AKI in the Western world.[576] Infections with Shiga toxin (Stx) producing *E. coli* can cause sporadic forms of HUS or can cause regional outbreaks.[576] HUS usually affects children between the age of 2 and 10 years; however, in outbreaks adults are also affected.[577] The overall incidence of STEC–HUS is about 2/100,000, with a peak of 6.1/100,000 in children younger than 5 years of age.[578,579]

The EHEC O157:H7 serotype was worldwide the main serotype causing HUS until recently.[578] In Europe and North America, other non-O157:H7 serotypes are now found to be as frequent as O157:H7.[578] However, O157:H7 remains the predominant (>70%) serotype in Latin America, where STEC–HUS occurs up to 10 times more frequently. In 2011, a large HUS outbreak in northern Germany was linked to the consumption of raw fenugreek sprouts. The unusual *E. coli* O104:H4 strain that combined the virulence potentials of enteroaggregative *E. coli* and typical Stx-producing EHEC was the causative agent.[577,580]

The main reservoir of STEC is the intestinal tract of (healthy) cattle, raw or undercooked beef, and unpasteurized milk.[581] Cattle manure–contaminated vegetables and fruit or water are major sources of STEC infection in humans, especially in outbreaks.[578,582] Person-to-person transmission and contact with infected animals after farm or petting zoo visits have also been reported.[581,583] The production of Stx is the main virulence factor of STEC and associated with the development of HUS. HUS symptoms occur typically 6 days after the start of diarrhea.[584] Host factors are thought to contribute to the risk of developing HUS after STEC infection; however, a genetic susceptibility polymorphism has not been identified yet.[585]

EHEC is ingested and causes secretory diarrhea followed by bloody diarrhea. Injured epithelial cells allow translocation of Stx from the gut into the bloodstream. LPS developed in the gut by STEC may also enter the systemic circulation and promote an inflammatory response that contributes to renal injury. Stx and LPS are thought to bind to various blood cells in order to reach the target organs, predominantly the kidney. Stx binds to the endothelial globotriaosylceramide type 3 (Gb3) receptor, is internalized, and undergoes retrograde transport to the endoplasmic reticulum. The Stx A subunit is proteolytically processed to form a 27-kDa A1-fragment, which upon translocation to the cytosol triggers molecular damage.[582] The A1 fragment also inactivates the host ribosomes by removing a single adenosine residue from the 28*S* ribosomal RNA of the 60*S* subunit, and subsequently inhibiting protein synthesis. Furthermore, multiple stress-induced signaling pathways are activated, leading to the induction of inflammatory gene expression and apoptosis. The cumulative effect is endothelial cell activation and loss of endothelial cell function. Leukocyte adhesion is a result of upregulation of adhesion molecules on the endothelial surface. Platelet adhesion might be the result of vWF release.[586]

In vitro evidence indicates that Stx can activate complement through the AP and results in formation of soluble C5b-9 in normal human serum exposed to Stx2. Stx binds to CFH, the main complement regulator in fluid phase, which results in endothelial C3b deposition.[587] Stimulation of whole blood with Stx2 increased formation of platelet–monocyte and platelet–neutrophil complexes, and the release of blood cell–derived microparticles, with surface-bound C3 and C9.[588] Stx2 also caused the release of red blood cell–derived microvesicles coated with C5b-9, indicating that complement plays a role in the hemolytic process that occurs during STEC–HUS.[589] Patients with STEC–HUS had alternative and TP activation at disease onset, which was not observed anymore after recovery.[389]

CLINICAL MANIFESTATIONS

The clinical manifestation of STEC–HUS is characterized by a typical sequence of (1) ingestion of the infectious agent, (2) diarrhea, abdominal pain, fever, and vomiting, (3) bloody diarrhea, and (4) spontaneous resolution (approximately 85%) versus manifestation of HUS (approximately 15%).[551] Ninety-four percent of patients show a diarrheal prodrome, with two-thirds of them having bloody diarrhea. HUS symptoms usually appear on days 6 to 7 after EHEC ingestion. Extrarenal manifestations are possible and include CNS involvement with neurological symptoms, pancreatitis, elevated transaminases, acute respiratory distress syndrome, and cardiac involvement. The kidney is the most commonly involved organ, and dialysis is required in two-thirds of the patients, with a median treatment duration of 10 days. Mortality in the acute phase of STEC–HUS manifestation is 1% to 4% with the main reason for death being multiorgan failure. CNS involvement is reported in 30% to 40% of patients. Prognostic markers for poor clinical outcome and persistence of long-term sequelae were longer time on dialysis (median 15 days), increased leukocyte count ($>20 \times 10^9$/L), and arterial hypertension at disease onset. Interestingly, the Stx serotype has no impact on long-term outcome.[576,590] The mortality in STEC–HUS is reported as 1% to 4% in sporadic cases; however, higher mortality rates occurred in outbreaks.[581,591]

For diagnostic purposes, EHEC can be isolated from stool specimen, and the serotype can be determined by PCR or enzyme immunoassay. In addition, Ig against LPS of O157 can be detected in the serum. Infection with EHEC belonging to one of the five traditional high-risk serogroups (O157, O26, O145, O111, and O103) or Stx2-producing EHEC has been shown to be associated with a higher risk of developing HUS. Nonsorbitol-fermenting O157:H7/H$^-$ is the most common serotype (50%), but non-O157 serotypes are emerging and occur more often in patients 1 year of age.[576,590]

In patients with positive testing for EHEC, Stx, or LPS antibodies the diagnosis of STEC–HUS is confirmed and no further investigations are required. In pediatric patients with bloody diarrhea but negative EHEC, Stx, or LPS testing, STEC–HUS still remains the most likely diagnosis. In patients without bloody diarrhea and negative testing (discussed earlier) other differential diagnoses for TMA should be considered.

TREATMENT

Supportive care is the main pillar of STEC–HUS treatment, including fluid management, treatment of arterial hypertension, RRT, and, if indicated, ventilatory support. Routine use of antibiotics is not recommended and is discouraged during the diarrheal phase often preceding the manifestation of STEC–HUS, unless secondary complications enforce such a treatment. Anticoagulants, antiplatelet drugs, plasma therapy, and steroids are of no use.[576,590,592] With more recent in vivo and in vitro data pointing toward an important role of complement in STEC–HUS pathogenesis, in 2011 three patients with STEC–HUS and severe neurological symptoms were treated with the TP inhibitor eculizumab and showed significant improvement.[593] During the 2011 STEC–HUS outbreak in Germany, eculizumab was administered in a noncontrolled trial in combination with concomitant treatments to mostly adult patients with the most severe disease course. In this instance, no clear treatment benefit was found.[594,595] Currently, there is not enough evidence to support eculizumab treatment in patients with STEC–HUS. However, individual patients with a severe disease course and signs of complement activation might benefit from complement inhibition.

Neurological sequelae are reported in 4% and proteinuria in 19% of patients after 1-year follow-up, arterial hypertension in 5% of patients after 5-years follow-up.[590] Long-term follow-up data indicate that some asymptomatic patients after 1 year will develop hypertension or proteinuria 2 to 5 years after first onset of HUS.[590] This important finding warrants yearly follow-up in all patients at least up to 5, probably up to 10 years, to monitor for proteinuria, decline in renal function, and hypertension. Patients with any sequelae need follow-up as indicated by their severity. Closer follow-up might be warranted in patients where—despite diarrhea—no EHEC or Stx was detected in stool or serum. Disease recurrence should trigger further investigations to exclude potential differential diagnoses such as aHUS and TTP (discussed later). Patients with ESRD after STEC–HUS receiving transplantation have a good long-term outcome, as no disease recurrence was reported so far.[590]

PNEUMOCOCCAL HUS

PATHOGENESIS

After EHEC-associated HUS, pneumococcal HUS is the most frequent form of postinfectious HUS, accounting for around 5% to 15% of all HUS cases.[366,596] TMA occurs 3 to 13 days after pneumococcal infection, with the main presenting features being pneumonia with empyema and bacteremia. Pneumococcal HUS is caused by infections with *S. pneumoniae* (SP HUS), and the most frequent serotype associated with TMA was *S. pneumoniae* 19A. *S. pneumoniae* produces neuraminidase, which cleaves N-acetyl neuraminic acid from cell surfaces and thereby exposes the so-called Thomsen–Friedenreich antigen (T-antigen) on the surface of various cell types including vascular endothelial cells and erythrocytes. Naturally occurring circulating antibodies bind the T-antigen, which results in Coombs positive hemolysis (i.e., erythrocytes), cellular injury (e.g., vascular endothelial cells), and subsequent thrombosis with the clinical picture of HUS.[366] In vitro work has also shown that neuraminidase causes desialylation, thus reducing the binding of complement regulator CFH to the endothelial surface.[597] Of note, investigation of the AP in patients with pneumococcal HUS (SP HUS) revealed mutations in complement regulators in three of five patients in one recent study.[366]

CLINICAL MANIFESTATIONS

The overall mortality rate of SP HUS is 2% to 12%; in patients who present with meningitis, mortality rates are as high as 37%. Dialysis was needed in 50% of the patients. Extrarenal manifestations occur. ESRD occurred in 10% to 16% of patients; proteinuria and hypertension persisted in others. No disease recurrence has been reported.

TREATMENT

Treatment of patients with pneumococcal HUS (SP HUS) includes infection-specific treatment (e.g., antibiotics, antiviral drugs) and supportive care (e.g., RRT, ventilatory support). PLEX was used in a proportion of patients and considered as beneficial. The use of complement-targeted treatment strategies such as commencing eculizumab in pneumococcal HUS (SP HUS) has not been reported, yet. However, the capacity of bacteria and viruses to enhance complement activation and induce endothelial damage, and the impaired endothelial surface binding of complement regulator CFH due to *S. pneumoniae*–induced change in the endothelial glycocalyx (discussed earlier) suggest enhanced complement activation on endothelial cell surfaces, and thus provide a rationale for the use of complement-targeted treatment strategies in patients with pneumococcal HUS (SP HUS). Immunization with the Prevnar 13 vaccine covers most of the serotypes associated with pneumococcal HUS (SP HUS), and is recommended in patients on eculizumab treatment (TP blockade), and after splenectomy.[598,599]

NONINFECTIOUS, NONCOMPLEMENT-MEDIATED FORMS OF THROMBOTIC MICROANGIOPATHY

Over the past 20 years, extensive genetic screening has identified the underlying genetic cause in about 50% of patients with TMA. In recent years, new techniques such as WES and WGS have allowed for a broader approach to genetic screening, and several mutations in genes not associated with the complement system were discovered in patients and families with aHUS/TMA. Although their prevalence is low, genetic testing applying next-generation sequencing should be considered in patients without complement alterations, being nonresponsive to complement-targeted therapy, and in patients presenting with specific clinical characteristics. Identification of the underlying genetic defect will guide treatment decisions and inform discussions about inheritability, long-term outcome, transplantation, and posttransplant recurrence risk.

DIACYLGLYCEROL KINASE EPSILON THROMBOTIC MICROANGIOPATHY

PATHOGENESIS

Recently, a subgroup of aHUS patients was identified, which was characterized by the manifestation of MAHA, thrombocytopenia, and acute renal failure during the first year of life. The further disease course was typically complicated by (nephrotic range) proteinuria, and the progression to ESRD during the second or third decade of life. Of note, treatment response to complement-targeted treatments (i.e., plasma; eculizumab) was incomplete, and disease recurrences occurred despite ongoing treatment.[600] While testing of complement mutations and anti-CFH autoantibodies was negative, WES identified homozygous and compound heterozygous mutations in the gene encoding for diacylglycerol kinase epsilon (*DGKE*).[600] Subsequently, the prevalence of *DGKE* mutations among aHUS patients was determined as 2% to 3%.[601]

DGKE is an intracellular lipid kinase that phosphorylates diacylglycerol to phosphatidic acid, securing recycling of phosphatidylinositol, an important second messenger in membrane signaling.[602] So far, 44 patients with a DGKE mutation (homozygous or compound heterozygous) have been reported, 35 of whom presented with the clinical picture of TMA, 9 with an MPGN-like picture. Additional complement mutations were only observed in three patients, although biochemical evidence for mild complement activation was detected in about 25% of patients. How *DGKE* mutations render the vascular endothelium to be more procoagulant and facilitate thrombus formation in particular in the small capillaries of the kidney is still unclear.[601] However, it has been shown that endothelial cells lacking DGKE show increased platelet binding capacity, impaired angiogenic response, and increased cell death.[603]

CLINICAL MANIFESTATIONS

As mentioned earlier, most TMA patients diagnosed with a *DGKE* mutation had a disease onset within the first year of life. Proteinuria, in most patients within nephrotic range, was a hallmark feature in the acute phase. Dialysis was needed in about 75% of patients. Disease recurrence was observed in 70% of patients. Distinctive for DGKE TMA, in 80% of patients proteinuria (up to nephrotic range in 24%) persisted beyond resolution of acute TMA. Of note, 88% of patients also showed persistent hematuria during follow-up. Progression to ESRD occurred in seven of 35 patients, with a median age of 11 years. Ten-year renal survival was reported as 89%.[601]

Nine patients with an MPGN-like picture on renal biopsy had a slightly later disease onset (median 2 years of age) with continuous nephrotic-range proteinuria in most of them. No TMA features were reported during follow-up. Three patients progressed to ESRD at a median age of 19 years.[601]

TREATMENT

The optimal treatment for DGKE TMA is unclear. Retrospectively collected data indicate spontaneous recovery in most patients without specific therapy (plasma or eculizumab for TMA, immunosuppression for MPGN). TMA recurrence was recorded while on eculizumab. As of now, patients with DGKE mutations will likely not benefit from either complement-targeted therapy (i.e., plasma and eculizumab) or immunosuppression. Renal transplantation is a safe option, as there is no risk of posttransplant recurrence.[601]

COBALAMIN C–ASSOCIATED THROMBOTIC MICROANGIOPATHY

PATHOGENESIS

Cobalamin C (CblC) deficiency is the most commonly inherited error of vitamin B_{12} metabolism. CblC deficiency is characterized by multisystem involvement with severe neurological, hematological, renal, and cardiopulmonary manifestations.[604,605] Mutations in the gene encoding for the methylmalonic aciduria and homocystinuria type C protein (MMACHC) are responsible for the most severe CblC disorder, which is characterized by methylmalonic academia (MMA) and hyperhomocysteinemia and can be associated with TMA.[606] Reports of CblC-associated TMA include patients with early (<4 years of age) and late-onset CblC disease, including two adults.[606–608] CblC deficiency causes hyperhomocysteinemia, which induces platelet aggregation and a procoagulant phenotype of the vascular endothelium, and is therefore thought to be the main contributor to the pathogenesis of CblC deficiency–associated TMA.[606]

CLINICAL MANIFESTATIONS

Infantile onset of CblC disease is associated with poor outcome and has a mortality rate of 30%. CblC-associated TMA increases the mortality rate to 56%.[606] Involvement of the lungs, the need for dialysis, and very early disease onset are negative prognostic markers.[605] The classical presentation of MMA with hyperhomocysteinemia, metabolic acidosis, hyperammonemia, and encephalopathy may not always be present, especially in patients with late onset of disease.[606,609] Furthermore, despite extended newborn screening, CblC deficiency might be missed[606] and therefore a high index of suspicion needs to be maintained. In patients with late-onset CblC-associated TMA, proteinuria and/or hematuria occasionally accompanied by renal failure might be the only symptom, and diagnosis has to be made via renal biopsy. Clinical overlap with C3G has also been reported.[366]

A recent case report highlighted the diagnostic challenge of CblC-deficiency TMA[604]: A 20-year-old female patient with CKD/ESRD developed severe pulmonary arterial hypertension (PAH). Ten years before, she had first presented with acute renal failure accompanied by malignant hypertension. Renal biopsy showed TMA, and—in light of decreased ADAMTS13 activity levels and after exclusion of other etiologies—a diagnosis of TTP was established. Despite chronic plasma treatment, renal function deteriorated progressively to CKD/ESRD. When presenting with PAH, CblC deficiency was—among other differential diagnoses—considered and supported by significantly elevated homocysteine and MMA levels. Genetic analysis confirmed the diagnosis via detection of two heterozygous pathogenic MMAACHC mutations. Treatment with hydroxocobalamin, betaine, folic acid, and carnitine resulted in dramatic improvements with resolution of PAH and improvement of renal function to a level allowing for discontinuation of dialysis.[604]

As highlighted by this case, timely diagnosis and initiation of specific treatment are crucial to prevent disease manifestation or slow down disease progression. Thus urine organic acids, serum MMA levels, total plasma homocysteine levels, plasma amino acid profile, and an acylcarnitine profile should be performed in all (pediatric and adult) patients with pulmonary hypertension or TMA with unclear etiology.[366] By contrast, complement mutations were described in several patients with CblC deficiency and TMA. Therefore genetic complement testing is also warranted in CblC-deficiency patients. The detection of an additional complement defect would alter the therapeutic approach and have a significant impact on risk assessment for renal transplantation and peritransplant patient management.[366]

TREATMENT

Treatment of patients with CblC deficiency includes systemic administration of hydroxocobalamin, folinic acid, betaine, and carnitine.[606] Complement-targeted treatment was reported to be nonbeneficial, except for patients with an additional complement abnormality.[366]

INF2-MEDIATED THROMBOTIC MICROANGIOPATHY

Atypical HUS and posttransplant TMA were recently also reported in two families with INF2 mutations and common aHUS risk haplotypes. INF2 is a ubiquitously expressed formin protein which accelerates actin polymerization and depolymerization, thus regulating a range of cytoskeleton-dependent cellular functions including the secretory pathway.[578,610] Mutations in INF2 are the commonest cause of familial autosomal dominant NS. In a minority of these cases the mutations cause FSGS associated with the demyelinating peripheral neuropathy Charcot–Marie–Tooth (CMT). One family with TMA was also affected with CMT. Of note, patients with INF2-induced TMA do not benefit from complement-targeted therapy including plasma and eculizumab.[610]

NONINFECTIOUS, COMPLEMENT-MEDIATED THROMBOTIC MICROANGIOPATHY

ATYPICAL HUS

PATHOGENESIS

The complement system (Fig. 72.8), in particular its AP, plays a critical role in the pathogenesis of aHUS. Different from IC-MPGN/C3G, where complement dysregulation mainly occurs in the fluid phase, complement dysregulation in aHUS occurs on the surface of microvascular endothelial cells and results in their activation and injury with initiation of a cascade of proinflammatory and procoagulatory events shared by all forms of HUS together with closing an amplification loop of injury to the microvasculature (Fig. 72.8).[579–581,611–613]

The constitutively active AP has the potential to cover cell surfaces with significant amounts of covalently bound C3b within seconds to minutes. Host cells, such as the vascular endothelial cells, require tight complement regulation to protect them from attack by uncontrolled complement activation and to limit complement activation to when and where needed. Complement regulation is provided by a combination of fluid-phase and membrane-bound regulators (Fig. 72.8). In principle, these regulators interfere with the progression of the complement cascade on the level of C3b activation or AP C3 convertase (C3bBb) activity. CFH is the principal fluid-phase regulator, which acts in three different ways: (1) as a cofactor for the serine protease complement factor I (CFI) in cleaving C3b to inactive C3b (iC3b), (2) by promoting disintegration (decay) and thus inactivation of the AP C3 convertase (C3bBb), and (3) by competitively preventing C3b deposition on cell surfaces.[614–617] By contrast, membrane-bound regulators include CR1 (CD35), MCP (CD46), and THBD (CD141), which serve like CFH as cofactors for CFI in C3b cleavage. DAF (CD55) facilitates the disintegration of both the AP C3 (C3bBb) and C5 (C3bBbC3b) convertases. Finally, CD59 interfering with the assembly of the MAC serves as last line of defense in the complement activation cascade.[360,618–620]

Tight regulation of complement is critical to maintain the integrity of the vascular endothelium. Loss of complement control, by contrast, results in activation and injury of the endothelial cells and induces the TMA phenotype. Complement dysregulation can be due to loss-of-function mutations and/or autoantibodies impairing the function of complement regulators, or gain-of-function mutations enhancing the level of activity of complement activators.[547,548,561] Beginning in 1998 with the seminal publication of a CFH mutation explaining for a familial case of HUS,[560] numerous reports on individual patients and registry-enrolled patient cohorts—most recently reporting on 851 patients enrolled in the global aHUS registry[621]—have identified mutations in additional complement regulators such as MCP (CD46),[622] complement factor I (CFI),[623] THBD (CD141),[624] and in complement activators such as CFB[625] and complement C3.[428,626]

Most insights in aHUS pathogenesis have been gained from the identification of complement mutations or anti-CFH autoantibodies in aHUS patients and subsequent studies unraveling their functional relevance in vitro. In-depth investigations of the aberrant functions of normal and mutant CFH[627,628] and of the nature and function of anti-CFH autoantibodies[629,630] have likely advanced our understanding of the principles of aHUS pathogenesis the most.[370] CFH is an abundant (about 200–300 μg/mL), mainly liver-made 155-kD protein composed of 20 SCRs; SCRs 1–4 (N terminus) govern the cofactor activity of CFH, and SCRs 19–20 (C terminus) govern the binding activity of CFH to C3b and to negatively charged glycosaminoglycans, phospholipids, and sialic acid residues of the plasma membrane of the vascular endothelial cells.[627–631] While IC-MPGN/C3G/CFH mutations typically impair CFH cofactor activity via mutations (e.g., affecting the CFH N terminus),[427] the vast majority of aHUS causing mutations affects the CFH C terminus, thus impairing CFH surface binding. Utilizing CFH mutant in SCR20 (truncation) purified from an aHUS patient, it was demonstrated in a seminal sequence of experiments that CFH cofactor activity was maintained in fluid phase but absent on the surface of endothelial cells, which was due to the lack of C-terminal binding of CFH to the endothelial cells.[632,633] Similarly, anti-CFH autoantibodies detected in aHUS patients exclusively recognize epitopes in the CFH C terminus, thus impairing CFH cofactor activity on surfaces via impaired binding—a scenario mimicking the presence of a C-terminal CFH mutation.[629,630,634]

Mutations in complement activators or regulators have been found in 50% to 70%, and anti-CFH autoantibodies in 10% to 20% of patients.[415,561,574,635–639] In patients with recurrent TMA postrenal transplantation complement mutations were found in up to 68% of cases[640]; in patients with a de novo TMA postrenal transplantation complement mutations were found in 29% of cases.[641]

In aHUS patients, complement mutations have also been described as combinations of two or even three defects.[621,642–646] In a US cohort, 12% of patients had mutations in more than one gene,[636] and in a Dutch (pediatric) cohort 9% of patients carried combined mutations.[645] In addition, genetic haplotypes and SNPs in complement genes have been identified in aHUS patients and may act as disease-susceptibility factors.[623] These variations can be additive,[647] and a single SNP—for example, in MCP (CD46)—is unlikely to be sufficient to cause disease.[648] Recent evidence has shown that the combination of complement gene risk haplotypes can increase disease penetrance.[637]

Complement mutations or autoantibodies have also been found in patients with secondary TMA (discussed later) in the context of hematopoietic stem cell transplantation (HSCT),[649,650] pregnancy,[651] CblC deficiency,[652] malignant hypertension,[653] exposure to certain medications,[654,655] and refractory antibody-mediated rejection (AMR).[656]

CLINICAL MANIFESTATIONS

aHUS is an ultra-rare disease with an incidence of about 0.5 pmp. aHUS affects patients in both the pediatric and the adult age group.[643,657] Data from the global aHUS registry demonstrate a female predominance of disease manifestation from about 30 years of age onward while disease manifestation in the younger age group is dominated by males (median age at disease manifestation 10.0 years in males vs. 25.6 years in females). This biphasic pattern found in females might reflect the enhanced risk for aHUS manifestation caused by pregnancy.[621] While age at disease manifestation was independent of any identified underlying complement abnormality, analysis of individual complement defects identified specific patterns: patients with CFI mutations typically presented during adulthood, whereas patients with MCP/CD46 mutations typically presented during childhood. Within the pediatric cohort, patients with anti-CFH autoantibodies presented at a significantly older age than patients with other complement abnormalities (median age 6.4 years). Of note, between 6 and 17 years, anti-CFH autoantibodies were the most frequent cause for aHUS.[621]

As demonstrated previously,[643,657] ESRD-free survival of aHUS patients was poor, and renal outcome was impacted by the underlying complement defect: patients with CFH mutations showed poorer and patients with MCP (CD46) mutations more favorable outcomes with longer ESRD-free survival compared with individuals who tested negative for mutations in these genes. Also relatively benign were anti-CFH autoantibodies with outcomes in antibody-positive patients being only marginally inferior as compared with antibody-negative patients.[648] Renal outcome was also dependent on age at first manifestation of disease with pediatric onset of aHUS providing a significant advantage with respect to decline in kidney function.[621]

aHUS causing mutations and autoantibodies are characterized by variable/incomplete penetrance. Incomplete penetrance has been described for mutations in genes encoding for MCP (CD46)[630] and other genetic and autoimmune forms of aHUS.[643,658,659] Following a current concept, genetic and autoimmune defects present a specific but variable susceptibility risk, which combines with epigenetic and/or environmental factors to possibly cross the threshold of disease manifestation (multiple-hit hypothesis).[366] While epigenetic factors other than the aforementioned risk haplotypes are to date widely unknown, a number of “environmental” factors triggering first manifestation or disease recurrence of aHUS have been recognized: among known events preceding the onset of aHUS are respiratory and gastrointestinal tract infections[637,660] and pregnancy.[651] Posttransplant recurrence of TMA is associated with ischemia-reperfusion injury, immunosuppressive medications, and AMR.[661–663]

TREATMENT

Treatment of aHUS includes supportive care (RRT, antihypertensive medications, fluid and electrolyte management), plasma therapy (PI, PLEX), and more specific complement-targeted therapy (anti-C5 antibody eculizumab).[563,565,664] Treatment of aHUS caused by anti-CFH autoantibodies (DEAP-HUS: deficiency of CFH-related proteins and anti-CFH autoantibody-positive HUS)[627] includes, besides eculizumab, also PLEX and immunosuppression (i.e., steroids, cyclophosphamide, MMF, rituximab, and combinations thereof).[563,565,639,665,666] Prior to the advent of eculizumab, plasma therapy was the mainstay of treatment for aHUS. However, treatment efficacy was limited (about 50%), and risk of progression to ESRD was high (about 50%).[410,561,643,657]

The advent of eculizumab marked a paradigm shift in the treatment of aHUS patients. Eculizumab is a humanized, monoclonal anti-C5 antibody, which specifically binds complement C5 and prevents its cleavage into the potent anaphylatoxin C5a, and C5b, which initiates the TP with the formation of the MAC, C5b-9.[485] While first successfully used in the treatment of PNH,[667,668] eculizumab has subsequently been commenced in pediatric and adult cases of acute and chronic aHUS, and proven highly efficient in blocking complement activation, reversing acute TMA, and preventing TMA progression and recurrence. Eculizumab was also effective in preventing disease recurrence postrenal transplantation.[669,670] Clinical trials suggest that eculizumab treatment not only results in recovery of renal function in acute TMA, but also could lead to some recovery of renal function even in chronic TMA, potentially allowing for discontinuation of dialysis.[599,671,672] Of note, success of eculizumab treatment was independent of the identification of an underlying genetic or autoimmune complement defect.[599,671,672]

The current recommended eculizumab treatment regimen consists of an induction followed by a maintenance phase. Eculizumab dosing and treatment intervals vary based on patient age and body weight, respectively. For adult aHUS patients a 4-week induction phase with weekly eculizumab doses is followed by a (timely nonlimited) maintenance phase with eculizumab doses administered every second week.[599] Of note, this treatment schedule applies regardless of an identified genetic defect. In patients with anti-CFH autoantibody-mediated aHUS, however, treatment options and recommendations are different and include (1) eculizumab monotherapy,[599,666] (2) immunosuppression via PLEX, steroids, cyclophosphamide, MMF, rituximab, or various combinations thereof,[639] and (3) a combination therapy with eculizumab and immunosuppression.[563,665] The overarching treatment principle is based on the appreciation of the autoimmune character of the disease identifying anti-CFH autoantibodies as treatment target and monitoring tool. A recent Kidney Disease Improving Global Outcomes (KDIGO) consensus conference has proposed a stepwise approach to the treatment of DEAP HUS including (1) eculizumab for the acute phase, (2) subsequent introduction of PLEX and/or immunosuppressants to remove circulating and prevent formation of new autoantibodies, and (3) discontinuation of eculizumab upon suppression of the autoantibody titer below the detection threshold.[563,665]

Eculizumab treatment is typically well tolerated with side effects being rare and mild.[599,671,672] In a small case series, liver enzyme elevation was observed resulting from eculizumab treatment—a complication, which was typically transient and on liver biopsy did not correlate with specific pathology.[673] As eculizumab-treated patients are vulnerable for infections by gram-negative encapsulated bacteria such as *Neisseria meningitides*, meningococcal vaccination is strictly required before eculizumab treatment can be safely commenced. Recently, additional chronic antibiotic prophylaxis with penicillin has been recommended to complete protection from infection with gram-negative bacteria.[599,652,674]

An international consensus has been achieved with respect to commencing complement-targeted treatment such as eculizumab in adult patients with confirmed diagnosis of aHUS in pediatric patients with suspected diagnosis of aHUS. Treatment recurrence in all confirmed cases of aHUS, and treatment of patients with a renal transplant at the time of transplantation or of aHUS recurrence as first-line treatment. However, the question of treatment duration has not yet been answered, and criteria for the safe discontinuation of eculizumab are lacking.[563,565] While successful discontinuation of eculizumab treatment has been reported in individual cases and small case series with urine testing for hematuria as reliable monitoring strategy,[675] recently the risk for aHUS recurrence after discontinuation of eculizumab varied depending on the type of the underlying genotype with MCP/CD46 versus CFH mutations spanning the risk spectrum of low versus high risk level, respectively.[547] Recent efforts aimed at defining reliable and practical biomarkers of aHUS disease activity and tests allowing for the monitoring of treatment efficacy (discussed earlier).[571,575]

SECONDARY FORMS OF THROMBOTIC MICROANGIOPATHY WITH POSSIBLE COMPLEMENT CONTRIBUTION

TRANSPLANT-ASSOCIATED THROMBOTIC MICROANGIOPATHY

TA-TMA can occur as de novo disease, disease recurrence, or in context of antibody-mediated recurrence.[640,641,676] Diagnostic approach and investigations are similar to the ones discussed for aHUS earlier. As complement dysregulation is central to most of these cases, complement-targeted treatment is the first choice. Weaning or treatment withdrawal needs to be carefully considered and should be guided by the evaluation of genetic findings and prior clinical presentation.

RECURRENT ATYPICAL HUS AFTER RENAL TRANSPLANTATION

After renal transplantation, the risk of aHUS recurrence is high and graft survival is poor. A recent study in 57 aHUS patients, who received a kidney transplant, highlights that TMA recurrence occurred during the early posttransplant course with a cumulative incidence of 43% (19/44) at 1 month, and 70% (31/44) at 1 year.[640] The type of complement mutation predicted the recurrence risk, and patients with complement mutations and low C3 had the highest risk of graft loss. The risk of recurrence was high in patients carrying CFH and gain-of-function (i.e., C3 or CFB) mutations, intermediate in patients carrying CFI mutations or homozygous at-risk CFH polymorphisms, and low in patients without identified mutations.[640] Several patients with low-titer CFH antibodies were successfully transplanted without aHUS recurrence.[415] Despite evidence of endothelial toxicity of CNIs in vitro, CNI immunosuppressive treatment was not associated with a higher risk of recurrence in these patients.[677] Thus currently there is no evidence supporting the discontinuation of CNI treatment in patients with posttransplant aHUS recurrence.

Renal transplantation in patients with aHUS needs to be carefully considered, and CFH antibody screening and complement genetic testing should be performed prior to transplantation. Recently, prophylactic treatment with complement-targeted treatment has improved patient outcome and enabled transplantation in high-risk patients. Eculizumab should be considered as first-line (prophylactic) treatment in patients with high recurrence risk.[678] In case of recurrence, eculizumab is the preferred choice and should be administrated as soon as possible after a diagnosis of aHUS recurrence has been established.[679] Except for one patient who lost the graft to an arterial thrombosis on day 1 after transplantation, eculizumab treatment was successful in 13 patients treated prophylactically, and in 17 in whom treatment was initiated after recurrence.[670,679] Although patients with recurrence after transplantation showed good response, prophylactic treatment with eculizumab should be considered especially in high-risk patients.

DE NOVO THROMBOTIC MICROANGIOPATHY AFTER RENAL TRANSPLANTATION

De novo TMA occurred in 4% to 15% of patients treated with cyclosporine, and 1% treated with tacrolimus after renal transplantation, and resulted in graft loss in one-third of the patients.[366] While not contributing to TMA recurrence, CNIs have a stand-alone role in de novo TMA after transplantation. In addition, complement mutations (i.e., CFH and CFI) were detected in 29% of patients with de novo TMA after transplantation.[641] This phenomenon remains one of the least understood of the CNI-related toxicities. CNI toxicity occurs primarily via its effects on the vascular endothelium.[645] CNIs are known to cause endothelial cell injury[680,681] related to cell mitochondrial dysfunction and reactive oxygen species formation.[682] Cyclosporine exposure also induces endothelial microparticle formation, leading to the activation of the AP.[677]

A consensus on treatment of de novo TMA after kidney transplantation does not exist. (Temporary) withdrawal of CNIs has been applied frequently.[683,684] A CNI-free, sirolimus-based immunosuppressive regimen was administered to 14 patients with aHUS or de novo TMA after transplantation. Treatment showed long-term success in 13/14 patients, but acute rejection was reported in 53% of patients.[711] Sirolimus treatment has recently also been associated with posttransplant TMA and CNI-free immunosuppressive regimens are not recommended.[366,679] Given the high rate of complement alterations in patients with de novo TMA after kidney transplantation, complement-targeted therapy is a reasonable option. Eculizumab treatment was reported in four patients with de novo TMA after combined lung–kidney or pancreas–kidney transplantation, or ABO-incompatible kidney transplantation. Administration of eculizumab led to a prompt hematologic and renal recovery.[366]

AMR-ASSOCIATED THROMBOTIC MICROANGIOPATHY

TMA was reported in 4% to 46% of patients with AMR and was associated with increased risk for graft loss.[685] Donor-specific antibodies have the potential to activate the complement system on the endothelium via the CP, the proposed mechanism of acute AMR, and facilitate chronic inflammatory processes, the proposed mechanism in chronic AMR.[686] Anaphylatoxins promote infiltration of inflammatory cells. In addition, C5b-9 might directly injure the allograft.[686] The ability of donor-specific human leukocyte antigen antibodies to bind C1q, and initiate the CP, was recently linked to an increased risk for AMR and graft loss.[687] In one reported patient, AMR was accompanied by severe TMA on the background of a homozygous CFHR3-CFHR1 deficiency.[656]

To date, there is no definitive treatment for AMR, irrespective of the finding of TMA on biopsy. Prophylactic eculizumab treatment in 26 highly sensitized kidney transplant recipients reduced the incidence of AMR from 41% to 8%.[688] In the same study, biopsies taken 1 year after transplantation were comparable between groups, and did not prevent chronic rejection in patients with a positive cross-match and high antibody titers. Several cases of successful treatment of AMR with eculizumab were reported.[366] On the contrary, nonresponders to eculizumab were also published in the literature.[689,690]

BONE MARROW/STEM CELL TRANSPLANT–ASSOCIATED THROMBOTIC MICROANGIOPATHY

TMA following allogeneic and autologous transplantation is estimated to occur in 0.5% to 15% and 0.1% to 0.25% of cases, respectively.[691] TMA after bone marrow transplantation (BMT)/HSCT is associated with a high mortality. Diagnosing TMA in the setting of BMT/HSCT is challenging, as thrombocytopenia, renal insufficiency, and the presence of schistocytes may also represent other complications, such as graft versus host disease (GvHD) and systemic infections.[692] It has been noted that lactate dehydrogenase, proteinuria, and hypertension are the first signs for TMA in this patient cohort, and a diagnostic risk score has been published recently.[649] Schistocytes may be absent in patients with TMA.[650]

Endothelial cell injury, induced by chemotherapy and other drugs such as cyclosporine, a general prothrombotic state, and additional risk factors such as intravenous catheters, infections, and GvHD may well contribute to TMA pathogenesis.[692] Endothelial cell regeneration is limited, as bone marrow endothelial progenitor cell populations, the source for endothelial cell repair, are depleted by conditioning treatments.[692]

Complement alterations were discovered in variants in 65% of HSCT recipients with TMA as compared with only 9% in those without TMA, including CFH antibodies and/or a heterozygous CFHR3-CFHR1 deletion and mutation in regulators. Complement alterations were associated with increased disease severity and mortality.[650] TMA after BMT or SCT might be multifactorial and requires the combination of genetic susceptibility as well as secondary endothelial cell activation.[366,650] Complement activation markers such as sC5b-9 were elevated in 70% of HSCT patients with TMA. Elevated blood sC5b-9 levels in conjunction with proteinuria are associated with the most severe TMA phenotype after HSCT, resulting in unfavorable outcomes if not treated.[649]

The optimal management of BMT/HSCT-related TMA has not been established yet. Treatment can be supportive or targeting the underlying condition. CNIs and sirolimus are discussed in the literature as important factors for endothelial injury.[650,692] Clinical decision for temporary discontinuation must balance the subsequent increased risk for GvHD and should only be considered in patients with severe kidney injury and/or severe hypertension. In patients with high risk for GvHD and well-controlled hypertension and kidney function no change in immunosuppressive regimen is needed.[650]

Historical data from two reviews involving 260 patients suggest a median response rate to plasma exchange (PE) to range from 37% to 55%, and a mortality rate of approximately 80% to 100%.[692] Prolonged time to resolution of TMA was associated with nonsurvival.[693] Other reported treatment options included rituximab, defibrotide, and intravenous Ig with variable success.[366] Recently, with the evolving role of complement, eculizumab has been used in several case reports and one single-center prospective study including 30 patients. The latter included patients with proteinuria, activated terminal complement, and multiorgan impairment. One-year survival was reported as 62% compared with 9% in patients with TMA not receiving eculizumab treatment. Patients received a median of 14 doses and were safely discontinued after resolution of the disease. Of note, these patients have a variable eculizumab clearance and need personalized drug dosing based on pharmacokinetics and treatment effect (i.e., CH50 suppression). Early treatment initiation was crucial for outcome. In the same study, patients with only hematologic signs of microangiopathy without proteinuria or elevated sC5b-9 fully recovered from TA-TMA without treatment.[649,650]

GLOMERULOPATHY AND VASCULITIS-ASSOCIATED THROMBOTIC MICROANGIOPATHY

TMA was reported in patients with SLE, ANCA vasculitis, APS, IgAN, MN, and IC-MPGN/C3G. Only a handful of patients are reported, and in-depth genetic testing was not performed in the majority. Treatment strategies include various immunosuppressive protocols and symptomatic therapy, for example, of hypertension and impaired renal function.

SLE-ASSOCIATED THROMBOTIC MICROANGIOPATHY

The prevalence of SLE-associated TMA is estimated as 1% to 4% with a higher prevalence in patients with renal involvement.[694,695] TMA lesions in renal biopsies of SLE patients were detected in 24%, 80% of whom were lacking the clinical diagnosis of TM.[696] Outcome (mortality rate 34%–52%) and treatment response were worse in patients with lupus nephritis and renal TM.[695,696] Infections were a major trigger of SLE-associated TMA and predicted poor outcome.[697]

Complement activation is a hallmark feature in both SLE with activation of the CP and aHUS activation of AP. In a large case–control study of 15,864 SLE patients, an association of CFHR1/CFHR3 deletions and SLE was established. Of note, different from DEAP HUS, these patients were anti-CFH autoantibody negative.[698] However, in a Swedish cohort anti-CFH autoantibodies were identified in 6.7% of patients with SLE.[697,699] Successful single case reports exist for the use of PLEX as well as eculizumab in patients with a syndrome combining TMA and SLE.

APS-ASSOCIATED THROMBOTIC MICROANGIOPATHY

APS or life-threatening catastrophic APS is an autoimmune disease characterized by arterial and venous thrombi due to autoantibodies directed against phospholipids.[700] APS nephropathy is common (71% of cases), histopathologically often presenting as TMA.[701] APS is associated with post-transplant recurrence, presenting as TMA especially in the early posttransplant phase. Of note, antiphospholipid antibodies lead to complement activation via the CP.

In biopsies of three patients with APS nephropathy that developed TMA after renal transplantation, intense C4d and C5b-9 staining on endothelial cells of injured vessels was detected.[702] In these patients, eculizumab treatment was commenced after failing PLEX and was successful in reversing clinical and histopathological findings of TMA. Despite terminal complement inhibition, C5b-9 deposition remained persistent over several months, and chronic vascular changes could not be prevented.[702]

IC-MPGN/C3G-ASSOCIATED THROMBOTIC MICROANGIOPATHY

Similar to aHUS, C3G is associated with AP activating (gain-of-function) and regulating (loss-of-function) mutations and—more frequently—with autoantibodies stabilizing the AP convertase C3bBb.[380] However, aHUS and C3G not only overlap pathogenetically, but also phenotypically, which became evident when patients previously diagnosed with aHUS developed C3G or vice versa, or when different individuals of one family carrying the same complement mutation developed either aHUS or C3G.[366] More recently, complement mutations and/or autoantibodies (i.e., C3NeF) were also detected in IC-MPGN patients.[437] Thus despite limited literature, there is a role for complement-targeted therapy in patients with IC-MPGN/C3G, especially with underlying complement alterations and TP activation.[487]

ANCA-ASSOCIATED VASCULITIS-ASSOCIATED THROMBOTIC MICROANGIOPATHY

TMA was also detected in 15% of patients with ANCA-associated vasculitis (AAV), which was associated with a negative outcome.[437] In AAV, endothelial cell activation and injury are induced by activated neutrophils and subsequent complement activation, which might result in TMA.[703]

Treatment for AAV includes immunosuppressive and antiinflammatory medications coupled with antibody depleting agents such as PLEX and rituximab. Unraveling the role of complement in AAV pathogenesis has led to the investigation of complement-targeted therapies, and C5a receptor (C5aR) blockage has shown to ameliorate the disease.[432,704]

MALIGNANT HYPERTENSION–ASSOCIATED THROMBOTIC MICROANGIOPATHY

Established nomenclature suggests a disease sequence beginning with (malignant) hypertension leading to TMA. Accumulating evidence in individual cases, however, today provides a rationale for the reversal of this concept with TMA being the initiating culprit, and being a secondary phenomenon arising secondary to microthrombus formation. In patients presenting with malignant hypertension and TMA, diagnostic and treatment algorithms should be followed as detailed for aHUS (Table 72.8). Hypertension causes endothelial cell dysfunction, platelet activation, and increased thrombin, thus generating a prothrombotic milieu.[705] While blood pressure normalization remains the main goal both in the acute and in the chronic phase, additional complement-targeted therapy was reported in some patients.[366]

DRUG-ASSOCIATED THROMBOTIC MICROANGIOPATHY

An association of drugs and TMA was found in 13% of patients,[706] and was mainly described in patients treated with CNIs (discussed earlier), and with mammalian target of rapamycin inhibitors (e.g., sirolimus).[707] Other drugs associated with TMA include quinine, chemotherapeutic agents, vascular endothelial growth factor (VEGF) inhibitors, antiplatelet agents, and cocaine.[708–712] ADAMTS13 mutations or anti-ADAMTS13 autoantibodies are prominent in the TMA associated with the antiplatelet agent ticlopidine.[708]

Mechanisms leading to drug-induced TMA include direct injury of the endothelium, vasoconstriction, impaired vasodilatation, procoagulant activity, and antiplatelet activity.[710,711] A genetic susceptibility might exist in some patients.[366] The association of TMA and VEGF inhibitors, such as bevacizumab, suggests VEGF as a critical player for glomerular endothelial cell integrity.[713]

Treatment of drug-associated TMA includes—as far as possible—the removal of the disease-inducing agent, and complement-targeted therapy in patients with underlying complement alterations.[366]

THROMBOTIC THROMBOCYTOPENIC PURPURA

PATHOGENESIS

The TMA TTP must be distinguished from (a)HUS. Like (a) HUS, TTP is a potentially life-threatening thromboembolic disorder of the microvasculature, which is characterized by organ ischemia in particular of the brain, but also the heart, the gastrointestinal tract, and the kidneys, as well as thrombocytopenia ($<30 \times 10^9$/L) and the appearance of fragmented red blood cells on blood film.[553–555] TTP was first described in 1924 by Moschcowitz;[714] however, its pathogenesis was only unraveled in 1982 by Moake who detected a defect in the processing of vWF in cases of chronic relapsing TTP.[715] Subsequently, impaired vWF cleavage was attributed to a functional deficiency in ADAMTS13 (a disintegrin and metalloproteinase with thrombospondin motifs 13).[716,717] Of note, a decrease in ADAMTS13 activity to less than 10% is considered pathognomonic.[555–557] ADAMTS13 deficiency can result from homozygous or compound heterozygous *ADAMTS13* mutations, which constitutes the hereditary or congenital form of TTP (cTTP; Upshaw–Schulman syndrome),[717–719] or from inhibiting anti-ADAMTS13 autoantibodies, which constitutes the acquired, immune-mediated form of TTP (iTTP).[716] TTP is more common in adults and occurs with a standardized annual incidence rate of 2.88×10^6 per year in adults versus 0.09×10^6 per year in children.[720] Of note, while adults are more likely diagnosed with iTTP, children are more likely diagnosed with cTTP.[555,557]

vWF is synthesized by endothelial cells and megakaryocytes as monomeric pre–pro-vWF, which is modified to become pro-vWF and form the typical ultra-large vWF (ULVWF; >10,000 kDa) multimers. The molecular architecture of the vWF monomers is characterized by their multidomain structure; vWF multimers are packaged and stored in granular compartments, that is, Weibel–Palade bodies in endothelial cells, and α-granules in platelets.[721,722] Besides storage, these granules also facilitate vWF release into the circulation.[721,723] Basal vWF release occurs at a low rate in a nonstimulated fashion, whereas regulated vWF release follows a stimulus such as elevated intracellular calcium or cyclic adenosine monophosphate levels.[721] Under the shear stress of blood flow, secreted vWF unfolds and elongates to form strings of 100–500 μm, thereby exposing otherwise cryptic platelet-binding (A1 domain; glycoprotein 1b platelet and collagen-binding site)[723,724] and ADAMTS13 cleavage (A2 domain)[725] sites.

Physiologically, vWF multimers are subject to proteolytic cleavage (A2 domain) by a zinc protease[726,727] termed ADAMTS13[717,728] to prevent them from becoming excessively large.[729] Alternatively, released vWF can also bind to collagen.[730] Secreted vWF differs from that in circulation, which is present in a globular state, thus maintaining the platelet GP1b (A1 domain) and the ADAMTS13 cleavage site (A2 domain) cryptic (i.e., hidden) to prevent platelet binding and clot formation.[699,731]

vWF multimers attached to endothelial cells undergo structural changes under shear flow that enable ADAMTS13 to cleave them.[732,733] However, when ADAMTS13 activity is decreased (<10%) via an *ADAMTS13* mutation or inhibiting autoantibodies, ULVWF multimers persist in the circulation, bind, accumulate, and activate platelets, and thus initiate the formation of vWF-rich platelet thrombi, the hallmark of TTP, which occlude the microcirculation.[555,557,723,734–737]

CLINICAL MANIFESTATIONS

The clinical manifestation of TTP is characterized by a nonimmune MAHA, thrombocytopenia, neurological symptoms (often transient; ranging from headache or mental changes to focal signs, seizures, and coma), variable degrees of renal dysfunction and fever, which together constitute the classical pentad of TTP historically found in 88% to 98% of patients.[555,557,738] Disease onset is often preceded by prodromal manifestations including fatigue, arthralgia, myalgia, and abdominal or lumbar pain. Cardiac involvement may include myocardial infarction, congestive heart failure, arrhythmias, cardiogenic shock, and sudden cardiac arrest. Digestive tract involvement may include abdominal pain, nausea, vomiting, and diarrhea. Autopsy studies revealed microvascular thrombi in almost all organs, particularly in the brain, heart, kidneys, digestive tract, spleen, pancreas, and adrenal glands. Of note, thrombi are typically rare in the lungs and liver.[555,557]

Pathognomonic are ADAMTS13 activity levels of less than 10%; the detection of inhibiting ADAMTS13 autoantibodies or genetic *ADAMTS13* mutations corroborate the diagnosis.[555,557] TTP is a chronic condition that often manifests as severe acute episodes of TMA, which are associated with a high (up to 20%) mortality and the continuing risk of relapse in survivors of a first acute disease episode.[555,557]

The clinical distinction of TTP from its major differential diagnosis (a)HUS was historically based on the involvement of the CNS and the relative sparing of the kidneys in TTP, versus the predominant renal phenotype in (a)HUS. However, although unusual in TTP, renal manifestation can also occur in up to 27% of TTP patients,[739,740] and while frequently occurring in TTP patients (up to 50%), neurological involvement has also been reported in 10% to 30% of (a)HUS patients.[741] The current standard of establishing the diagnosis of TTP involves measuring plasma ADAMTS13 activity.[742–744] ADAMTS13 activity assays using multimeric vWF substrate are difficult to perform and laborious. Today's fluorescence resonance energy transfer–based assay uses instead a minimal functional ADAMTS13 substrate composed of 73 amino acid residues (D1596–R1668) in the vWF A2 domain.[742,744] Activity levels of less than 10% are considered highly suggestive of TTP and are generally used as a diagnostic guide.[679] Additional diagnostic criteria include the degree of thrombocytopenia with platelet levels in TTP of typically less than 30×10^9/L (as opposed to aHUS with $>30 \times 10^9$/L), and serum creatinine levels reflecting the degree of renal impairment being less elevated in TTP than in (a)HUS.[555,557]

TTP is associated with the life-long risk of disease recurrence, and recurrent episodes can be life-threatening. Most relapses occur during the first or second year of the previous disease manifestation, but can occur as late as after 10 to 20 years.[740] As at first onset of disease, TTP relapses are characterized by decreased ADAMTS13 activity, an observation which points toward a potential for ongoing monitoring of ADAMTS13 activity to allow for preemptive treatment.[557]

TREATMENT

In adult patients fulfilling the diagnostic criteria for TMA, STEC–HUS, aHUS, and TTP are—besides the numerous forms of secondary TMA—the main differential diagnoses. In the absence of clinical or diagnostic evidence for STEC–HUS and secondary TMA, TTP and aHUS remain the key differential diagnoses. In adult patients in whom the likelihood of TTP is significantly higher than in pediatric patients (in whom aHUS would be the more likely diagnosis), immediate treatment initiation is critical.[563,679,745]

Acute TTP episodes are medical emergencies and require immediate and intensive treatment.[555] The historically very poor prognosis of TTP with survival rates of <10% improved dramatically to >80% with the implementation of the consequent use of plasma treatment.[746]

Plasma treatment is appropriate for both iTTP and cTTP and can be delivered in form of PI or PLEX. While both treatment modalities were suitable to supplement or replace decreased or missing function of ADAMTS13, PLEX removes circulating inhibiting ADAMTS13 autoantibodies as well as high-molecular-weight vWF multimers. In 1991, a randomized controlled trial of PI versus PLEX for the treatment of adult TTP patients established a plasma volume-dependent response (9 days: 47% in PLEX vs. 27% in PI) and survival (6 months: 78% in PLEX vs. 49% in PI) rate.[747,748] Compared with fresh frozen (individual donors) and solvent detergent (multiple pooled donors) plasma, cryo-supernatant plasma is depleted of higher-molecular-weight proteins including high-molecular-weight vWF multimers. In iTTP, plasma treatment in form of PLEX—in the acute phase possibly even twice daily—continues until recovery of the platelet count, resolution of hemolysis, and resolution of organ dysfunction.[555,557] By contrast, in cTTP, plasma treatment in form of chronic periodical PI might be sufficient.[555,557,749]

For iTTP, additional treatment strategies include the use of immunosuppressants in the form of corticosteroids and/or biologics such as rituximab. Prior to the systematic use of PLEX, corticosteroids were reported to be successful in 55% of TTP patients with mainly hematological manifestation,[750] and the use of high-dose corticosteroids resulted in remission in 77% of patients.[751] In addition, administration of corticosteroids in combination with PLEX reduced the number of PLEX sessions required to achieve remission.[752]

The use of the monoclonal anti-CD20 antibody rituximab in iTTP patients is still controversial. As a B-cell depleting drug, the use of rituximab has been suggested for iTTP, and has been proven beneficial in several trials.[753–759] Rituximab was successful when used in combination with PLEX (daily PLEX + 4 weekly doses of rituximab) to achieve more frequent recovery of ADAMTS13 activity and more effective depletion of anti-ADAMTS13 autoantibodies, but also when used as frontline therapy in reducing number and frequency of TTP relapses.[757–760]

For TTP patients who were refractory to plasma cell depletion via rituximab, treatment strategies developed for patients with multiple myeloma were considered, and the successful use of bortezomib as rescue therapy in rituximab refractory TTP patients has been reported.[761–763]

Recently, ADAMTS13 replacement therapy in the form of recombinant ADAMTS13 has become available as a promising treatment strategy for both cTTP and iTTP. Recombinant ADAMTS13 effectively restores vWF cleaving activity in cTTP.[764] In iTTP, the required amount of recombinant ADAMTS13 was dependent on the anti-ADAMTS13 antibody titer.[765] In a recent first-in-human phase 1 clinical study of recombinant ADAMTS13 (BAX 930) in 15 cTTP patients, treatment was found to be safe, nonimmunogenic, and well tolerated. The pharmacokinetic profile of recombinant ADAMTS13 was comparable to PIs, and there was evidence for pharmacodynamic activity. A dose-related increase in ADAMTS13 antigen and activity levels was observed with a maximum effect after 1 hour. Besides, with increasing doses, a dose-dependent persistence of vWF cleavage products and reduced vWF multimeric size were observed.[745]

Other treatment approaches currently under investigation include compounds (1) modifying vWF affinity for platelets, and (2) interfering with vVF–platelet interaction.

The reducing agent *N*-acetylcysteamine has been shown to depolymerize vWF multimers and prevented clot formation in *Adamts13*$^{-/-}$ mice. In addition, *N*-acetylcysteamine reduced vWF disulfide bonds, thus inhibiting vWF–platelet interactions via GP1b.[766] *N*-acetylcysteamine was also successful in preventing clot formation in 5/8 TTP patients, and is currently tested in a clinical trial (ClinicalTrials.gov identifier NCT01808521).[555,557]

Caplacizumab is a bivalent humanized llama Ig variable region that recognizes the vWF A1 domain, thus preventing vWF–platelet binding via GP1b.[94] Caplacizumab was recently tested in a phase 2 randomized trial (TITAN) during PLEX and for 30 days afterward, and was successful in shortening the time to platelet count normalization. TTP relapse occurred quickly after caplacizumab discontinuation in patients with persistently decreased ADAMTS13 activity. A follow-up trial (HERCULES) is currently investigating the extension of caplacizumab treatment in patients with ADAMTS13 activity of 10% after 30 days of treatment.

SYSTEMIC DISEASES WITH RENAL INVOLVEMENT

HENOCH–SCHÖNLEIN PURPURA AND IGA NEPHROPATHY

INTRODUCTION

IgAN is the most common type of chronic glomerulonephritis in children,[767] while HSP is the most common type of vasculitis in children, with glomerulonephritis being its most common chronic manifestation.[768] First described by Berger and Hinglais in 1968 among patients with macroscopic hematuria during viral upper respiratory infections, IgAN is now recognized as the most common primary glomerulopathy in the world.[769,770] While early on it was considered a benign disease, subsequent studies revealed that between 20% and 50% of adults and children ultimately developed ESRD.[770–775] In contrast to IgAN which is limited to the kidneys, HSP nephritis is a multisystem disease primarily affecting the skin, joints, gastrointestinal tract, and kidneys.[776,777] Because these diseases share many pathogenic and clinical features, however, they will be discussed together.

EPIDEMIOLOGY

Although IgAN occurs worldwide, its incidence varies widely among different regions. In North America it accounts for only approximately 5% to 10% of all cases of primary glomerulonephritis, whereas in Europe it accounts for 20% to 30% of cases, and in the Pacific Rim it accounts for almost 50% of cases.[778] It is generally thought that both genetic and environmental factors account for much of this diversity, with differing approaches among various countries to screening urinalyses also likely playing a role in the detection of disease.[779–781] While IgAN occurs in both children and adults, it presents most commonly during the second and third decades of life.[770] Among children, IgAN appears to generally occur with a male-to-female ratio of approximately 2 : 1.[773,775,782]

In comparison to IgAN, HSP occurs almost 20 times more commonly.[767] However, because only a minority (~30%) of children with HSP develop glomerulonephritis, which is most commonly only mild, the incidence of clinically significant HSP nephritis has been estimated to be potentially lower than that of IgAN.[277,783–785] In addition, HSP appears to occur with a male-to-female ratio of approximately 1 : 1, although HSP nephritis has a slight male predilection with a male-to-female ratio of approximately 1.5 : 1.[775,786,787] While HSP affects children far more commonly than adults, with a peak incidence at approximately 4 to 5 years of age, it is rare among children younger than 2 years of age.[293,788–790] There is also a notable seasonal variation in the incidence of HSP, with its peak incidence occurring during the winter months.[310,783,787]

PATHOPHYSIOLOGY

The molecular pathogenesis of IgAN has yet to be fully clarified, but it is understood to be an IC-mediated disease. The disease is characterized by granular electron-dense deposits in the glomerular mesangium that are comprised at least in part of IgA and C3. In addition, circulating IgA ICs have been well described, and conditions known to induce IgA ICs have been reported in patients with IgAN. The mesangial deposits have been reported to contain primarily the IgA1 subclass, and to be present in the polymeric form, rather than the monomeric form.[791,792] Although not yet proven, it is now suspected that incomplete galactosylation of O-linked glycans in the hinge region of IgA1 may lead to increased serum levels as well as a propensity for IgA1 complexes to bind to the glomerular mesangium, stimulating the production of cytokines and growth factors by both glomerular cells and circulating lymphocytes that results in the mesangial cell proliferation and deposition of extracellular matrix that characterize IgAN.[793–796] Serum levels of IgA are increased in 50% to 70% of patients with IgAN during active disease, but return to normal during recovery.[787,797,798] In addition, various IgA-derived autoantibodies and circulating ICs have been frequently reported in patients with IgAN.[799,800] More recently, microRNAs have also been suggested to have potential roles in the pathogenesis of IgAN and/or disease progression.[801–804] Lastly, the available evidence suggests that the precipitating event in IgAN, abnormal glycosylation of IgA1, appears to be an inherited trait rather than an acquired abnormality, but that different families may have differing molecular mechanisms of disease development.[805]

CLINICAL PRESENTATION

IgAN in children initially presents with macroscopic hematuria in almost 75% of cases in the United States, although only in 26% of cases in Japan.[772,773,806,807] The hematuria usually occurs during an upper respiratory infection and resolves within 2 to 4 days. Less commonly it occurs in association with other infections associated with the mucosal system, such as sinusitis or diarrhea. Other less common presentations include microscopic hematuria with proteinuria, isolated microscopic hematuria, and isolated proteinuria. Even less commonly (~10% of cases), IgAN can present with macroscopic hematuria with reversible AKI, although it is rare for a child to present with CKD,[772,773] and even rarer to present with RPGN.[808,809] Hypertension is present in approximately 25% of children at presentation, although it is usually not severe unless AKI is also present.

Children with HSP nephritis typically present far earlier (~4–6 years) than those with IgAN (~10–30 years), although like IgAN, HSP nephritis can present at any age. The typical presentation includes a lower extremity purpuric rash, often associated with abdominal pain, and sometimes arthritis. In addition, approximately 25% of such children will also have macroscopic hematuria, although the renal manifestations of HSP nephritis at presentation may range from nonexistent to severe nephritic and/or NS.[767,810] Notably, children with HSP nephritis are approximately twice as likely to develop NS than those with IgAN.[773,810–813]

LABORATORY EVALUATION

Because no specific serologic biomarker exists to diagnose IgAN, confirmation of the diagnosis depends on the identification of IgA deposits by renal biopsy in the glomerular mesangium. When IgAN is suspected, however, laboratory evaluation generally includes serum creatinine, serum albumin, serum C3 level, serum antinuclear antibody (to exclude possible SLE), as well as a complete urinalysis and urine protein-to-creatinine ratio determination. Diagnosis of HSP nephritis includes identifying the clinical features of HSP noted earlier in combination with evidence of renal involvement on urinalysis and urine protein-to-creatinine ratio.

Serum IgA levels are often elevated in both IgAN and HSP nephritis, but have not been found to be sufficiently sensitive and specific to serve as a clinically useful biomarker of disease.[772,814,815] Serum C3 and C4 levels are usually normal in both diseases as well, helping to distinguish IgAN from the hypocomplementemia seen in PIGN, MPGN, or SLE nephritis. Although a renal biopsy for children with HSP nephritis is usually reserved only for more severe cases, the renal biopsy findings for HSP nephritis and IgAN are virtually indistinguishable.[787,816]

TREATMENT

Because of its variable disease progression, and its likely multifactorial pathogenesis, effective treatment of children with IgAN represents a significant challenge and remains somewhat controversial.[817] In addition, the published evidence to guide the treatment of children with IgAN differs significantly from the published evidence to guide the treatment of adults. Specifically, suggested treatments for adults with IgAN tend to be generally more passive, focusing more on nonimmunosuppressive treatment with ACE inhibitors and ARBs, whereas suggested treatments for children with IgAN tend to be more active, often including immunosuppression. In this context, Table 72.12 lists several of the current and emerging treatments reported for children with IgAN.

For children presenting with only mild IgAN or HSP nephritis, reflected by a first-morning urine protein-to-creatinine ratio of less than 0.5 mg/mg, most often no treatment is necessary. For those children who remain asymptomatic but who have more active disease, as reflected by a first-morning urine protein-to-creatinine ratio of greater than 0.5 mg/mg, initiation of ACE inhibitors and/or ARBs is generally indicated. By contrast, children who present with nephrotic-range proteinuria or NS require more aggressive treatment. This should include ACE inhibitors and/or ARBs to which oral prednisone (2 mg/kg PO daily; maximal dose = 80 mg/day) is added. If this treatment successfully induces

Table 72.12 Current and Emerging Treatment Options for Children With IgA Nephropathy or Henoch–Schönlein Purpura Nephritis

Immunosuppressive	Nonimmunosuppressive
Oral glucocorticoids	Angiotensin-converting enzyme inhibitors
Pulse intravenous methylprednisolone	Angiotensin receptor blockers
Mycophenolate mofetil (and mizoribine)	Fish oil[a] (omega 3 fatty acids)
Alkylating agents[a] (cyclophosphamide, chlorambucil)	Tonsillectomy[a]
Calcineurin inhibitors[a] (cyclosporine, tacrolimus)	Chinese herbal medicine[a] (Sairei-to; TJ-114)
Budesonide[b]	Coagulation-modifying drugs[a] (warfarin, urokinase, antiplatelet drugs)

[a]Less commonly used treatment.
[b]Emerging treatment.

remission of disease, the prednisone dose can be gradually tapered over 2 to 3 months and discontinued. If significant proteinuria persists, however, additional immunosuppressive treatment in indicated. Moreover, most pediatric nephrologists perform renal biopsies in children who present with NS, nephritic syndrome, or nephrotic-range proteinuria. If the biopsy reveals a significant percentage of crescents (typically >10%–20%), more aggressive immunosuppression with pulse IV methylprednisolone (typically 10–30 mg/kg over 1 hour; maximal dose = 1000 mg) is administered daily for 3 consecutive days, followed by initiation of daily oral prednisone as described earlier.

Interestingly, there is a fairly high rate of spontaneous remissions among children with mild forms of IgAN. One report of a large cohort of 555 children with only mild glomerular abnormalities on renal biopsy documented spontaneous remissions (i.e., without drug therapy) in the range of approximately 55% at 5 years and approximately 75% at 10 years from diagnosis.[818]

As noted earlier, the most widely used initial therapy for IgAN and HSP nephritis in both adults and children is treatment with ACE inhibitors and ARBs. While there are robust data demonstrating their effectiveness in reducing proteinuria and preserving renal function in adults,[819,820] there are fewer data supporting these same beneficial effects in children.[821,822] In addition, combination therapy with ACE inhibitors and ARBs has been reported to have beneficial and potentially additive effects in both adults and children.[823–826]

The second most widely used therapy for IgAN and HSP nephritis in children is treatment with oral and/or pulse IV glucocorticoids. While very widely used, the many historical variations in the reported routes of administration, as well as the doses and durations of treatment, have made it difficult in the past to accurately assess their efficacy.[785] However, there are now several randomized, controlled trials in adults and children demonstrating their effectiveness in reducing proteinuria and preserving renal function.[827–832]

In addition to ACE inhibitors/ARBs and glucocorticoids, fish oil has been utilized to treat IgAN. Notably, while randomized trials of adults have reported that fish oil for 2 years slowed the progression of renal disease, a randomized trial of children treated with omega 3 fatty acids failed to demonstrate a clear benefit in preserving renal function compared with placebo treatment.[833–836]

Additional commonly used immunosuppressive treatments for IgAN include cyclophosphamide, MMF or mizoribine, and cyclosporine. There are published reports suggesting some efficacy for each of these agents, but their use is generally less favored than the treatments described earlier, largely due to a relative lack of clear evidence to support more widespread use.[827,837–840]

There are also some nonimmunosuppressive treatments for IgAN that have reported some success. Tonsillectomy has been reported to have some marginal efficacy, although the studies were complicated by its use in combination with other simultaneous treatments, and the overall beneficial effects were uncertain.[841,842] Thus clear evidence to support its more widespread use in IgAN remains lacking. Another nonimmunosuppressive treatment is the Chinese herbal medicine Sairei-to (TJ-114). Although not widely used, a large randomized, controlled trial in children with mild IgAN by the Japanese Pediatric IgA Nephropathy Treatment Study Group reported that treatment early in disease for 2 years was more effective than no treatment.[843] In addition, coagulation-modifying treatments including warfarin, antiplatelet drugs, and urokinase have also been used to treat IgAN.[332,831,844] The data to date do not provide much support for their use as primary treatments for IgAN, although their inclusion as adjunctive treatments may have some potential.

Several new approaches to the treatment of IgAN are under investigation, at various stages of development. To date, however, only budesonide, a corticosteroid targeting the mucosal production of IgA1, has undergone a phase II trial demonstrating a reduction in proteinuria.[845]

Finally, in consideration of future clinical trials to improve the care of children with IgAN and HSP nephritis, it should be noted that given the prolonged clinical disease course for many children, the design of rigorous clinical trials to compare potential new treatments with current approaches suffers from the reality that most children will transfer their care to an adult nephrologist before a clinically meaningful endpoint for a clinical trial can be reached. Thus enhanced collaboration between pediatric and adult nephrology programs will likely be critical to demonstrate the long-term efficacy and safety of any future treatments for children with IgAN and HSP nephritis.

OUTCOME

The long-term outcome for children with IgAN is highly variable. Historical reports show outcomes ranging from complete remission of disease to development of ESRD. While approximately one-third of children will experience a complete remission of disease, long-term studies among children have reported the development of progressive CKD in 9% at 15 years from diagnosis, and ultimately ESRD in 20% to 27% of children, whereas adult studies have reported progressive CKD among 30% to 35% of patients at 20 years from diagnosis.[298,767,770,772,806,807,846–849]

In general, poor clinical prognostic indicators have included persistent severe proteinuria and persistent hypertension, while poor histologic prognostic indicators have included higher percentages of glomeruli with crescents, adhesions or sclerosis, significant tubulointerstitial changes, subepithelial electron-dense deposits, or lysis of the GBM.[296,782,848,850–852]

For children with HSP nephritis, long-term outcomes are less clear, due to marked variability in disease severity among the reported cases. While some reports have noted ESRD developing in 22% to 25% of children, others have estimated the risk for ESRD among all children with HSP nephritis at only approximately 3%.[308,769,786,787,810,813,844,853–855]

RENAL DISEASES WITH VASCULITIS

Vasculitis, or inflammation of the blood vessel walls, is an uncommon cause of glomerular injury in children, but can result in severe glomerulonephritis and even death. Vascular inflammation can lead to stenosis and/or aneurysm of the vessel walls, and result in various types of glomerulonephritis, depending of the size of the vessel walls and the organs most affected by disease. As such, a major effort was undertaken recently to update the nomenclature describing these various lesions.[856] Table 72.13 summarizes the current classification of vasculitides affecting children. The most common types of vasculitis seen in children are Kawasaki disease (KD) and IgA vasculitis (IgAV; i.e., HSP). IgAV and IgAN have already been discussed in detail earlier in this chapter, while the large vessel vasculitides, Takayasu arteritis, and giant cell arteritis are extremely rare in children and are thus discussed elsewhere in the text.

MEDIUM VESSEL VASCULITIDES

The medium vessel vasculitides include two primary diseases that include KD, which is common but only rarely has renal manifestations, and polyarteritis nodosa (PAN), which is rare in both children and adults, but commonly has renal manifestations (Table 72.13).

KAWASAKI DISEASE

KD is a vasculitis that typically involves the mucosa and skin, and is known to also preferentially affect coronary vessels. This common vasculitis is defined by a child having at least 5 days of fever, accompanied by four of the following five criteria: (1) edema, erythema, or desquamation of the hands, feet, or perineal area, (2) bilateral nonpurulent conjunctival injection, (3) polymorphous exanthema (primarily truncal), (4) oral and pharyngeal mucosal injection, and (5) cervical lymphadenopathy (typically unilateral). KD only rarely has renal manifestations, which can include sterile pyuria, proteinuria, AKI with tubulointerstitial changes, and rarely renal artery aneurysms. NS has also been reported, although it appears to remit spontaneously without glucocorticoid therapy.[857–861]

POLYARTERITIS NODOSA

PAN is now defined as a necrotizing arteritis involving small or medium arteries, without evidence of antineutrophil cytoplasmic antibodies or vasculitis in venules, capillaries, or arterioles.[856] While very uncommon in both children and adults, specific criteria for diagnosing pediatric PAN have been published.[862,863] This disease is specifically distinguished from microscopic polyangiitis (MPA) pathologically by its lack of ANCAs or necrotizing glomerulonephritis, and clinically by its higher incidence of gastrointestinal symptoms and peripheral neuropathy.[864] Clinical presentation typically includes fever, skin lesions, and/or myalgia, with involvement of the kidneys, heart, and/or lungs occurring in 15% to 20% of children.[865–867] Notably, the major renal manifestations result from involvement of medium size vessels, and include renovascular hypertension and aneurysms with the potential for rupture or hemorrhage. Treatment generally includes oral and/or IV glucocorticoids combined with cyclophosphamide.[868–870]

SMALL VESSEL VASCULITIDES

The small vessel vasculitides are now divided into two main categories (Table 72.13). The first category is AAV, and includes granulomatosis with polyangiitis (GPA; Wegener), MPA, and

Table 72.13 Classification of Pediatric Vasculitides by Primary Vessel Involvement

Small Vessel Vasculitis	Medium Vessel Vasculitis	Large Vessel Vasculitis[a]
• Antineutrophil cytoplasmic antibody (ANCA)–associated vasculitis (AAV) (i.e., pauci-immune small vessel vasculitis) • Granulomatosis with polyangiitis (Wegener, GPA; can also involve medium vessels) • Microscopic polyangiitis (MPA) - Eosinophilic granulomatosis with polyangiitis (Churg–Strauss syndrome, EGPA)[a]	• Kawasaki disease (KD)	• Takayasu arteritis[a]
• Immune complex small vessel vasculitis • IgA vasculitis (Henoch–Schönlein) (IgAV) • Antiglomerular basement membrane (anti-GBM) disease[a] • Hypocomplementemic urticarial vasculitis (anti-C1q vasculitis)[a] - Cryoglobulinemic vasculitis (CV)[a]	• Polyarteritis nodosa (PAN, can also involve small vessels)[a]	• Giant cell arteritis[a]

[a]Rarely seen in children.

Modified from Jennette JC, Falk RJ, Bacon PA, et al. 2012 revised International Chapel Hill Consensus Conference nomenclature of vasculitides. *Arthritis Rheum*. 2013;65:1–11.

Table 72.14 Clinical and Pathologic Features Distinguishing Selected Small Vessel Vasculitides

Feature	IgA Vasculitis (IgAV; Henoch–Schönlein Purpura Nephritis)	Granulomatosis With Polyangiitis (GPA; Wegener Granulomatosis)	Microscopic Polyangiitis	Eosinophilic Granulomatosis With Polyangiitis (EGPA; Churg–Strauss Syndrome)	Antiglomerular Basement Membrane (Anti-GBM) Disease	Systemic Lupus Erythematosus Nephritis (SLEN)
Vasculitis of small vessels	YES	YES	YES	YES	YES	YES
Immune complexes	YES	NO	NO	NO	YES (linear IgG deposits along GBM due to in situ immune complex formation)	YES
Necrotizing granuloma presence	NO	YES (usually extrarenal)	NO	YES (usually extrarenal)	NO	NO
Predominant IgA glomerular immune deposits	YES	NO	NO	NO	NO	NO
PR3 antibody presence	NO	Sometimes (more likely)	Sometimes (less likely)	Sometimes	NO	NO
MPO antibody presence	NO	Sometimes (less likely)	Sometimes (more likely)	Sometimes	Sometimes (overlap between anti-GBM and MPO-ANCA disease)	NO
Eosinophilia and/or asthma presence	NO	NO	NO	YES	NO	NO

eosinophilic granulomatosis with polyangiitis (EGPA; Churg–Strauss syndrome). The second category is IC small vessel vasculitis, and includes IgAV (HSP), anti-GBM disease, hypocomplementemic urticarial vasculitis (anti-C1q vasculitis), and cryoglobulinemic vasculitis. The clinical and pathologic features that distinguish those small vessel vasculitides seen more commonly in children from each other are compared in Table 72.14. Far and away the most common of these diseases seen in children is IgAV, and this topic was discussed earlier in this chapter.

While vasculitis is an uncommon cause of glomerular injury in children, presentation with vasculitis can be associated with the development of an RPGN. This is a clinical diagnosis characterized by rapid deterioration in renal function over a few days to weeks, as measured by a consistently rising serum creatinine. In this clinical setting, consideration needs to be given to the possibility of either an AAV or an IC small vessel vasculitis (Table 72.13). The ANCA-associated vasculitides are typically "pauci-immune," meaning that there are few or no immune deposits affecting the arterioles, capillaries, or venules (i.e., small vessels). From a clinical perspective, any of the small vessel vasculitides noted in Table 72.13 may be associated with ANCAs directed against either proteinase 3 (PR3) or myeloperoxidase (MPO). However, PR3–ANCA are more commonly associated with GPA, whereas MPO–ANCA are more commonly associated with MPA. In contrast to the pauci-immune ANCA-associated vasculitides, anti-GBM disease is associated with linear IgG staining of the GBM, and is almost always associated with a positive serum anti-GBM antibody test. In very rare cases where the serum anti-GBM antibody is negative, it is suspected that disease may be due to an anti-GBM antibody not detected by the current serum anti-GBM antibody test (i.e., "atypical anti-GBM disease"). Of significant clinical interest, a recent publication from the Midwest Pediatric Nephrology Consortium of a large multicenter cohort of approximately 260 children with glomerular disease who had more than one crescent on renal biopsy has reported a significantly worse 1-year renal survival among children whose biopsies had 43% or more crescents versus those with less than 43% crescents, regardless of the specific histologic diagnosis.

GRANULOMATOSIS WITH POLYANGIITIS (WEGENER GRANULOMATOSIS)

GPA is rare in children, with an incidence of 1 to 2 per million, compared with 1 to 2 per 100,000 adults.[848,871] It is a necrotizing granulomatous vasculitis of small- and medium-sized vessels that primarily affects the kidneys, the upper and lower respiratory tracts, and multiple other organs. Diagnosis in a child requires three of the following six features: (1) renal involvement, (2) upper airway involvement, (3) pulmonary involvement (infiltrates, nodules, or cavities), (4) laryngo-tracheobronchial stenosis, (5) granulomatous inflammation, and (6) ANCAs.[862] On renal biopsy, while granulomata are only rarely detected, approximately 90% of children have crescents, and more than half of children have crescents in more than 50% of their glomeruli.[872] Although the etiology remains unclear, recent reports have suggested that the initial injury is to endothelial cells, and that this injury is induced by inflammatory cells that have been primed with ANCAs.[873,874] However, the factor(s) precipitating the initial production of ANCAs remain unclear.[875–877]

From a clinical perspective, the vast majority of children with GPA (~90%) present initially with upper and/or lower respiratory findings, including epistaxis, sinusitis, nasal ulceration, hemoptysis, cough, subglottic stenosis (hoarseness), or dyspnea.[201,878,879] Additional common constitutional symptoms include weight loss, fever, and abdominal pain.[806] The presence of renal disease at presentation varies greatly, but can include hematuria, proteinuria, and AKI. Other organ systems that can also be affected include the eyes, CNS, skin, musculoskeletal system, and the heart. In 70% to 90% of cases, ANCAs are detected, usually with specificity for PR3 (c-ANCA).[201,862,878]

Treatment of GPA is generally separated into induction and maintenance phases. Induction phase treatment generally includes three daily initial IV methylprednisolone boluses, followed by oral glucocorticoids (1–2 mg/kg/day). In addition, IV cyclophosphamide (500–750 mg/m^2) is also administered every 2 to 4 weeks, although oral cyclophosphamide is sometimes used instead of IV treatments.[874] The anti-CD20 mAb rituximab has also been used as a newer alterative to cyclophosphamide for induction therapy, and has been reported to be as effective as cyclophosphamide in inducing remission in new patients, and superior to cyclophosphamide in inducing remission in relapsing patients.[880–882] Another alternative to cyclophosphamide for induction therapy is methotrexate, which has been reported to induce remission by 6 months at a similar rate in patients with mild vasculitis (serum creatinine < 150 μmol/L and without critical organ manifestations), but was associated with delayed induction of remission and more frequent relapses compared with cyclophosphamide.[861] Methotrexate should be avoided in patients with glomerulonephritis and decreased GFR. PLEX is used to treat patients with diffuse pulmonary hemorrhage, as well as those with RPGN approaching a need for dialysis, and has been reported to improve both patient survival and renal survival.[883–886] A large randomized controlled trial is ongoing to evaluate the role of plasmapheresis in patients with more moderate degrees of renal dysfunction (PEXIVAS trial, ClinicalTrials.gov identifier NCT00987389##). This trial is expected to be completed by the end of 2018. Lastly, an emerging induction treatment is the use of a C5a receptor blocker (Avacopan) as a steroid-sparing agent in addition to either cyclophosphamide or rituximab.[704] Table 72.15 summarizes the current and emerging induction phase and maintenance phase treatments used to treat GPA, which is very similar to the treatment of MPA described in the following section.

Maintenance phase treatment is initiated once a patient is declared to be in remission, and generally includes a gradual taper of prednisone, after the initial 4 weeks of therapy, over several months to a final maintenance dose of 5 to 10 mg every other morning. In addition, options for maintenance treatment also include azathioprine, methotrexate, MMF, or more recently rituximab. In a randomized controlled trial, maintenance therapy with azathioprine was found to be associated with lower rates of relapse than MMF.[887–890]

Relapses are relatively common among children with GPA, as they are among adults.[871] While the time to relapse ranges widely, from 4 months to 20 months, the median time to relapse is approximately 28 months from diagnosis.

Long-term outcomes for GPA appear to be somewhat better among children than adults. Approximately 80% to 90% of children achieve a remission of disease, although 40% of them have subsequent relapses.[891,892] Overall mortality rates are approximately 3% to 10%.[806]

Table 72.15 Current and Emerging Treatment Options for Children With Granulomatosis With Polyangiitis and Microscopic Polyangiitis

Induction Phase Treatment	Maintenance Phase Treatment
Pulse intravenous (IV) methylprednisolone (daily for 3 days)	Oral glucocorticoids (taper over several months to 5–10 mg qod)
Oral glucocorticoids (daily at 1–2 mg/kg/day)	Azathioprine
IV cyclophosphamide (500–750 mg/m^2 every 2–4 weeks; PO dosing sometimes substituted)	Mycophenolate mofetil
IV rituximab	IV rituximab
Plasma exchange (plasmapheresis; especially indicated in pulmonary hemorrhage)	Methotrexate[a]
C5a receptor blocker (Avacopan) as steroid-sparing agent in addition to cyclophosphamide or rituximab[b]	
Methotrexate[a]	

[a]Less commonly used treatment.
[b]Emerging treatment.

MICROSCOPIC POLYANGIITIS

MPA is also rare in children. Like GPA, it is also a necrotizing glomerulonephritis, with arteritis involving small- and medium-sized vessels of the kidneys and lungs. Unlike GPA, however, patients with MPA do not have granulomatous lesions, and typically do not have subglottic stenosis or severe, destructive involvement of the upper respiratory tract.[862,893] Children typically present with AKI due to necrotizing crescentic glomerulonephritis, and approximately 50% of them will also present with a pulmonary–renal syndrome, making a chest X-ray mandatory in all suspected cases.[891,894–898] For diagnosis of MPA, these children also have ANCAs, which are most often anti-MPO antibodies that show a perinuclear pattern (i.e., p-ANCA positivity) by indirect IF staining.[829] While the etiology of MPA is not entirely clear, reports from several experimental studies support the concept that the anti-MPO antibodies in MPA have a pathogenic role in disease induction, and that ANCA-associated disease has a genetic component.[874,899,900]

Treatment of MPA is very similar to GPA, and indeed many of the studies developing evidence for treatment of these vasculitides have included both patients with MPA and GPA. Table 72.15 summarizes the current and emerging treatments used to treat both GPA and MPA. Like children with GPA, the long-term outcomes for children with MPA appear to be generally better than those reported in adults.

Eosinophilic Granulomatosis With Polyangiitis (Churg–Strauss Syndrome)

EGPA, also known as Churg–Strauss syndrome, is even rarer than GPA or MPA. Like these diseases, it is a pauci-immune small vessel necrotizing vasculitis, but it is distinguished from them by the presence of extravascular granulomatous inflammation combined with asthma and hypereosinophilia.[856,901]

Clinically, children typically present with a long history of asthma, and are often found to also have histories of allergies, sinus disease, or peripheral nervous system abnormalities. In addition, vasculitic skin involvement appears to be relatively common, and similar to that seen in children with PAN.[902,903] Although relatively few data are available related to renal involvement specifically among children, renal involvement based on adult studies occurs in approximately 20% of patients, and typically manifests as acute glomerulonephritis with or without AKI.[902]

Laboratory evaluation for EGPA is generally similar to that for GPA or MPA, and children with EGPA are typically found to have ANCAs, most often p-ANCA indicating the presence of anti-MPO antibodies.[901] Histologic confirmation of disease usually includes findings of a necrotizing vasculitis of small- and/or medium-sized vessels, and granulomatous inflammation of the respiratory tract.

There are relatively few data on optimal treatment or long-term outcomes for children with EGPA, given its rarity. However, treatment is generally similar to that for the other small vessel vasculitides GPA and MPA, and is summarized in Table 72.15.

ANTI-GLOMERULAR BASEMENT MEMBRANE (ANTI-GBM) DISEASE

Anti-GBM disease is yet another rare cause of necrotizing small vessel vasculitis. Indeed, among children presenting with RPGN, anti-GBM comprised only 12% of cases, which was far less common than either pauci-immune RPGN (42% of cases) or IC-mediated RPGN (45% of cases).[872,904] This disease results from circulating antibodies which bind the noncollagenous globular domain (NC1 domain) at the C terminus of the alpha-3 chain of type IV collagen (COL4A3) in the GBM. This is seen in IF staining of renal biopsies as intense linear IgG staining along the GBM of glomerular capillary loops, with or without the presence of C3. Notably, these autoantibodies can also cross-react with alveolar basement membranes, resulting in lung injury and pulmonary hemorrhage (Goodpasture syndrome). Such pulmonary involvement is thought to result from exposure of the alveolar basement membranes to these circulating autoantibodies as a result of prior pulmonary injury, potentially due to hydrocarbons, glue, or cigarette smoke.[905,906]

Anti-GBM disease occurs more commonly among males and can occur at any age. However, there are two known peak incidences; at 30 to 40 years and again at 70 to 80 years.[806] Children with anti-GBM often present with constitutional symptoms, and in approximately 50% of cases will also have pulmonary symptoms.[904] For those with pulmonary involvement, presenting symptoms can include cough, anemia, hemoptysis, fever, as well as pulmonary infiltrates or frank pulmonary hemorrhage. Overall, approximately 50% of children present with isolated renal disease, whereas the remaining 50% of children present with combined renal and pulmonary disease.[872,907,908] Diagnosis of anti-GBM is confirmed with a combination of intense linear IgG staining of the GBM in glomerular capillary loops noted earlier, and a positive serum anti-GBM antibody test.

Treatment of anti-GBM is generally similar to the other vasculitides described earlier, with pulse IV methylprednisolone for several days followed by daily oral prednisone during the initiation phase of treatment, in conjunction with IV cyclophosphamide (500–1000 mg/m^2 monthly for 3–6 months), or less commonly oral cyclophosphamide (2 mg/kg daily for several months). In addition, PLEX is also widely used in an attempt to reduce circulating anti-GBM autoantibodies and proinflammatory cytokines.[872,909] Following induction of disease remission, which occurs in as many as 90% of patients, maintenance phase treatment typically includes azathioprine, methotrexate, or MMF.

Few long-term outcome data for children with anti-GBM are available. Despite this, unlike the pauci-immune vasculitides described earlier, anti-GBM is generally not considered a relapsing disease.[806] Known poor prognostic indicators for renal survival include severe AKI and/or extensive fibrous crescents at initial clinical presentation.[872]

SYSTEMIC LUPUS ERYTHEMATOSUS NEPHRITIS

SLE is a common systemic inflammatory autoimmune disease that can affect many organ systems. While renal involvement may not be present at the initial disease presentation, most children with SLE will develop renal involvement at some point in their disease, and SLEN often becomes among the most significant predictors of morbidity and mortality among children with SLE.[910–914] For example, data from adults have reported that approximately 10% of patients who develop SLEN will progress to ESRD.[915] In this light, this section will not address the initial diagnosis of SLE, but focus instead on the diagnosis and treatment of SLEN among children previously diagnosed with SLE.

The epidemiology of SLE is notable for its strong predilection for females, with the female-to-male ratio ranging from 1.3:1 among prepubertal children to 4.5:1 among adolescents to 8 to 15:1 among adults.[916–919] In addition, the prevalence of SLE among children has also been estimated to be as frequent as 10 per 100,000 persons.[920] Furthermore, higher risks for SLE compared with Caucasians have been reported for Asian (7×), African-American (4.5×), and Hispanic (3×) females, as well as more severe disease with poorer prognoses.[806,921]

The pathogenesis of SLEN among patients with SLE is multifactorial. While a detailed review of the genetics and pathogenesis is beyond the scope of this chapter, such a review has recently been published.[922] Briefly, recently published data suggest that the nephritis is initiated by an interferon response, possibly due to compromised clearance of apoptotic cells, that leads to differentiation of B cells into plasmablasts, which then progresses to activation of neutrophils and myeloid cells that results in both renal and systemic inflammation.[923–925] In addition, complement system activation is also common in SLEN and is thought to mediate kidney injury both directly via the terminal complement pathway and indirectly via recruitment of leukocytes to the kidney.[924]

The diagnosis of SLEN in patients with SLE may be clinically subtle, making screening urinalyses (at least yearly) of all SLE patients important. The most common clinical manifestations of SLEN include proteinuria (100%), which may range from mild to NS (50%), followed by microscopic hematuria (80%), tubular abnormalities (70%), and AKI (60%).[924] In children with SLE in whom SLEN is suspected, the renal evaluation should include a urinalysis, renal function panel, and consideration of a renal biopsy if more than mild disease appears present. The histologic classification of SLEN is described in detail elsewhere in the text.

The treatment of SLEN has been primarily guided by trials in adults due to the relatively low numbers of children with SLEN. In general, children with class I or class II SLEN generally do not need kidney-specific treatment, although close observation to ensure they do not convert to a more severe class of SLEN is important. Children who develop mild class III SLEN (<20% glomerular involvement) can usually also be managed conservatively, although more aggressive treatment is indicated when moderate or severe class III disease is present. For patients with class IV SLEN, aggressive treatment is clearly indicated, although the optimal treatment approach remains uncertain. Like some of the other small vessel vasculitides described earlier (i.e., GPA and MPA), treatment is usually divided into an induction phase and a maintenance phase. The current and emerging treatment options for each of these phases for SLEN in children are shown in Table 72.16.[926]

Induction phase treatment typically includes pulse IV methylprednisolone for 3 or more days, followed by daily oral prednisone at 1 to 2 mg/kg/day, in addition to treatment with IV cyclophosphamide (500–1000 mg/m^2 monthly for ~6 months), or less commonly oral cyclophosphamide (2 mg/kg daily for ~6 months). Such combined therapy has been reported to reduce the risk for subsequent disease progression, although it is not clear whether cyclophosphamide given IV is superior to oral administration.[910,927–931] A more recent alternative to cyclophosphamide for induction phase treatment has been MMF. Several reports in both children and adults have found MMF to be as or more effective than cyclophosphamide in inducing complete and partial remissions of disease, often with fewer side effects.[932–941] Dosing for MMF in children with SLEN is typically approximately 1200 mg/m^2/day, during both the induction phase and the maintenance phase of treatment. However, titration of dosing based on trough mycophenolic acid levels and/or area under the curve (AUC) has been associated with improved clinical efficacy.[942–944]

Table 72.16 Current and Emerging Treatment Options for Children With Systemic Lupus Erythematosus Nephritis

Induction Phase Treatment	Maintenance Phase Treatment
Pulse IV methylprednisolone (daily for 3 days)	Oral glucocorticoids (taper over several months)
Oral glucocorticoids (daily at 1–2 mg/kg/day)	Mycophenolate mofetil
IV cyclophosphamide (500–1000 mg/m^2 monthly for ~6 months; PO dosing sometimes substituted at 2 mg/kg daily for ~6 months)	Azathioprine
Mycophenolate mofetil	IV cyclophosphamide[a] (500–750 mg/m^2 every 3 months)
Calcineurin inhibitors (tacrolimus [preferred], cyclosporine, voclosporin[b])	Cyclosporine[a]
IV rituximab[b]	
Belimumab[b] (anti-BLyS mAb) (in addition to standard of care with steroids + cyclophosphamide or mycophenolate mofetil)	Hydroxychloroquine (adjunctive therapy)
Plasma exchange (plasmapheresis)[a]	
Azathioprine[a]	

[a]Less commonly used treatment.
[b]Emerging treatment.
From Dooley MA, Houssiau F, Aranow C, et al. Effect of belimumab treatment on renal outcomes: results from the phase 3 belimumab clinical trials in patients with SLE. *Lupus.* 2013;22:63–72.

Another newer alternative to cyclophosphamide for induction phase treatment of SLEN is the use of CNIs. While tacrolimus has become preferred over cyclosporine, there have now been several uncontrolled and a few controlled studies in both children and adults that have collectively reported that tacrolimus (or cyclosporine) is as or more effective than either cyclophosphamide or MMF in inducing partial or complete remissions of disease.[945–950] For induction phase treatment, voclosporin, a new CNI with less variable plasma concentrations, is under investigation in a phase III trial (ClinicalTrials.gov identifier NCT03021499).[949]

Yet another emerging alternative to cyclophosphamide for induction phase treatment is rituximab. This anti-CD20 mAb targets B lymphocytes, which have been implicated in the pathogenesis of SLE and SLEN due to their ability to both present (auto)antigens and produce (auto)antibodies. Several uncontrolled studies in adults and children and one controlled study in adults have reported that rituximab can be relatively effective in inducing partial or complete disease remissions, although variable frequencies of serious side effects were also noted.[951–956]

A less commonly used induction phase treatment is azathioprine. Although there are relatively few data to support its use in this setting, one study reported comparable partial and compete remissions to that of cyclophosphamide, although those patients who received azathioprine experienced more relapses than those who received cyclophosphamide.[957]

Finally, PLEX has also been used infrequently as an induction phase treatment for SLEN, with the rationale that removal of circulating autoantibodies should be clinically beneficial. One randomized trial compared standard treatment with oral glucocorticoids + oral cyclophosphamide with that same treatment combined with 12 plasmapheresis treatments over 4 weeks, but found no differences in either short- or long-term outcomes compared with standard treatment.[958] To date, there is no established role for PLEX or plasmapheresis in the routine induction therapy of SLEN. PLEX may have a role in the treatment of severe SLEN associated with thrombotic microangiopathies or APS.[959–961] Whether it has a role as an adjunctive therapy for severe SLEN refractory to conventional therapy is unclear.[962,963]

Maintenance phase treatment almost always includes oral glucocorticoids, with the dosage minimized to maintain remission while reducing the risk of long-term glucocorticoid side effects. Based on the available data from primarily adult studies, MMF and azathioprine both appear to be more effective than IV cyclophosphamide in preventing relapses, CKD, and death, with MMF being somewhat more effective than azathioprine.[964–969] Lastly, although not widely used, cyclosporine has also been used for maintenance phase treatment, where it has been reported to have similar long-term outcomes to azathioprine.[970] The optimal duration for maintenance phase treatment of SLEN remains uncertain, although suggested durations have ranged from approximately 2 years to as long as 8 years.[971]

Long-term outcomes for children with SLEN have improved significantly over the last few decades. Improved treatments have resulted in 10-year survival rates that now approach approximately 90%.[972–974] Among those patients with childhood onset of SLE, almost 50% of patients developed cumulative organ damage in one report, and those at highest risk for this were those who presented with CNS involvement and those who had more prolonged disease and received more IV cyclophosphamide.[972] Specifically among children, a recent study of long-term outcomes among 16 children with SLEN reported a cumulative 10-year renal relapse-free survival of 73%, and a cumulative probability that hospitalization would not be required of 94% at 1 year and 71% at 10 years.[975]

TUBULAR DISORDERS

FANCONI SYNDROMES—WITH EMPHASIS ON CYSTINOSIS

ETIOLOGY

Fanconi syndrome is the result of global proximal tubular dysfunction. It is characterized by the constellation of renal tubular acidosis (RTA), low-molecular-weight proteinuria, amino aciduria, glucosuria, and hypophosphatemia (secondary to renal phosphate wasting). Causes can be subdivided into primary/genetic and secondary. Table 72.17 provides a broad list of causes. Of note, Dent disease and the oculocerebral syndrome of Lowe are described in more detail in the next section discussing urolithiasis and nephrocalcinosis. The most common genetic cause of Fanconi syndrome in childhood is cystinosis, which will be focused on here. Cystinosis is a lysosomal storage disorder. It is the result of a mutation in the *CTNS* gene, encoding the protein cystinosin.[976] Defective cystinosin fails to transport cystine, a product of protein degradation, out of the lysosome into the cytoplasm.[977] Cystine therefore accumulates and precipitates in the lysosome, leading to cell toxicity and the clinical manifestations of the disease. Although cystinosis is the most common genetic cause of Fanconi syndrome, Fanconi syndrome is more likely to be seen as a consequence of drug toxicity, particularly in patients receiving chemotherapy. Finally, there are a number of rheumatological causes of acquired Fanconi syndrome that should be considered in the differential diagnosis (Table 72.17).

Table 72.17 Causes of Fanconi Syndrome

Genetic	
	Cystinosis
	Dent disease
	Galactosemia
	Glycogen storage disease (type 1)
	Hereditary fructose intolerance
	Lowe syndrome
	Mitochondrial diseases
	Tyrosinemia
	Wilson disease
Acquired – Drugs	
Nucleoside reverse transcriptase inhibitors	Tenofovir, adefovir
Nucleoside analogs	Didanosine, lamivudine, stavudine
Chemotherapeutics	Ifosfamide, cisplatin, streptozocin
Anticonvulsants	Valproic acid
Antibiotics	Aminoglycosides, expired tetracyclines
Antivirals	Cidofovir
Antiparasitics	Suramin
Miscellaneous/toxins	Fumaric acid, paraquat
Acquired – Other conditions	
Amyloidosis	
Heavy metals	For example, lead, cadmium, mercury
Membranous nephropathy	
Multiple myeloma	
Paroxysmal nocturia	
Postrenal transplantation	
Tubulointerstitial nephritis	
Vitamin D deficiency	

PRESENTATION

Patients with genetic causes of Fanconi syndrome typically present within the first year of life with manifestations of tubular transport dysfunction including polyuria, polydipsia, hypokalemia, metabolic acidosis, hypophosphatemia, dehydration, and subsequent failure to thrive.[978] This is also the typical presentation for the infantile form of cystinosis, the most common form of the disease. Less commonly, patients with cystinosis present later in childhood with other manifestations of the Fanconi syndrome, often rickets.[979] Because the proximal tubule is the primary site of vitamin D activation, patients with Fanconi syndrome often display vitamin D–deficient rickets.[980] This later presenting form of cystinosis has been referred to as the “late-onset” or “juvenile” form.[978,979] Although the clinical manifestations are the same, juvenile cystinosis is typically less severe, and loss of renal function proceeds less rapidly. Cystinosis is a multiorgan disease; over time patients typically develop a number of other sequalae including hypothyroidism, photophobia, chronic renal failure, myopathy including difficulty swallowing, diabetes mellitus, male hypogonadism, and pulmonary dysfunction.[978,981] Interestingly, an ocular variant of the disease typically manifests in adults[982] and is characterized by eye involvement only. Clinical manifestations include photophobia due to cystine crystal accumulation.

DIAGNOSIS

When the clinical features of Fanconi syndrome are identified on urine analysis and blood work (i.e., hypokalemia, hypophosphatemia, nonanion gap metabolic acidosis, low-molecular-weight proteinuria, amino aciduria, and glucosuria), patients should be screened for cystinosis by examining polymorphonuclear cell cysteine concentration.[978] Alternatively, in older patients, examination of the cornea for cystine crystals by slit lamp can be completed. Crystals are universally present after 16 years of age. Finally, molecular genetic testing demonstrating biallelic variants in the *CTNS* gene is also consistent with the diagnosis. In children presenting with Fanconi syndrome in the first year of life, the differential diagnosis should also include tyrosinemia, hereditary fructose intolerance, galactosemia, glycogen storage disease (type 1) mitochondrial disorders, Wilson disease, oculocerebral syndrome of Lowe, and Dent disease.

TREATMENT

Therapy should be aimed at correcting metabolic abnormalities arising from tubular transport defects. This should include the administration of alkali, typically as potassium or dicitrate, volume replacement, administration of vitamin D, and phosphate replacement.[978] Children who are severely affected may require growth hormone (GH) as well. Patients with cystinosis should receive cysteamine to reduce cellular cysteine levels as soon as the diagnosis is made.[983] This can prevent/delay the onset of other symptoms and protect renal function. Cysteamine eye drops should also be administered to prevent the onset or progression of ocular disease.[984] When children progress to renal failure, renal transplantation will prevent further renal disease. However, cysteamine therapy should be continued in order to prevent nonrenal disease. Adherence to cysteamine therapy has been facilitated by the introduction of the delayed-release cysteamine bitartrate.[985,986]

DISORDERS OF RENAL TUBULAR NACL TRANSPORT

BARTTER SYNDROME

Etiology

Bartter syndrome is typified by hypochloremic, hypokalemic, metabolic alkalosis, and both hyperreninemia and hyperaldosteronism secondary to volume contraction. Clinically, Bartter syndrome can be subdivided into antenatal Bartter syndrome, which presents shortly after birth; classical Bartter syndrome presenting in childhood or adolescence; and Bartter syndrome with deafness.[987] The etiology of each of these various clinical presentations is a mutation in a single gene affecting transport of sodium and chloride across the thick ascending limb (TAL) of the loop of Henle (Table 72.18). Antenatal Bartter syndrome is the result of a mutation in either the *SLC12A1* gene encoding the furosemide-sensitive sodium–potassium–chloride cotransporter, NKCC2, or *KCNJ1*, the gene encoding the renal outer medullary potassium channel, ROMK.[988–990] Classical Bartter syndrome is caused by mutations in the *CLCKNB* gene encoding the chloride voltage-gated channel CLCKb.[991] Bartter syndrome with sensorineural deafness is the result of mutations in either *BSND*, which encodes the accessory protein Barttin required for CLCKb (and CLCKa) activity or disease-causing mutations

Table 72.18 Gene Defects and Description of Disorders of Tubular Salt Handling

Name	Gene Affected[a]	Clinical Description
Primary Disorders of the Thick Ascending Limb		
Bartter syndrome	*SLC12A*	Polyhydramnios, premature delivery, hypokalemic metabolic alkalosis
	KCNJ1	Polyhydramnios, premature delivery, hypokalemic metabolic alkalosis, transient postnatal hyperkalemia
	CLCKNB	Childhood/adolescent presentation, hypokalemia, metabolic alkalosis
	BSND[b]	Polyhydramnios, premature delivery, hypokalemic metabolic alkalosis, and sensorineural hearing loss
	CASR	Metabolic alkalosis, hypokalemia, hypercalciuria, hypocalcemia with low parathyroid hormone
Primary Disorders of the Distal Convoluted Tubule		
Gitelman syndrome	*SLC12A3*	Metabolic alkalosis, hypokalemia, hypomagnesemia
EAST (epilepsy ataxia sensorineural-deafness tubulopathy) syndrome	*KCNJ10*	Hypokalemic metabolic alkalosis, epilepsy, ataxia, sensorineural deafness
Pseudohypoaldosteronism type 2	*WNK1, WNK4, CUL3,* and *KLH3*	Hyperkalemia, metabolic acidosis, high plasma aldosterone
Primary Disorders of the Collecting Duct		
Pseudohypoaldosteronism type 1	*SCNN1A, SCNN1B, SCNN1G*	Hyperkalemia, metabolic acidosis, high plasma aldosterone, autosomal recessive inheritance
	NR3C2	Hyperkalemia, metabolic acidosis, high plasma aldosterone, autosomal dominant inheritance
Liddle syndrome	*SCNN1A, SCNN1B, SCNN1G* (activating)	Hypertension, hypokalemia, suppressed renin and aldosterone

[a]Unless stated otherwise genetic alterations are inactivating.
[b]Mutations in both *CLCKNB* and *CLCKNA* genes cause a similar phenotype.

in both CLCKa and CLCKb.[992–994] Both CLCKb and CLCKa are expressed in the inner ear and their activity is required for hearing. Activating mutations in the calcium sensing receptor (CaSR), which is expressed in the basolateral membrane of the TAL, inhibit sodium chloride reabsorption from this segment, causing a Bartter-like syndrome.[995,996] However, the phenotype of these patients also includes low plasma calcium and inappropriately low parathyroid hormone (PTH) levels. Finally, polyhydramnios and transient antenatal Bartter syndrome have been ascribed to mutations in the *MAGED2* gene.[997]

Bartter syndrome is classified as type 1 to type 6, which reflects the gene affected. Type 1 is caused by mutations in *SLC12A1*, type 2 by mutations in *KCNJ1*, type 3 by *CLCKNB* mutations, type 4 *BSND* mutations, type 5 by activating mutations in the *CASR*, and type 6 by mutations in both *CLCKNA* and *CLCKNB*. Some references refer to Bartter-like syndromes and include the previous as well as Gitelman syndrome and the EAST (epilepsy ataxia sensorineural-deafness tubulopathy) syndrome (Table 72.18).[998] As these latter two disease processes affect a different part of the nephron and can often be separated clinically by the presence of hypomagnesemia and the absence of hypercalciuria, they are discussed separately in the following sections.

Presentation

Antenatal Bartter syndrome manifests in utero with polyhydramnios that commonly results in premature delivery. Infants typically present shortly after birth with polyuria, episodes of severe dehydration, recurrent vomiting, and failure to thrive. Significant hypercalciuria and nephrocalcinosis are often present, as 25% of filtered calcium is reabsorbed from the TAL by a process dependent on sodium chloride reabsorption. The presence of hyperkalemia in the first few weeks shortly after birth, which then changes to the classic hypokalemia typical of this syndrome, is seen in patients with mutations in *KCNJ1*.[999] This interesting phenotype has been proposed to be the result of the developmental expression pattern of potassium channels. Specifically, the BK potassium channel of the collecting duct is not expressed until approximately 6 weeks. As such, children with *KCNJ1* mutations cannot effectively excrete potassium until then and therefore become hyperkalemic until after this channel is expressed.[151]

Classical Bartter syndrome manifests itself later in infancy or throughout childhood. Presenting symptoms are varied but can include muscle weakness and cramps, fatigue, constipation, recurrent vomiting, polyuria, and/or significant volume depletion. Hypercalciuria and nephrocalcinosis are not always present. It is noteworthy that *CLCKNB* is also expressed in the basolateral membrane of the distal convoluted tubule (DCT), where it contributes to magnesium reabsorption. Thus in some cases mild hypomagnesemia has been reported.[1000]

Children with antenatal Bartter syndrome and deafness typically present in a similar fashion to antenatal Bartter syndrome, with severe polyhydramnios, premature delivery, polyuria, severe volume contraction, and failure to thrive.[993,994] Hyperkalemia is not a feature. However, auditory testing distinguishes these subtypes as sensorineural hearing is severely affected. Patients with activating mutations in the CaSR can present as previously or secondary to alterations in divalent cation handling, specifically with cramps or seizures. These patients display hypocalcemia, hypercalciuria, hypomagnesemia, and inappropriately low plasma PTH levels.[995,996]

Diagnosis

Diagnosis is based on analysis of specific genes based on the clinical presentation. A history of significant antenatal polyhydramnios, postnatal polyuria, and typical features of hypokalemic, hypochloremic metabolic acidosis should prompt hearing testing. If this is normal, then examination of *SLC12A1* should be considered. An immediate postnatal history of hyperkalemia shifting to hypokalemia several weeks later should prompt investigation of *KCNJ1*. If abnormalities in sensorineural hearing are detected, examination of the *BSND* gene should ensue, and if normal, then *CLCKB* and/or *CLCKA*. Later presentation without a significant antenatal history should lead one to suspect isolated abnormalities in the *CLCKB* gene. Finally, if serum calcium is low in the presence of low PTH, one needs to consider abnormalities in the CaSR resulting in its inappropriate activation.

Treatment

There is no cure for Bartter syndrome currently. Treatment should focus on replacing electrolyte losses and mitigating the effects of hyperaldosteronism and increased prostaglandin release.[998] Most patients will require potassium supplementation and younger children might need additional sodium in order to achieve adequate growth. Administration of an NSAID (2–4 mg/kg/day of indomethacin) and a potassium sparing diuretic (spironolactone or amiloride) can help to attenuate the effects of increased prostaglandin release and help to raise the serum potassium level. The addition of an ACE inhibitor should be considered to block the secondary activation of the RAAS; however, some children will have difficulty tolerating this due to blood pressure effects. Long term, approximately one-quarter of patients will develop some degree of renal impairment. However, with attention to electrolyte replacement and administration of indomethacin and amiloride, growth velocity can be recovered and maintained.[1001]

GITELMAN SYNDROME

Gitelman syndrome is the result of a failure to reabsorb sodium and chloride from the DCT. Patients with this syndrome display hypochloremic, hypokalemic metabolic alkalosis. In contrast to Bartter syndrome–affected children, patients manifest hypocalciuria and hypomagnesemia.[1002] The only identified etiology of this disease to date is a mutation in *SLC12A3*, which encodes the sodium-chloride cotransporter (NCC), expressed in the apical membrane of the DCT (Table 72.18).[989] Presentation of Gitelman syndrome is similar to classical Bartter syndrome, although patients often present later in life, including as adults. The clinical picture varies and can range from growth retardation through milder manifestations such as muscle weakness, cramping, and electrolyte abnormalities.[997]

Molecular diagnostics should be considered in patients with typical electrolyte abnormalities, including hypomagnesemia. Treatment requires replacement of electrolytes, in particular potassium and magnesium. The long-term outcome is excellent for these patients; however, potassium and magnesium supplementation at higher doses can lead to some gastrointestinal upset.[997] Because urinary prostaglandin excretion is not elevated in this syndrome, administration of prostaglandin synthesis inhibitors is not indicated.[1003]

EAST SYNDROME

EAST syndrome is a rare multiorgan disease that also includes renal salt losing due to a failure of sodium and chloride reabsorption from the DCT.[1004] The renal phenotype is thus very similar to Gitelman syndrome. Patients display hypokalemic metabolic alkalosis and hypomagnesemia due to renal wasting of this divalent cation. In contrast to Gitelman syndrome, patients with EAST syndrome also display epilepsy, ataxia, sensorineural deafness, and a tubulopathy. This condition is caused by mutations in the *KCNJ10* gene, which encodes the Kir4.1 potassium channel that is expressed in the basolateral membrane of the DCT (Table 72.18).[1004] This syndrome has also been referred to as the SeSAME syndrome, an acronym for seizures, sensorineural deafness, ataxia, mental retardation and electrolyte imbalance.[1005] Treatment of the renal manifestations includes replacement of electrolytes as in Gitelman syndrome.

PSEUDOHYPOALDOSTERONISM

Pseudohypoaldosteronism (PHA) is a collection of rare heterogeneous syndromes characterized primarily by mineralocorticoid resistance, resulting in reduced potassium and proton excretion from the collecting duct.[1006] The clinical constellation of hyperkalemia, metabolic acidosis, and increased plasma aldosterone levels is universally present. In some subtypes there is also significant salt wasting. PHA has been subdivided into three subtypes, which are described in the following sections (Table 72.18).

PHA Type I

PHA type I is characterized by presentation in infancy with life-threatening hyperkalemic metabolic acidosis and severe salt wasting tubulopathy leading to dehydration and failure to thrive.[1007] Both plasma renin and aldosterone levels are extremely high. PHA type I can be further subdivided into an autosomal recessive form that is the result of abnormalities in the genes encoding the different subunits of the ENaC, *SCNN1A*, *SCNN1B,* and *SCNN1G*.[1008] This autosomal recessive form can be distinguished from the autosomal dominant form as ENaC is also expressed in sweat glands, the lung, and the colon, and manifests as a multiorgan disease including the skin, respiratory as well as renal dysfunction.[1009] The autosomal dominant form is limited to renal resistance to mineralocorticoids and typically displays a less severe phenotype. Metabolic acidosis may be absent, but salt losing is a persistent feature. Autosomal dominant PHA type I is due to mutations in *NR3C2*, the gene encoding the mineralocorticoid receptor.[1010] Treatment of both varieties of PHA type I requires attention to sodium replacement enabling volume expansion. Potassium binders are also frequently necessary.

PHA Type II

In contrast to PHA type 1, patients with PHA type II, which has also been called Gordon syndrome, do not display a salt-losing phenotype. Rather, patients with this disorder display hyperkalemia and hypertension.[1011] This heterogeneous disorder is the result of increased sodium chloride reabsorption from the DCT, leading to volume expansion and suppressed renin levels. It is the result of alterations in a variety of genes including *WNK1*, *WNK2*, *CUL3,* and *KLHL3*.[1012,1013] Given that this disorder consists of increased sodium chloride transport from the DCT, simply blocking this pathway with a thiazide diuretic generally ameliorates the symptoms of this disease.[1012]

LIDDLE SYNDROME

The inappropriate increase of sodium and chloride transport from the collecting duct results in volume expansion and hypertension in the presence of suppressed renin and aldosterone levels, a syndrome referred to as Liddle syndrome. This constellation of clinical features is due to activating mutations in one of the genes encoding the ENaC, *SCNN1A*, *SCNN1B*, or *SCNN1G* (Table 72.18).[1014–1017] As in Gordon syndrome, therapy is directed at inhibiting the primary mechanism and involves treatment with the ENaC inhibitor amiloride. Although required life-long, this therapy completely prevents the manifestations of this disease.

RENAL TUBULAR ACIDOSIS

A nonanion gap metabolic acidosis is consistent with the presence of an RTA. However, especially in young children, it is more common to demonstrate a nonanion gap metabolic acidosis as a result of diarrhea. Therefore diarrheal illnesses must first be excluded as a cause of the nonanion gap metabolic acidosis. The nephron segment affected has been used to classify different RTAs.[1018] Abnormalities in proton excretion from the distal nephron are classified as distal or type I RTA. A failure to reabsorb bicarbonate from the proximal tubule is referred to as a proximal or type II RTA. Finally, type IV, which is also called hyperkalemic RTA, is due to aldosterone resistance or absence. Type III, or mixed RTA, is a rare disease that displays both a failure to reabsorb bicarbonate from the proximal tubule and the inability to secrete protons from the distal nephron.

Examination of plasma electrolytes is useful in identifying the subtype of the RTA. Types I and II typically display normal or low serum potassium levels while patients with type IV have high serum potassium.[1019] Consideration of the etiology of the disease helps to further distinguish between proximal and distal RTAs. Patients with proximal RTA maintain an ability to secrete protons into the urine. Thus when not receiving treatment, they should display a urine pH less than 5.5. Children with proximal RTA display a bicarbonate reabsorption threshold above which they will spill bicarbonate into the urine.[1020] Thus when administered alkali they have an alkaline urine. Another approach to determine urine acidification is the urine anion gap.[1021] Protons secreted into the urine are trapped by ammonia, forming ammonium (NH_4^+). While ammonium is difficult to measure in the urine, a surrogate is to calculate the difference between urine cations ($Na^+ + K^+$) and urine Cl^-. When acidotic, the patient should be secreting protons into their urine, forming ammonium, and therefore display a negative anion gap (urine $Na^+ + K^+ - Cl^- <0$). If this is not the case, then there is a failure to secrete protons by the distal nephron and the patient has a distal RTA. If the previous does not clarify the etiology of the disease, then more definitive testing can be employed. The traditional means of diagnosing proximal RTA is to bicarbonate load the patient in order to correct the acidosis and demonstrate a significant increase in fractional excretion of bicarbonate, that is, to greater than 15%.[1020] Distal RTA is definitively demonstrated by either an acid load with ammonium chloride (100 mg/kg), which is poorly tolerated, or by administrating furosemide and fludrocortisone and monitoring urine pH over the next 8 hours.[1022,1023] Patients with distal RTA are not able to acidify their urine below pH 5.3.

Type I, Distal Renal Tubular Acidosis

The most common form of RTA is type I, or distal RTA. Clinically patients with this disorder display hypercalciuria, hypocitraturia, nephrolithiasis, and nephrocalcinosis.[1024] Untreated disease often results in failure to thrive, bone demineralization, and rickets.[1025] The etiology of this disorder is either primary, that is, genetic, or acquired secondarily from drug toxicity, or a rheumatological condition. Genetic causes of distal RTA include mutations in the genes encoding machinery required for the secretion of protons from alpha intercalated cells of the collecting duct, specifically the plasma membrane H^+ATPase, that is, *ATP6V1B1* or *ATP6V0A4*,[1026–1029] or the basolateral chloride–bicarbonate exchanger AE1 (gene name *SLC4A1*; Table 72.19).[1026,1030–1032] The gene affected results in slightly different clinical manifestations. Mutations in *SLC4A1* produce a milder phenotype, which might not manifest itself clinically until the teenage years or adulthood, and is often inherited in an autosomal dominant fashion. By contrast, patients with mutations in the subunits of the H^+ATPase typically have more severe disease, which often presents in infancy and is commonly associated with deafness.

Type II, Proximal Renal Tubular Acidosis

Proximal RTA rarely occurs in isolation and is much more frequently seen as a part of the Fanconi syndrome, that is, with hypophosphatemia, glycosuria, aminoaciduria, and low-molecular-weight proteinuria. Common causes of the Fanconi syndrome in childhood have been described earlier and in Table 72.17. Patients with isolated RTA rarely have renal calculi or nephrocalcinosis, potentially due to the increased urinary citrate excretion that typically accompanies this disorder.[1024] They also tend not to display abnormalities in bone mineral density or rickets. Isolated proximal RTA is caused by mutations in the basolateral sodium-bicarbonate exchanger NBCe1 (gene name *SLC4A4*), which is essential to the reclamation of filtered bicarbonate from the proximal tubule (Table 72.19).[1033–1037] Patients with mutations in this gene also display ocular abnormalities and mental retardation.

Table 72.19 Renal Tubular Acidosis Characteristics and Single-Gene Defects Causing Renal Tubular Acidosis

Gene	Inheritance	Associated Clinical Features
Type I – Distal Renal Tubular Acidosis		
ATP6V1B1	AR	Nephrocalcinosis/ nephrolithiasis, sensorineural hearing loss
ATP6V0A4	AR	Nephrocalcinosis/ nephrolithiasis, late-onset sensorineural hearing loss
SLC4A1	AD (rarely AR)	Nephrocalcinosis, osteomalacia with AR inheritance can be associated with hemolytic anemia
Type II – Proximal Renal Tubular Acidosis		
SLC4A4		Cataracts, glaucoma, band keratopathy
Type III – Mixed Renal Tubular Acidosis		
CA2		Osteopetrosis

Causes of type IV, hyperkalemic metabolic acidosis, are described in Table 72.1.
AD, Autosomal dominant; *AR,* autosomal recessive.

Type IV, Hyperkalemic Renal Tubular Acidosis

Patients with type IV RTA display resistance to, or an absence of, aldosterone effect on the distal nephron. Patients typically display a milder degree of acidosis and hyperkalemia, which is proportionate to the degree of aldosterone deficiency. This disease can be the result of drug administration such as NSAIDs, ACE inhibitors, CNIs or potassium-sparing diuretics, obstructive uropathy, or due to a single gene defect such as occurs in PHA, which is described earlier.

A Note on Type III, Renal Tubular Acidosis

A failure to reabsorb bicarbonate from the proximal tubule and acidify the urine is known as a mixed, or type III RTA. This rare autosomal recessive disorder is a consequence of mutations in carbonic anhydrase II (CAII, gene name *CA2*), which is required for both processes (Table 72.19).[1038,1039] Type III RTA is a complex syndrome; in addition to RTA, patients also demonstrate osteopetrosis, cerebral calcifications, and mental retardation.[1040] Further features that have been associated with the disorder include facial bone overgrowth resulting in facial dysmorphism, conductive hearing loss, and blindness.

Treatment of RTA involves supplementation with alkali to normalize plasma pH. In the hypokalemic forms of the disease this is best supplied as potassium citrate.[1041] The amount of supplementation required to treat the disease differs depending on the cause, with distal RTA requiring much less supplementation (typically 1–2 mEq/kg/day of alkali) than proximal RTA (which can require upward of 10 mEq/kg/day of alkali). In patients with distal RTA, treatment should also include minimizing the risk of stone formation and renal calcification as this can lead to renal insufficiency.[1042] To this end, thiazide diuretics and increased fluid intake should be considered.

NEPHROGENIC DIABETES INSIPIDUS

Etiology

A failure to respond to the action of arginine vasopressin (AVP) results in nephrogenic diabetes insipidus (NDI).[1043] AVP is released in response to increased plasma osmolarity and results in the phosphorylation and insertion of aquaporin-2 into the apical membrane of principal cells of the collecting duct, allowing water to be reabsorbed.[1044] NDI can be either congenital (i.e., genetic) or acquired. The majority (90%) of congenital diabetes insipidus is the result of mutations in the AVP receptor, V2R, encoded by the gene *AVPR2*.[1045,1046] The other roughly 10% are due to mutations in aquaporin-2, encoded by *AQP2*.[1047] *AVPR2* mutations are inherited in an X-linked recessive fashion, whereas AQP2 mutations occur in an autosomal recessive manner. Acquired NDI can be the result of renal injury (i.e., AKI, obstructive uropathy, renal dysplasia, tubulointerstitial nephritis, cystic kidney diseases, or nephronophthisis) or drug toxicity (i.e., lithium, ifosfamide, foscarnet, and amphotericin B).

Presentation

Congenital forms of NDI tend to be more severe and present in infancy with polyuria, hypernatremic dehydration, fever, irritability, constipation, and failure to thrive.[1048] In contrast to Bartter syndrome, polyhydramnios is not a feature. Infants who are breastfed have less severe features, likely due to their reduced solute load. Untreated patients can present later in childhood with enuresis and mental retardation.[1049] Overall the symptoms tend to improve with age.

Diagnosis

The diagnosis of NDI requires the demonstration that urine osmolarity cannot be concentrated in the presence of desmopressin (1-deamino-8-D-arginine vasopressin [ddAVP]). This is demonstrated by performing a ddAVP challenge, and monitoring urine concentration. ddAVP can be administered intranasally, orally, intramuscularly, or intravenously. Administration of ddAVP by the intramuscular or intranasal route requires longer monitoring than oral or IV administration (6 vs. 4 hours). A diagnosis of NDI is established by the failure of the urine osmolarity to increase above 200 mOsm in this time frame. The development of a urine osmolarity greater than 200 mOsm but less than 800 mOsm in children (and <500 mOsm in infants) is consistent with a partial NDI. It is noteworthy that activation of V2R mediates peripheral vasodilation, thus patients with mutations of *AQP2* display tachycardia and reduced blood pressure in response to IV ddAVP, in contrast to children with mutations of V2R.

Treatment

It is important to make the diagnosis of NDI early and treat patients aggressively to avoid complications arising from severe and frequent hypernatremia. Adequate fluid intake should be insured (at least 150 mL/kg/day), which becomes easier as children age and can respond to thirst. A thiazide diuretic (i.e., hydrochlorothiazide 2 mg/kg/day) can be employed to increase proximal sodium and consequently water reabsorption and decrease urine output. Amiloride (0.2 mg/kg/day) can also be employed to reduce urine output, via a similar mechanism. Indomethacin (1–3 mg/kg/day) has also been demonstrated to be useful in reducing urine volume, perhaps by reducing GFR. Typically, a combination of these drugs are employed initially to reduce urine output. The combination of amiloride, a potassium sparing diuretic, and hydrochlorothiazide is beneficial to maintain normal plasma potassium levels.

There are several promising new therapies for this disease. Vaptans such as tolvaptan are small-molecule agonists that bind V2R. Many mutations in V2R cause protein misfolding and intracellular retention. These membrane-permeant drugs are proposed to rescue the receptor to the cell surface, thereby ameliorating the disease.[1050] Further, as the majority of patients have functional aquaporin-2 water channels, different attempts have been made to stimulate their phosphorylation and trafficking to the cell surface by activating the pathway downstream of V2R.[1051] Most promising has been the administration of phosphodiesterase inhibitors such as sildenafil[1052] or agonists of other G-protein coupled receptors expressed in the principal cell such as the prostanoid receptor butaprost.[1053] Finally, statins have been proposed to alter the principal cell membrane cholesterol content in a way that promotes the recruitment of aquaporin-2 to the apical plasma membrane.[1054,1055]

UROLITHIASIS AND NEPHROCALCINOSIS IN CHILDREN

EPIDEMIOLOGY

Although a common disorder in adults, affecting up to 10% of adult males, the frequency of kidney stone formation in childhood is much lower. However, the incidence of urolithiasis in children is increasing, accounting for a greater number of emergency room visits and hospitalizations.[1056,1057] The reason for this increasing incidence is unclear, although it is speculated to be the result of increased sensitivity of radiographic techniques, the increased incidence of childhood obesity, increased consumption of processed foods contributing to increased sodium chloride consumption, or the result of a greater number of surviving premature infants who are treated with medications causing urolithiasis.[1056,1057] In contrast to adults, males are not more frequently affected; indeed, emerging evidence suggests that girls more commonly suffer from kidney stones, especially during adolescence.[1058] Ultimately, when the mechanisms preventing the precipitation of solutes in urine are overwhelmed, calculi are precipitated, causing the signs and symptoms of urolithiasis

RISK FACTORS

HYPERCALCIURIA

Most kidney stones are composed of calcium oxalate, and the next most frequent type of stone is calcium phosphate. Perhaps not surprisingly then, the most common risk factor for kidney stone formation is the generation of urine containing an increased concentration of calcium.[1059,1060] Because of the difficulty in collecting a 24-hour urine in younger children, hypercalciuria is often diagnosed by a spot urine collection and then normalizing the concentration of urine calcium to urine creatinine. Importantly, urinary calcium excretion decreases with age.[1061,1062] Normal values are listed in Table 72.20.[1061,1063–1066] The majority of calcium filtered by the renal glomerulus is reabsorbed from the proximal tubule and TAL by a passive paracellular process.[1067] Therefore a common cause of hypercalciuria is excess sodium chloride ingestion. Metabolic acidosis causes hypercalciuria via liberation from bone.[1024] Multiple drugs induce hypercalciuria including loop diuretics, vitamin D (in high doses), and corticosteroids.

The most common etiology of hypercalciuria in childhood is idiopathic. However, the genetics of this “unknown” cause of hypercalciuria have begun to be unraveled with the discovery of the CaSR–claudin-14 axis (Table 72.21). This axis refers to the ability for the TAL to sense circulating plasma calcium levels via the basolaterally expressed CaSR and respond by increasing the expression of the tight junction protein claudin-14.[1068] This claudin blocks paracellular calcium reabsorption from the TAL, increasing urinary calcium excretion. Subtle variations in the *CaSR* gene as well as intronic

Table 72.20 Normal Pediatric Values for Urinary Excretion of Solutes

Solute	Ratio; Solute/Creatinine		24-Hour Urinary Excretion
Calcium	**mol/mol**	**g/g**	
<1 year	<2.2	<0.8	<0.1 mmol/kg
1–3 years	<1.5	<0.53	<4 mg/kg
3–5 years	<1.1	<0.4	
5–7 years	<0.8	<0.3	
>7 years	<0.6	<0.2	
Citrate	**mol/mol**	**g/g**	
0–5 years	0.12–0.25	0.20–0.42	>0.8 mmol/1.73 m^2
>5 years	0.08–0.15	0.14–0.25	>0.14 g/1.73 m^2
Oxalate	**mmol/mol**	**mg/g**	
<1 year	15–260	12–207	<0.5 mmol/1.73 m^2
1–5 years	11–120	9–96	<45 mg/1.73 m^2
5–12 years	60–150	47–119	
>12 years	2–80	2–64	
Uric Acid	**mol/mol**	**g/g**	
<1 year	<1.5	<2.2	>1 year; <815 mg/1.73 m^2
1–3 years	<1.3	<1.9	
3–5 years	<1.0	<1.5	
5–10 years	<0.6	<0.9	
>10 years	<0.4	<0.6	
Cystine	**mmol/mol**	**mg/g**	
<1 month	<85	<180	<10 years: < 55 µmol/1.73 m^2
1–6 months	<53	<112	>10 years: < 200 µmol/1.73 m^2
>6 months	<18	<38	>18 years: < 250 µmol/1.73 m^2

Table 72.21 Genetic Conditions Displaying Hypercalciuria

Condition Name	Gene(s)	Clinical Features
Dent disease	*CLCN5*	Nephrocalcinosis, low-molecular-weight proteinuria, X-linked inheritance
Oculocerebral syndrome of Lowe	*OCRL*	Congenital cataracts, mental retardation, hypotonia, rickets, partial or complete Fanconi syndrome
Bartter syndrome	*SLC112A1, KCNJ1, CLCKNB, BSND*	Hypokalemic metabolic alkalosis, nephrocalcinosis
Autosomal dominant hypocalcemia and hypercalciuria	*CASR*	Hypocalcemia, suppressed parathyroid hormone, hypokalemic metabolic alkalosis
Familial hypomagnesemia hypercalciuria and nephrocalcinosis	*CLDN16, CLDN19*	Hypomagnesemia, hypercalciuria, nephrocalcinosis, progressive renal impairment, macular colobomata, myopia horizontal nystagmus (CCLDN19 mutations only)
Distal renal tubular acidosis	*ATP6V0A4, ATP6V1B1, SLC4A1*	Nephrocalcinosis, kidney stones, renal tubular acidosis
Williams syndrome	???	Cardiovascular disease, triangular facies, connective tissue abnormalities, intellectual disabilities

variations in the *CLDN14* gene appear to inappropriately cause increased claudin-14 expression and calciuria.[1069–1072] Further, mutations in the TAL paracellular calcium pore forming claudin-16 also contribute to idiopathic hypercalciuria.[1073] Finally, there are multiple genetic syndromes, which have as a component of their phenotype hypercalciuria. They are discussed in the next section on the genetics of pediatric urolithiasis and nephrocalcinosis.

HYPOCITRATURIA

Citrate binds calcium in the urine and prevents the precipitation of calcium salts. Thus a consequence of hypocitraturia is the formation of calcium containing urolithiasis. Although affecting approximately 10% of children with urolithiasis, the etiology of reduced urinary citrate excretion is unclear, except in specific genetic conditions such as distal RTA.[1074]

Another approach to detecting low urinary citrate excretion has been to evaluate the calcium-to-citrate ratio (rather than citrate-to-creatinine ratio). This has helped to identify children who might benefit from therapy.[1075]

HYPERURICOSURIA

A small fraction of children with nephrolithiasis (approximately 5%) will display hyperuricosuria. Like urinary calcium excretion, urinary urate excretion is highest in infants and decreases with age. In fact, the concentration of uric acid in the urine of infants can be so high that it precipitates as crystals and can be confused with blood. Normal values for urinary uric acid excretion are presented in Table 72.20. Traditionally, uric acid stones have been seen in children with overproduction due to tumor lysis syndrome, although the frequency of this has been greatly reduced with the introduction of urate oxidases such as rasburicase. Other causes include high dietary purine intake and hemolysis.[1076] There are also a number of rare genetic disorders, which result in increased urinary uric acid secretion, which are outlined later.

HYPEROXALURIA

After hypercalciuria, hyperoxaluria is the next most common urinary abnormality seen in children with urolithiasis.[1077] The etiology of hyperoxaluria is most frequently idiopathic, although a number of primary and secondary causes should be considered. Secondary causes are much more frequent than primary genetic disorders. Increased ingestion of oxalate or oxalate precursors such as vitamin C can lead to hyperoxaluria as can pyridoxine deficiency or ingestion of methanol. Alternatively, hyperabsorption of oxalate from the intestine, which can be due to a variety of diseases that lead to fat malabsorption including inflammatory bowel disease, post-bowel resection, pancreatic insufficiency, or biliary disease, can cause hyperoxaluria. The mechanism mediating this is via excess intestinal luminal fat binding calcium, leading to increased free oxalate in the intestinal lumen that can be more readily absorbed, and subsequently excreted in the urine. Therapies aimed at improving the underlying disorder, decreased oxalate ingestion, increasing free water intake, and added calcium administration help to treat this form of hyperoxaluria.[1040,1078]

ALTERED URINARY FLOW AND INFECTIONS

A nonmetabolic risk factor for kidney stone formation is urinary stasis or abnormal urinary flow. Children with congenital abnormalities of the kidney and urinary tract are more likely to develop kidney stones. UTIs with urea-splitting organisms such as *Proteus* spp. can alkalinize the urine and increase ammonium production. This in turn favors the formation of struvite stones (magnesium ammonium phosphate).[1079] These stones typically fill multiple renal calyces in a staghorn configuration. Patients with neurogenic bladders who have undergone surgical procedures including ileal augmentation are at increased risk of struvite stone formation.[1076]

PRESENTATION

Infants and younger children often do not present with typical flank pain radiating to the groin.[1058,1080] Further, a significant fraction of children may not even display hematuria. Instead young children can present with nonspecific abdominal pain, dysuria, frequency, urgency, emesis, diarrhea, or a UTI.[1081] Of note, patients with hypercalciuria are at increased risk of developing a UTI independent of urinary tract abnormalities.[1082,1083] Why hypercalciuria predisposes to UTI is not clear; perhaps crystals in the urinary tract form nidus for bacteria to adhere to. Adolescents more commonly present like adults with renal colic. However, as noted earlier, there is not a clear male preponderance.

DIAGNOSIS

Having a high index of suspicion is key in younger children. The radiologic test of preference is a renal ultrasound as it avoids exposure to ionizing radiation. Unfortunately, despite significant advances in ultrasound technology, smaller stones and stones in the calyces, papillae, or ureter may be missed by this technique.[1084] Large radiopaque stones are visible on plain films of the abdomen; however, helical noncontrast computed tomography (CT) examination is the most sensitive test and should be considered when ultrasound is negative and there is a high index of suspicion.[1085] Helical CT has the added benefit of providing further information with regard to anatomy and obstruction.

Examining urine for the presence of crystals is also useful. Identification of colorless hexagonal crystals are diagnostic of cystinuria. Large rectangular crystals are consistent with hyperoxaluria and calcium oxalate stone formation. Further clues can be provided by urine examination for hypercalciuria, hyperoxaluria, cystinuria, and hyperuricosuria. Serum calcium, phosphorus, uric acid, and electrolytes should be measured and can help identify the underlying cause.

TREATMENT

Renal calculi greater than 7 mm are unlikely to pass and elective surgical removal should be considered.[1080] Other indications for surgical removal include single kidney, prolonged obstruction, significant pain, or infection. Surgical options include extracorporeal shock wave lithotripsy (for stones >2 cm), percutaneous nephrolithotomy, and ureteroscopy. It may be necessary to leave a ureteral stent in some patients in order to dilate the ureter before extraction can be accomplished. For stones impacted in a ureter, medical expulsive therapy can be considered. This includes hydration and use of either an α-adrenergic blocker or a calcium channel blocker. While significant evidence of the effectiveness of these therapies exists for adults, there is only anecdotal evidence available for children.[1086,1087]

Dietary advice should otherwise be provided to decrease the risk of recurrence, which is significant in infants and children.[1088] Fluid intake should be increased to approximately 2 L/m^2 to both dilute precipitating solutes and help with the passage of stones in situ. In patients with hyperoxaluria, a low oxalate diet should be recommended. Hypercalciuric patients (and indeed most patients) benefit from a normal sodium diet (i.e., at the minimum <2000 mg sodium chloride per day). Restriction of calcium should be avoided,[1089] as this can be paradoxically associated with hypercalciuria due to the dissolution of calcium from bone. Instead attention should be paid to ensuring adequate calcium intake. High fructose containing beverages should be avoided.[1090] Finally, in contrast to adults, a low protein diet is not recommended due to potential adverse consequences on growth.

Medical therapy should target underlying metabolic risk factors. Patients with hyperuricosemia typically benefit from allopurinol. Patients with hypocitraturia should receive potassium citrate. It is not recommended to alkalinize the urine of all stone formers as this simply changes the forming calcium oxalate stones to calcium phosphate. Patients with hypercalciuria, besides the many dietary recommendations suggested earlier, may also benefit from administration of a thiazide (i.e., 1–2 mg/kg per day of hydrochlorothiazide). Thiazides, in blocking sodium reabsorption from the DCT, augment sodium and consequently calcium reabsorption from the proximal nephron, thereby contributing a hypocalciuric effect.[1091]

GENETICS OF PEDIATRIC UROLITHIASIS AND NEPHROCALCINOSIS

HYPERCALCIURIA-INDUCING CONDITIONS

WILLIAMS SYNDROME

Williams syndrome is characterized by a constellation of cardiovascular disease (typically peripheral pulmonary stenosis but can include supravalvular aortic stenosis or hypertension), a distinctive triangular facies, connective tissue abnormalities, intellectual disability (typically not severe), unique personality features, early puberty, and in up to 50% of cases hypercalcemia in infancy, hypercalciuria, and nephrocalcinosis.[1092] The etiology of this disorder is a 7q11.23 contiguous deletion of the Williams–Beuren syndrome critical region encompassing the elastin gene (Table 72.21). It is unclear why a deletion of this region results in hypercalcemia, hypercalciuria, and nephrocalcinosis. Hypercalcemia tends to resolve by the age of 4.[1093] The hypercalciuria responds to typical treatments including increased fluid intake, a reduced sodium diet, and thiazide diuretics, and this treatment should be considered, especially when nephrocalcinosis is present.

DENT DISEASE/OCULOCEREBRAL SYNDROME OF LOWE

Dent disease is typically an X-linked recessive disorder characterized by hypercalciuria, nephrocalcinosis, low-molecular-weight proteinuria, and renal insufficiency with some patients progressing to renal failure.[1094] Dent disease can also be associated with the complete Fanconi syndrome. It is most commonly caused by a mutation in *CLCN5,* which encodes a chloride-proton exchanger expressed at high levels in the kidney, with the highest levels in the proximal tubule and TAL (Table 72.21).[1095,1096] The exact molecular pathophysiology is unclear, especially with regard to the pathogenesis of hypercalciuria and nephrocalcinosis. However, defects in CLCN5 appear to cause an impairment of endocytosis, which somehow results in the clinical phenotype observed.

A smaller fraction of patients with the Dent phenotype, described earlier, do not have mutations in CLCN5, but instead display variations in the *OCRL* gene, which encodes the oculocerebral syndrome of Lowe protein 1.[1097] This protein is an inositol polyphosphate 5-phosphatase, which regulates endomembrane lysosomal sorting. More typically individuals with mutations in *OCRL* have congenital cataracts, mental retardation, muscle hypotonia, and rickets as well as evidence of partial or complete Fanconi syndrome.[1098,1099] Renal insufficiency can occur in more severely affected individuals, which are predominantly boys as this is also an X-linked disorder.

BARTTER SYNDROMES

Disorders of TAL sodium and chloride reabsorption are often accompanied by hypercalciuria and nephrocalcinosis. This association is the direct consequence of transcellular sodium-chloride transport being necessary for passive paracellular calcium (and magnesium) reabsorption from this nephron segment. Hypercalciuria and nephrocalcinosis tend to be more severe in neonatal Bartter syndrome patients with mutations in *SLC12A1* and *KCNJ1* than in patients with classical Bartter syndrome due to mutations in *CLCKNB*.[1100]

AUTOSOMAL DOMINANT HYPOCALCEMIA WITH HYPERCALCIURIA

Children with activating mutations in either the CaSR or the G protein coupled to CaSR display a phenotype characterized by hypokalemia, hypercalciuria, and hypoparathyroidism.[996,1101] Interestingly, there is at least one other gene responsible for this disease that has yet to be identified.[1102] The CaSR resides both in the parathyroid gland and in the nephron, including the basolateral membrane of the TAL. In the TAL, activation of the CaSR not only prevents paracellular calcium reabsorption by increasing claudin-14 expression in this segment, but also inhibits sodium and chloride reabsorption. Hence patients with these mutations can also demonstrate a Bartter phenotype (Table 72.21).[996] This is why these mutations have also been referred to as Bartter syndrome type 5 causing mutations. Treatment of these defects is slightly different than classical Bartter syndrome as administration of active vitamin D is required to elevate serum calcium levels as is a thiazide diuretic to reduce urinary calcium excretion. Unfortunately, this can lead to hypokalemia, which can be remedied by addition of a potassium-sparing diuretic in a combined diuretic such as aldactazide.

FAMILIAL HYPOMAGNESEMIA HYPERCALCIURIA AND NEPHROCALCINOSIS

The clinical constellation of hypomagnesemia (secondary to renal wasting), hypercalciuria, and nephrocalcinosis is due to a mutation in either claudin-16 or claudin-19.[1103–1105] Both these tight junction proteins are expressed in the TAL of the loop of Henle where they are essential to forming a paracellular divalent cation permeable pore. Consequently, disruption in the structure of one of these pore forming claudins prevents the reabsorption of calcium and magnesium from the TAL, leading to the clinical phenotype. Unfortunately, a majority of patients tend to progress to renal impairment with upward of 50% requiring RRT.[1106] Clinically, mutations in claudin-19 can be distinguished from claudin-16 alterations by the presence of ocular abnormalities including macular colobomata, myopia, and horizontal nystagmus.[1105] Claudin-19 is also expressed to a high level in the eye consistent with this phenotype. Interestingly, milder mutations in claudin-16 have been proposed to cause isolated hypercalciuria in the presence of only mild hypomagnesemia or without any alteration in plasma magnesium (Table 72.21).[1073]

DISTAL RENAL TUBULAR ACIDOSIS

Type 1, or distal RTA, is due to a defect in the ability to secrete protons into the urine. This is a result of mutations in either genes encoding the plasma membrane H^+ATPase (i.e., *ATP6V1B1* or *ATP6V0A4*) or the basolateral chloride bicarbonate exchanger, AE1 (gene name *SLC4a1*; Table 72.21). In contrast to proximal RTA, distal RTA is often accompanied by hypercalciuria and nephrocalcinosis, perhaps due to hypocitraturia, which is not seen with proximal RTA.[1024] Nephrocalcinosis has been proposed to contribute to renal dysfunction in these patients, thus treatment of the hypercalciuria is recommended to prevent renal insufficiency. Finally, it should be considered that patients with incomplete distal RTA due to mutations in *SLC4a1* may display hypercalciuria and kidneys stones.[1107]

HEREDITARY HYPOPHOSPHATEMIC RICKETS WITH HYPERCALCIURIA

This syndrome presents with rickets, short stature, and increased renal phosphate excretion despite hypophosphatemia and hypercalciuria. It is the result of mutations in the type 2, sodium phosphate transporter isoform C, encoded by the gene *SLC34a3*.[1108] This transporter contributes significant phosphate reabsorption from the proximal tubule, and when not functioning leads to a primary renal leak. As a consequence, serum phosphate levels are reduced, driving the suppression of PTH and increasing 1,25-dihydroxyvitamin D synthesis. This leads to increased intestinal phosphate and calcium absorption, resulting in hypercalciuria.

NONCALCIUM WASTING DISORDERS

CYSTINURIA

This is a rare autosomal recessive cause of kidney stone formation. The primary defect is in the reabsorption of dibasic amino acids, specifically cysteine, ornithine, arginine, and lysine.[1109] These amino acids are reabsorbed from the glomerular filtrate in the proximal tubule via the amino acid transporters encoded by the *SLC3A1* or *SLC7A9* genes. Mutations in *SLC34A1* cause type A cystinuria and *SLC7A9* type B cystinuria.[1110,1111] They are clinically indistinguishable and do not appear to correlate with disease severity or response to therapy. Moreover, compound heterozygotes with monoallelic mutations in both genes do not display altered urinary dibasic amino acid excretion.[1112]

Patients can present at any age, but typically present in the early teens. Severe cases may present in the first year of life. Diagnosis is suggested by family history and confirmed by isolation of pathognomonic hexagonal crystals on urinalysis.[1109] Incident patients with kidney stones can be screened with the cyanide-nitroprusside test. A positive result of this screening study should be confirmed with high-performance liquid chromatography for urine amino acids. Urinary cystine excretion increases with age and normal values can be found in Table 72.20. Patients with cystinuria generally display more than five times the upper limit for age. Patients with Fanconi syndrome and heterozygotes will have intermediate excretion values; however, these patients tend not to form stones with this amount of cysteine excretion.

First-line therapy should include increased fluid intake, urinary alkalization, and a lower sodium containing diet. In cases where significant cystinuria persists despite these measures a number of therapies have been proposed. Alpha-mercaptopropionylglycine forms a soluble dimer with cysteine and can be administered at a dose of 10 to 15 mg/kg/day.[1113] This therapy reduces new stone formation and may dissolve existing calculi. D-penicillamine is a chelating agent that binds to cysteine and homocysteine increasing their solubility. This compound has the potential for significant side effects including renal and monitoring is required. Other potential therapies that bind cysteine increasing its solubility include captopril and bucillamine.[1114]

HYPERURICOSURIA (INCLUDING 2,8-DIHYDROXYADENINE UROLITHIASIS)

A number of rare single-gene defects of purine metabolism result in hyperuricosuria and uric acid lithiasis. Defects in hypoxanthine guanine phosphoribosyl-transferase (HPRT1) cause Lesch–Nyhan syndrome which is an X-linked disorder characterized by dystonia, gout (including hyperuricosuria), intellectual disability, and self-mutilation.[1115] Patients with glycogen storage disease type 1a have mutations in the enzyme glucose-6-phosphatase.[1116] They demonstrate a severe phenotype typically in the first year of life, which includes growth retardation, lactic academia, hypoglycemia, hepatomegaly, hyperlipidemia, and hyperuricemia leading to hyperuricosuria and uric acid stone formation. Another rare enzymatic deficiency is adenine phosphoribosyl-transferase deficiency, which leads to the overproduction of 2,8-dihydroxyadenine that is excreted in the urine and can precipitate causing calculi.[1117] Urolithiasis is the most common manifestation of this disorder and patients can present at any age with calculi. A urinalysis will display characteristic round, brown crystals suggestive of the disorder, which can be tested by enzymatic activity testing or demonstration of a biallelic pathogenic variant in the *APRT* gene. A low purine diet and ample food intake is the recommended treatment for the renal manifestations of these disorders. Medical therapy can include allopurinol even for patients with APRT deficiency. Alternatively, for APRT deficiency, the xanthine dehydrogenase inhibitor febuxostat can be employed.

PRIMARY HYPEROXALURIAS

This is a group of rare autosomal recessive disorders that display increased generation and urinary excretion of oxalate (i.e., hyperoxaluria).[1064] Three different subtypes have been described and are the result of mutations in three different enzymes that participate in oxalate metabolism. Type 1 is the most severe form and type 3 the least. Type 2 can be difficult to distinguish from type 1 by clinical features alone. Instead, measurement of urinary glycolate and L-glyceric acid is beneficial. Patients with type 1 hyperoxaluria will display elevated urinary glycolate, whereas patients with type 2 will demonstrate elevated urinary L-glyceric acid. Confirmation of the diagnosis can be made by DNA analysis of the affected enzyme and demonstrating a deleterious genetic alteration.

Hyperoxaluria Type 1

Many patients with type 1 hyperoxaluria will develop progressive severe nephrolithiasis and nephrocalcinosis leading to renal failure. However, there is significant phenotypic variation, with some patients developing renal failure in infancy and others not developing their first renal calculi until adulthood.

The majority of patients with type 1 hyperoxaluria typically present with symptoms of their renal calculi before the age of 5 years, and in one cohort these patients accounted for 1% of children with ESRD.[1118] Mutations in alanine-glyoxalate aminotransferase are responsible for hyperoxaluria type 1.[1119] The variability of disease severity is due to mutations that permit residual enzyme activity. Importantly, residual alanine-glyoxalate aminotransferase enzyme activity in some cases can be stimulated by the cofactor pyridoxine.[1120] Patients with these mutations tend to display a more benign course. In the most severe cases infants can progress to renal failure within the first year of life; other cases might not be diagnosed until presentation of ESRD.

Systemic oxalosis occurs when plasma oxalate levels rise in patients with primary hyperoxaluria with decreasing GFR, due to reduced oxalate clearance. Systemic oxalosis results in the precipitation of calcium oxalate crystals in multiple tissues including the skeleton, heart, blood vessels, joints, retina, and skin. This is a severe disease with significant functional sequalae. Treatment is difficult and relies on aggressive dialytic therapy bridging to kidney–liver transplantation. Diagnosis is suggested by calcium oxalate nephrolithiasis, nephrocalcinosis, and significantly increased urinary oxalate excretion. Urinary glyoxalate will be elevated in the majority of patients; however, urinary glyoxalate can also be increased in patients with type 3 hyperoxaluria. Systemic oxalosis is not described in patients with a GFR greater than 60 mL/min/1.73 m^2, and as such there is no need to monitor plasma oxalate levels until GFR is below this level. Historically measurement of reduced alanine-glyoxalate aminotransferase activity was performed on liver biopsy to definitively demonstrate the diagnosis of primary hyperoxaluria type 1. However, this this is no longer necessary if one can demonstrate a deleterious mutation in the alanine-glyoxalate aminotransferase gene.

Therapy involves increasing fluid intake (to at least 2.5 L/m^2), avoidance of high oxalate-containing foods (i.e., chocolate, tea, rhubarb, and spinach), and administration of potassium citrate to both ensure adequate urinary citrate excretion and alkalinize the urine. Both of these urinary alterations tend to prevent calcium oxalate precipitation. A trial of high-dose pyridoxine (5–10 mg/kg/day) should be administered and urine oxalate followed to identify responders. Specific mutations are known to be sensitive to this therapy. Thus genetic analysis is helpful for therapeutic decision-making and prognostication. The earlier in the course of the disease therapy is instituted, the better the reported outcome and the longer renal function can be preserved.[1121] Unfortunately, even very intensive dialytic therapies (i.e., up to 40 hours of hemodialysis per week) are often ineffective at preventing the consequences of systemic oxalosis. Isolated kidney transplantation is not recommended, as it does not correct the primary metabolic problem because alanine-glyoxalate aminotransferase is expressed exclusively in the liver. However, in some patients who are responsive to pyridoxine, who present late with renal dysfunction, it might be considered.[1122] Isolated liver transplantation should be considered when GFR is greater than 40 mL/min/1.73 m^2.[1123] In patients with more severe renal dysfunction a combined kidney–liver transplantation is the treatment of choice.[1123] Unfortunately, administration of the oxalate degrading intestinal bacteria *Oxalobacter formigenes* has failed to demonstrate significant reduction in oxalate absorption and urinary excretion.

Hyperoxaluria Type 2

Primary hyperoxaluria type 2 is the result of mutations in the enzyme glyoxylate reductase/hydroxypyruvate reductase (GRHPR).[1124] This form of primary hyperoxaluria is typically less severe than type 1. It traditionally presents in adolescence with repeated calcium oxalate stone formation. A minority of patients develop chronic renal impairment. Diagnosis is suggested by the demonstration of significantly elevated urinary oxalate excretion and the presence of increased urinary L-glycerate. Definitive diagnosis is made via mutation analysis of the *GRHPR* gene, or traditionally by demonstration of reduced liver GRHPR enzyme activity. Pyridoxine is not beneficial. Liver transplantation has not been demonstrated to be beneficial in patients with type 2 primary hyperoxaluria. Patients with this disease may instead benefit from isolated renal transplantation.

Hyperoxaluria Type 3

Primary hyperoxaluria type 3 is the result of mutations of 4-hydroxy-2-oxoglutarate aldolase (*HOGA1* gene).[1125,1126] This enzyme is localized to the mitochondria in both the liver and the kidney and catalyzes 4-hydroxy-2-oxoglutarate to glyoxylate and pyruvate. Of the three subtypes, this is the most mild, clinically, and consists of recurrent calcium oxalate stone formation. Diagnosis is suspected by increased urinary oxalate excretion and can be confirmed by genetic analysis of the *HOGA1* gene. Fortunately, nephrocalcinosis and renal impairment are rarely seen.

PEDIATRIC ASPECTS OF CHRONIC KIDNEY DISEASE

GROWTH, NUTRITION, AND DEVELOPMENT

Childhood CKD impairs statural growth in a manner related to the age of CKD onset. Approximately 50% of children requiring RRT before their 13th birthday have a final height below the normal range.[1127] Results from the Chronic Kidney Disease in Children Study (CKiD, a US registry of children with a GFR <75 mL/min/1.73 m^2) demonstrated that height is below the normal range (median height standard deviation score [HtSDS] −0.55, fall in HtSDS of 0.12 to 0.16 for each decline of 10 mL/min/1.73 m^2) in children with predialysis CKD as well.[1128] One-third of all postnatal growth occurs during the first 2 years of life. Birth with CKD and inhibition of growth during this critical 2-year period can have major effects on the total amount of growth achieved.[1129] This particular age-related risk to final statural growth is illustrated by the 2006 North American Pediatric Transplant Cooperative Study (NAPRTCS) report, which demonstrated a mean HtSDS at under 2 years of age being −2.3 (increasing to −1.7, −1.4, and −1.0 at 2 to 6, 6–12, and >12 years, respectively).[1130] Between the second year of life and onset of puberty, growth inhibition is more subtle. Growth patterns typically remain stable along the normal percentiles as long as GFR remains greater than 25 mL/min/1.73 m^2 but gradually deviate from the normal range in CKD stages 4 and 5[1131] (Fig. 72.13). Even in children who appear to be growing normally successful kidney transplantation can result in accelerated or catch-up growth as metabolic homeostasis is restored, which suggests persistent suppression of growth potential in the uremic state.

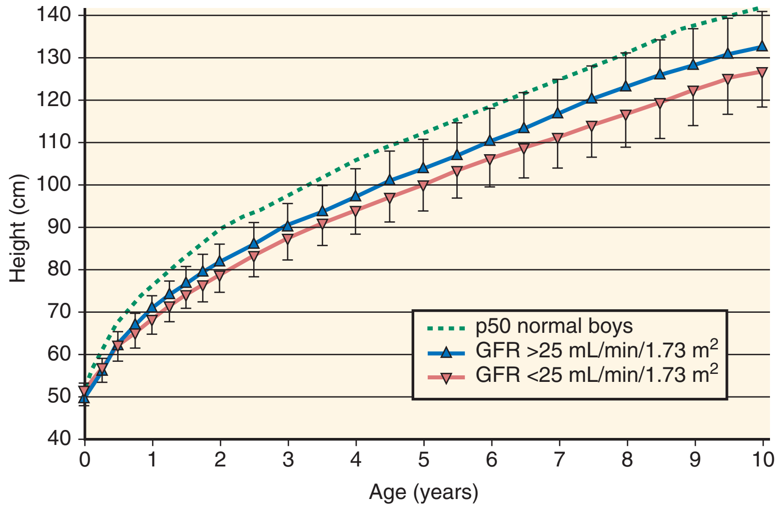

Fig. 72.13 Glomerular filtration rate (GFR)–dependent growth pattern in children with chronic renal failure due to hypodysplastic renal disorders. Approximately 100 children per age interval were evaluated. Mean ± 1 standard deviation of height is shown for children with an average GFR more or less than 25 mL/min/1.73 m^2. (Modified from Schaefer F, Wingen AM, Hennicke M, et al. Growth charts for prepubertal children with chronic renal failure due to congenital renal disorders. European Study Group for Nutritional Treatment of Chronic Renal Failure in Childhood. *Pediatr Nephrol.* 1996;10:288–293.)

Renal dysfunction can also delay the onset of puberty, as well as the start of the pubertal growth spurt, in a manner dependent on the degree of renal dysfunction.[1132] Pubertal height gain is diminished by up to 50% (from 30 to 15 cm) in adolescents with advanced stages of CKD. Final adult heights below the normal range are attained by 30% to 50% of children with CKD, although a trend of improving final heights has been noted during the past decade. The duration of CKD, presence of a congenital nephropathy, and male gender are among the most important predictors of a reduced final height.[1129]

Growth failure in infants with CKD usually is caused by inadequate spontaneous intake of nutrients due to anorexia and frequent vomiting. Numerous factors contribute to anorexia in CKD. These include accumulation of circulating satiety factors, retarded psychomotor development, slow gastric emptying, fluid and electrolyte losses due to tubular dysfunction in renal dysplasia, and catabolic episodes related to intercurrent infections. Metabolic acidosis, which usually occurs when GFR is below 50% of normal, contributes to CKD-associated growth failure by various mechanisms, including increased protein breakdown, suppressed secretion of GH and insulin-like growth factor-1 (IGF-1), as well as impaired GH and IGF-1 receptor expression in target tissues.[1129] Independent of metabolic acidosis, the endocrine systems show a complex dysregulation in the uremic state. Circulating GH levels are normal or increased due to impaired metabolic clearance, but the actual rate of GH secretion from the pituitary is diminished.[1133] GH-induced IGF-1 synthesis is impaired in uremia due to impaired activation of the GH-dependent JAK2/STAT5 (Janus kinase 2/signal transducer and activator of transcription-5) signaling pathway.[1134] This postreceptor signaling defect may be caused by upregulation of inhibitory suppressors of STAT signaling (SOCS [suppressor of cytokine signaling] proteins), which are induced by inflammatory cytokines in the uremic state. In addition, accumulation of IGF-1–binding proteins results in a molar excess of IGF-binding proteins relative to circulating IGFs, which results in reduced IGF bioactivity. Finally, normal or reduced GH secretion in the presence of markedly reduced IGF-1 bioactivity is consistent with an insufficient feedback activation of the somatotropic hormone axis at the hypothalamic and pituitary levels.

The gonadotropic hormone axis is subject to a similarly impaired activation in CKD.[1133] Peripubertal patients with advanced renal disease or ESRD exhibit normal or low sex steroid levels in the presence of elevated levels of circulating gonadotropins. The elevation of circulating gonadotropins is explained by impaired metabolic hormone clearance, whereas pituitary secretion rates are low, possibly due to impaired hypothalamic gonadotropin-releasing hormone (GnRH) secretion related to increased inhibitory neurotransmitter tone. In addition to deficient central nervous activation, there is accumulation of circulating factors that inhibit GnRH release from the hypothalamus and testosterone release from Leydig cells. Finally, crosstalk between gonadotropic and somatotropic hormones during puberty appears to be impaired, as evidenced by a blunted surge of GH secretion in response to rising sex steroid levels. Together, advanced stages of CKD cause a state of multiple endocrine resistance, effectively inhibiting longitudinal growth and sexual development.

PREVENTION AND TREATMENT OF GROWTH FAILURE IN CHRONIC KIDNEY DISEASE

In infants with CKD, the most important measures for avoiding growth failure are the provision of adequate energy intake, correction of metabolic acidosis, and maintenance of fluid and electrolyte balance.[1135] Supplementary feedings via nasogastric tube or gastrostomy are frequently required to achieve these targets. Caloric intake should be targeted to provide 80% to 100% of the regular daily allowance for healthy children. Increasing caloric intake above 100% of the recommended daily allowance does not induce further catch-up growth but

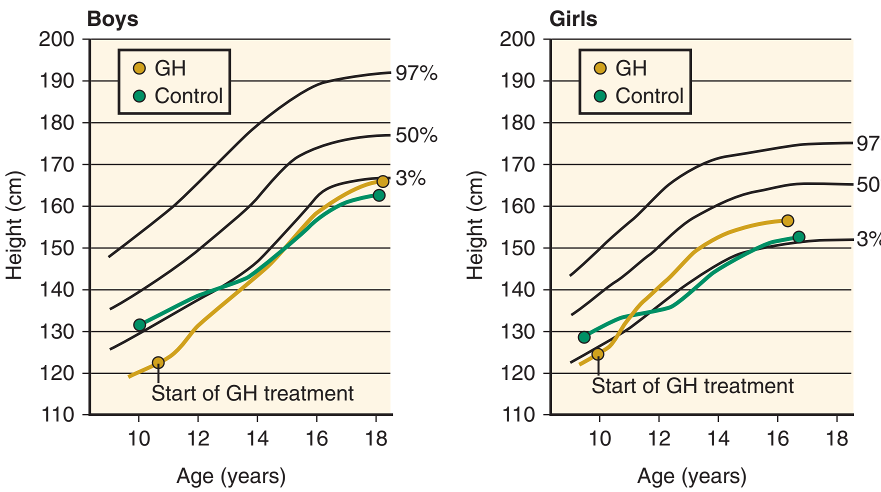

Fig. 72.14 Favorable effects of long-term recombinant human growth hormone (GH) therapy on final adult height in children with chronic kidney disease (CKD). Synchronized mean growth curves are shown for 38 children (32 boys and 6 girls) with CKD who were treated with GH and for 50 control children with CKD who did not receive GH. Normal values are indicated by the 3rd, 50th, and 97th percentiles. The *circles* indicate the time of the first observation (the start of GH treatment in the treated children) and the end of the pubertal growth spurt. (Haffner D, Schaefer F, Nissel R, et al. Effect of growth hormone treatment on the adult height of children with chronic renal failure. German Study Group for Growth Hormone Treatment in Chronic Renal Failure. *N Engl J Med.* 2000;343:923–930.)

rather results in obesity. Protein intake should be at least 100% of the dietary reference intake but should not exceed 140% of the dietary reference intake in patients with CKD stage 2 or 3 and 120% of the reference intake in children with CKD stage 4 or 5. Excessive protein intake should be avoided in advancing CKD to limit phosphorus and acid load in patients with failing kidney function.[1136] Metabolic acidosis should be rigorously treated by oral alkaline supplementation. In addition, the supplementation of water and electrolytes is essential for children with polyuria and/or salt-losing nephropathies, as observed in tubulopathies and renal dysplasia. Early and consistent provision of supplementary nutrients, fluids, and electrolytes, with delivery ensured by enteral feeding via a nasogastric tube or percutaneous gastrostomy whenever necessary, has markedly improved the growth and development of young children with CKD.[1129,1137] In the postinfantile phase of childhood, nutrition, fluid, and electrolyte balance continue to be permissive factors for adequate longitudinal growth, but catch-up growth can rarely be provided by dietary and supplementary measures alone.

If growth failure is imminent and/or has already occurred, despite adequate provision of nutrients, salts, and fluid, then treatment with recombinant GH (rGH) may be indicated. The efficacy and safety of rGH therapy in children with CKD have been demonstrated in numerous short- and long-term trials and has been comprehensively reviewed.[846] The administration of rGH at pharmacologic dosages (0.05 mg/kg/day subcutaneously) overcomes endogenous GH resistance and markedly increases systemic and local IGF-1 production, with only a slight effect on IGF-binding proteins. This restores normal IGF-1 bioactivity and stimulates longitudinal growth. In children with predialytic CKD, height velocity typically doubles in the first treatment year, and a steady albeit less marked catch-up growth is observed during subsequent years of therapy. Long-term studies have demonstrated mean standardized height increases of −2.6 to −0.7 SD in North American children, −3.4 to −1.9 SD in German children, and −3.0 to −0.5 SD in Dutch children after 5 to 6 treatment years.[1129] The therapeutic response to rGH in CKD patients is superior to that observed in children who are undergoing dialysis or who have undergone kidney transplantation, probably due to a more marked uremic GH resistance in ESRD and the growth-suppressive effect of glucocorticoids in renal allograft recipients. In patients receiving rGH studied at around 9 to 10 years of age and followed through puberty until attainment of final height, catch-up growth was largely limited to the prepubertal period (Fig. 72.14).[1138] Final height was markedly improved in comparison with an untreated control group, with an overall benefit attributable to rGH treatment of 10 to 15 cm. Total height gain was positively correlated with the duration of rGH therapy and was negatively affected by the time spent on dialysis. These studies support a conclusion that rGH therapy should be initiated as early as possible in the predialytic CKD period, preferentially before severe growth retardation has occurred.

CARDIOVASCULAR COMORBIDITY IN PEDIATRIC CHRONIC KIDNEY DISEASE

CKD-associated cardiovascular disease manifests relatively early in the course of CKD,[1139] leading to significant morbidity and mortality (Fig. 72.15).[1140] Among individuals younger than 19 years of age with ESRD, cardiac disease is responsible for 22 deaths/1000 patient-years, accounting for 16% of all deaths in whites and 26% in African Americans. This represents up to a 1000-fold increase in risk compared with that in the general population.

HYPERTENSION

The prevalence of hypertension is 40% to 50% in children with CKD stages 2 to 4 and approaches 70% by the time ESRD has developed. High blood pressure is associated with target organ damage during childhood and is closely related to the rate of progression of renal failure. CKD-associated

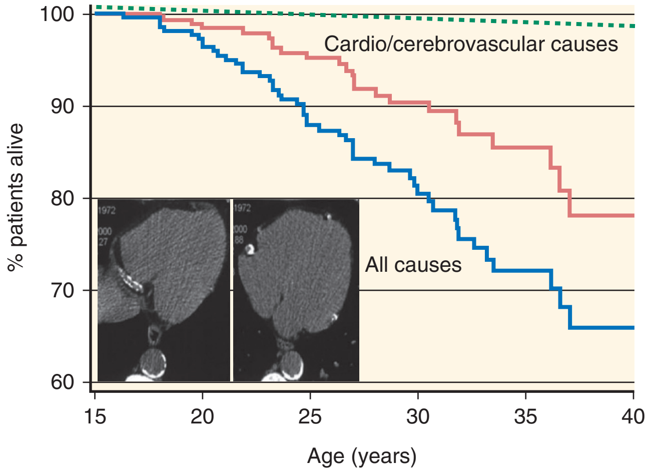

Fig. 72.15 Long-term survival of 283 patients with childhood-onset end-stage kidney disease. *Green dotted line,* Survival rate considering all causes of death. *Red line,* Survival rate considering cardiovascular and cerebrovascular causes of death only. *Blue line,* Survival rate in the general population. *Inset,* Computed tomography sections from a 27-year-old male hemodialysis patient with extensive calcification in all three coronary arteries and aorta. (From Oh J, Wunsch R, Turzer M, et al. Advanced coronary and carotid arteriopathy in young adults with childhood-onset chronic renal failure. *Circulation.* 2002;106:100–105.)

hypertension develops by a variety of pathophysiologic mechanisms including fluid overload, increased peripheral vascular resistance, activation of the RAAS, sympathetic hyperactivation, endothelial dysfunction, and chronic hyperparathyroidism.[1141]

The American Academy of Pediatrics Fourth Report on the Diagnosis, Evaluation, and Treatment of High Blood Pressure in Children and Adolescents defines arterial hypertension in the pediatric population.[1142] According to these guidelines, childhood hypertension is defined based on the distribution of blood pressure in the general population matched for age, gender, and height. Hypertension is defined as a systolic and/or diastolic blood pressure that on at least three occasions is greater than or equal to the 95th percentile for age, sex, and height. Values from the 90th to 95th percentile at any blood pressure level at or above 120/80 mm Hg are termed prehypertension.[1142] Altered circadian blood pressure patterns, as described in adult patients with CKD, can be detected by ambulatory blood pressure monitoring in children.[1143] Analysis of the CKiD cohort showed that up to 35% of children with CKD (eGFR 30–90 mL/min/1.73 m^2) have so-called masked hypertension – that is, elevated blood pressure with 24-hour monitoring despite normal office readings, a condition associated with an increased prevalence of left ventricular hypertrophy (LVH).[1144,1145] Various online calculators are available to compare the blood pressure of a child of a given gender, age, and height with that in the normal population (e.g., www.pediatriconcall.com/fordoctor/pedcalc/bp.aspx).

Current management guidelines recommend regular blood pressure screening for all children with CKD.[1146] In addition to clinic measurements, ambulatory blood pressure monitoring should be performed at least once a year and within 1 to 2 months of modification of therapy in patients receiving antihypertensive medication. Children with CKD should be considered at increased risk for cardiovascular complications, and therapeutic interventions should be initiated when blood pressure exceeds the 90th percentile for age and height. The therapeutic target for blood pressure in CKD varies according to different published guidelines.[1146] The American Academy of Pediatrics Fourth Report recommends a target blood pressure of less than 90%. Both the European Society of Hypertension and KDIGO recommend a target of less than 75% for nonproteinuric CKD and less than 50% for proteinuric CKD.[1146,1147] Lifestyle changes such as increased physical activity, avoidance of salty foods, and reduction of weight in obese patients are also recommended for children with CKD but are rarely effective as sole measures.[560,1141] ACE inhibitors and ARBs should be the first choice of antihypertensive medications (see the "Progression of Chronic Renal Failure in Children" section) and, if given at appropriate dosages (e.g., ramipril 6 mg/m^2/day[565,1148] or candesartan 0.2–0.4 mg/kg/day[1149]), will normalize blood pressure in most patients. If the blood pressure–lowering effect is insufficient, a loop diuretic should be added (e.g., furosemide, 2–4 mg/kg/day), followed by a calcium channel antagonist (e.g., amlodipine, 0.2 mg/kg/day). The timing of drug dosing should be adapted to achieve optimal blood pressure control throughout 24 hours as verified by ambulatory blood pressure monitoring.

ANEMIA

Anemia is a common and serious complication in both adult and pediatric patients with CKD. Severe anemia is associated with increased morbidity and mortality and development of cardiovascular disease in patients with CKD. In children, growth, neurocognitive development, school attendance, and exercise capacity are important clinical outcomes related to anemia.

Erythropoiesis-stimulating agents (ESAs) are central to the treatment of anemia. While the efficacy of ESAs is clear in childhood CKD,[1150] the desired therapeutic target of blood hemoglobin has not been defined in children via randomized trials. Studies in adults with CKD have raised important concerns regarding an increased cardiovascular morbidity and mortality in patients with hemoglobin greater than 120 g/L.[1151] A recent study[1152] provides new insights into the association of hemoglobin levels with cardiovascular morbidity in pediatric patients receiving hemodialysis. Using data from the Centers for Medicare and Medicaid Services ESRD Clinical Performance Measure project and United States Renal Data System, the authors studied 1569 children receiving hemodialysis. Examination of outcomes including mortality, hospitalization, and cardiovascular event over 1 year follow-up showed that the hazard ratio of all-cause mortality and the adjusted relative rate of all-cause hospitalization were significantly lower in the hemoglobin 12 g/dL or greater group, and cardiovascular hospitalizations were significantly higher in the hemoglobin under 10 g/dL group. Inpatient and outpatient visits for congestive heart failure, cardiomyopathy, and valvular heart disease were most common in the hemoglobin under 10 g/dL group, and the frequency of these diagnoses decreased with increasing hemoglobin level.

INTERMEDIATE CARDIOVASCULAR ENDPOINTS IN CHILDREN WITH CHRONIC KIDNEY DISEASE

LVH and increased carotid intima-media thickness have been demonstrated in hypertensive children with advanced CKD, in whom alterations of mineral metabolism and volume

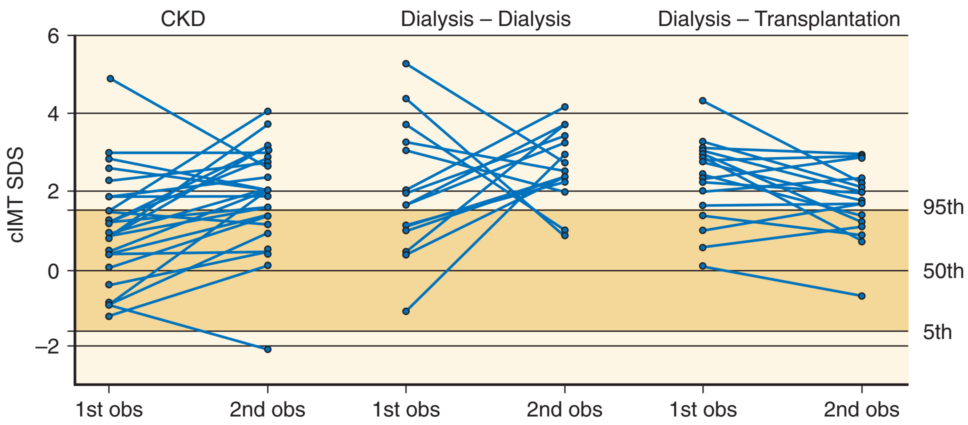

Fig. 72.16 Change in sonographically determined carotid intima-media thickness (cIMT) in children with chronic kidney disease, children undergoing dialysis, and children who underwent kidney transplantation. Mean time interval between the first and second observations *(obs)* was 12 months. IMT is expressed as a standard deviation (SD) score normalized for age and gender. *Shaded area* denotes normal range, with SDs of –1.64, 0, and +1.64 corresponding to the 5th, 50th, and 95th percentiles. *cIMT SDS,* Change in intima media thickness standard deviation score. (From Litwin M, Wühl E, Jourdan C, et al. Evolution of large-vessel arteriopathy in pediatric patients with chronic kidney disease. *Nephrol Dial Transplant.* 2008;23:2552–2557.)

overload are important superimposed risk factors, but also in children with early essential hypertension and in children with masked hypertension. LVH is the most common identifiable cardiac alteration in CKD and the most important indicator of cardiovascular risk in adult and pediatric patients with ESRD.[1153–1156] LVH is believed to contribute to the high risk of sudden cardiac death in children with CKD due to lethal arrhythmias via underlying myocardial fibrosis and cellular hypertrophy. LVH is present in approximately 15% to 30% of children with stage 2 to 4 CKD,[1157,1158] in 50% to 75% of children undergoing dialysis,[1159,1160] and in 50% to 67% of renal allograft recipients.[1161,1162] Factors that promote LVH could include increased circulating volume, hyperactivation of the RAAS and sympathetic nervous system, hyperparathyroidism, and proinflammatory cytokines. In addition to the morphologic change of LVH, a subclinical impairment of left ventricular systolic function is found in about 25% of children with mild to moderate CKD.[1163] Systolic dysfunction is most common in patients with concentric LVH and is associated with a low GFR and anemia.

Increasing evidence in children, as in adults, has suggested that the CKD-associated bone mineral disorder and its treatment not only affects bone and mineral metabolism but also contributes to the development of calcifying uremic vasculopathy. This is a consequence of a redistribution of mineral salts from the skeleton to the large arteries and soft tissue compartments. Coronary artery calcifications are found in individual adolescent patients undergoing dialysis[1164] and in more than 90% of young adults with childhood-onset CKD.[1140] Early signs of vasculopathy, such as an increased intima-media thickness and stiffness of the carotid artery, can be detected as early as the second decade of life.[1165] Both morphologic and functional alterations are progressive over time and are most marked in adolescents undergoing dialysis, but they also may be seen in children with moderate CKD and appear to improve partially after successful renal transplantation (Fig. 72.16). Intima-media thickness and stiffness of the carotid artery correlate with the degree of hyperparathyroidism, serum calcium–phosphorus ion product, and cumulative dose of calcium-containing phosphate binders.

PROGRESSION OF CHRONIC RENAL FAILURE IN CHILDREN

COURSE OF RENAL FUNCTION IN CHILDREN WITH CHRONIC KIDNEY DISEASE

The course of renal function in pediatric CKD is mainly determined by the age and degree of renal failure at the time of first disease manifestation. Two-thirds of children with CAKUT progress to ESRD during the first 2 decades of life.[6] The physiologic early increase in GFR is typically extended to the first 3 to 4 years of life, which potentially reflects an adaptive hypertrophy of the reduced number of functioning nephrons.[173] In approximately 50% of children, this early incremental phase is followed by a period of stable or very slowly deteriorating renal function, which usually lasts for 5 to 8 years. Around the onset of puberty, the GFR loss tends to accelerate, typically leading to ESRD in late adolescence or early adulthood (Fig. 72.17). The reasons for this nonlinear progression of renal failure are incompletely understood and may include an insufficient capacity of the nephrons to adapt to the rapidly growing metabolic needs during the pubertal growth spurt, adverse renal effects of puberty-related increases in sex steroid production, and/or an accelerated sclerotic degeneration of a diminishing number of remnant hyperfiltering nephrons. In children with very severe renal hypoplasia, the early increase in GFR may be blunted, and early progression to ESRD may occur instead. About 20% of patients with renal hypoplasia maintain a stable GFR, even beyond puberty. Follow-up studies have shown that in patients with mild bilateral renal hypoplasia, progressive deterioration of GFR frequently ensues in the third decade of life.

RISK FACTORS FOR PROGRESSION OF RENAL FAILURE

Among the numerous factors predicting progressive renal failure in adult populations and in animal models, hypertension and proteinuria have also been identified as consistent, modifiable risk factors in children. In the European Study

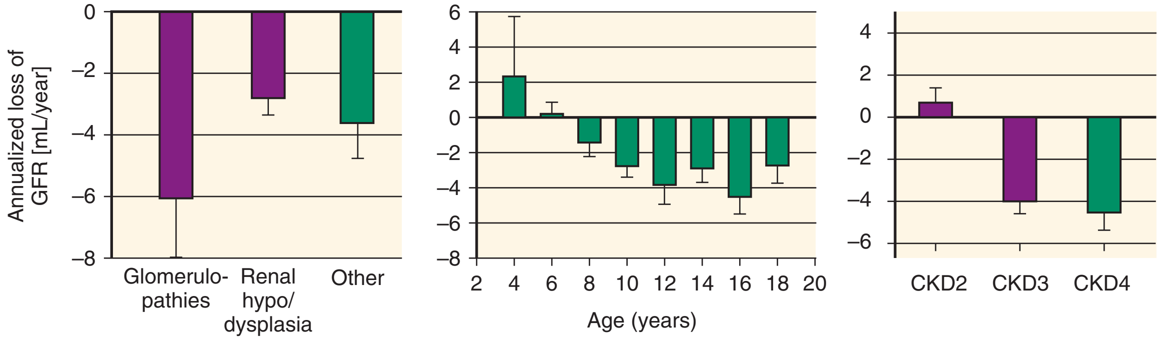

Fig. 72.17 Mean spontaneous change in estimated glomerular filtration rate (GFR) in 385 children with chronic kidney disease (CKD) randomly assigned to treatment conditions in the ESCAPE trial (measured during the run-in period). The rate of renal failure progression in children depends on underlying renal disease, age, and CKD stage.

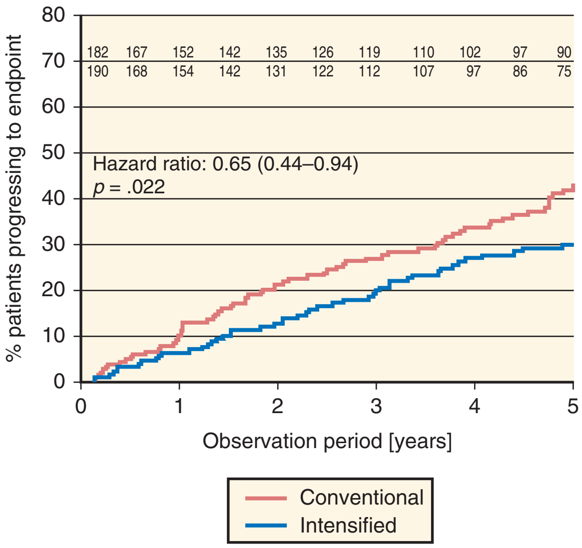

Fig. 72.18 Effect of intensified blood pressure control on renal survival in children with chronic kidney disease. Within 5 years, 30% of patients randomly assigned to intensive control with a low-normal, 24-hour blood pressure target reached the endpoint of 50% glomerular filtration rate loss or end-stage kidney disease, compared with 42% of patients assigned to standard control with a conventional blood pressure target. (Data from Wühl E, Trivelli A, Matteucci MC, et al. Strict blood pressure control and progression of renal failure in children. *N Engl J Med.* 2009;361:1639–1650.)

for Nutritional Treatment of Chronic Renal Failure in Childhood, a systolic blood pressure greater than120 mm Hg was associated with a significantly faster decline of GFR.[1136] The randomized, prospective ESCAPE trial (Effect of Strict Blood Pressure Control and ACE Inhibition on the Progression of CRF in Pediatric Patients) demonstrated that in children with CKD due to various disorders, intensified antihypertensive treatment forcing blood pressure into the low-normal range (<50th percentile for age) was associated with improved long-term renal survival (Fig. 72.18).[208] Proteinuria is predictive of CKD progression in children, as it is in adult populations. Proteinuria predicts renal disease progression in children with renal hypodysplasia and, even in children with normal kidney function, persistent nephrotic-range proteinuria is a risk factor for progressive renal injury. In the ESCAPE trial, for children receiving fixed-dose ACE inhibitors, residual proteinuria proved to be predictive of CKD progression, despite treatment. These findings provide a strong rationale for the early and consistent application of RAAS antagonist treatment protocols.[208] Both ACE inhibitors and ARBs have been shown to be safe and effective in children with CKD. Ramipril, administered at a dosage of 6 mg/m^2/day, normalized blood pressure and lowered proteinuria by approximately 50% in the ESCAPE trial patients.[1148] Similar results have been obtained with the ARBs losartan,[1166,1167] valsartan,[1168] and candesartan.[1149] It should be noted, however, that the renoprotective superiority of RAAS antagonists over other antihypertensive agents has not been formally demonstrated in pediatric CKD. Data from the ItalKid registry did not show significant modification of CKD progression by ACE inhibitor treatment in children with hypodysplastic kidney disease compared with matched untreated cases.[1169] However, no information regarding the type and dosages of ACE inhibitors used and prevailing degree of proteinuria was available, and the baseline progression rate was very slow. Another important observation in the ESCAPE trial was a gradual return of proteinuria, despite ongoing ACE inhibitor treatment. This effect was dissociated from persistently excellent blood pressure control and might limit the long-term renoprotective efficacy of ACE inhibitor monotherapy in pediatric CKD.[208]

Complete reference list available at ExpertConsult.com.

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BOARD REVIEW QUESTIONS

1. The following are features characteristic of renal dysplasia:
   a. Abnormal tissue patterning
   b. Small size for age
   c. Epithelial cysts
   d. Metaplastic tissue
   e. All of the above

   Answer: e

   Rationale: While these features are not observed in every dysplastic kidney, they are characteristic features across the spectrum of dysplastic kidneys.

2. A 9-year-old boy presents with a 3-day history of nasal congestion without fever and a 1-day history of brown urine, but no dysuria. Physical exam is unremarkable. Urinalysis reveals 3+ blood, 3+ protein, trace leukocyte esterase, and no nitrate. Serum studies are notable for normal electrolytes, creatinine 0.6 mg/dL, albumin 2.7 g/dL, and normal C3 level. Renal ultrasound reveals no structural abnormalities or stones, and normal sized kidneys for age. A spot urine protein/creatinine ratio is 2.8. A urine culture confirms no growth.

   Which of the following is the most appropriate next step in the treatment of this child's condition?
   a. Initiate ACE inhibitor treatment
   b. Observe with follow-up clinic visit in 8 weeks
   c. Initiate daily oral prednisone at 2 mg/kg/day + ACE inhibitor treatment
   d. Arrange for renal biopsy
   e. Initiate IV solumedrol at 20 mg/kg daily × 3 days, then oral prednisone at 2 mg/kg/day

   Answer: d

   Rationale: The clinical presentation of an apparent upper respiratory infection with the concurrent appearance of gross hematuria and nephrotic-range proteinuria is most likely due to IgA nephropathy. While presentation with overt nephrotic syndrome would warrant initiation of IV solumedrol, and only modest proteinuria (protein/creatinine ratio <0.5) would warrant only ACE inhibitor treatment, in this setting of nephrotic-range proteinuria without nephrotic syndrome most pediatric nephrologists would arrange for a renal biopsy (typically in conjunction with initiating oral prednisone) to exclude the possibility of a significant percentage of glomerular crescents that would warrant more intensive treatment with IV solumedrol.

3. A 9-year-old boy with a history of nephrotic syndrome since 3 years of age presents with a recent relapse. He initially presented with steroid dependent nephrotic syndrome, but after multiple relapses underwent a renal biopsy which confirmed minimal change nephrotic syndrome 3 years ago. He has been taking prednisone 60 mg PO daily for the last 8 weeks. His exam is notable for BP 147/85, moderate pedal edema, and moderate Cushingoid facies. Serum studies reveal normal electrolytes, albumin 1.3 g/dL, BUN 26 mg/dL, and creatinine 0.7 mg/dL. A spot urine protein/creatinine ratio is 8.7.

   Which of the following is the most appropriate diagnosis and next step in the treatment of this child's condition?
   a. Steroid-dependent nephrotic syndrome; initiation of cyclosporine A
   b. Steroid-resistant nephrotic syndrome; initiation of tacrolimus
   c. Steroid-dependent nephrotic syndrome; initiation of mycophenolate
   d. Steroid-resistant nephrotic syndrome; initiation of cyclophosphamide
   e. Steroid-resistant nephrotic syndrome; initiation of IV solumedrol

   Answer: b

   Rationale: The clinical presentation of a child with relapse of nephrotic syndrome that fails to achieve complete remission after 8 weeks of daily oral steroids meets the definition for steroid resistant nephrotic syndrome. Although not rigorously proven, the majority of available evidence supports the initiation of calcineurin inhibitors (cyclosporin A or tacrolimus) as the best initial alternative treatment for steroid resistant nephrotic syndrome.

4. The most common genetic cause of Fanconi syndrome in childhood is:
   a. Dent disease
   b. Oculocerebral syndrome of Lowe
   c. Cystinosis
   d. Wilson's disease

   Answer: c

   Rationale: Whereas chemotherapy more commonly causes Fanconi syndrome in childhood, the most common genetic ethology is due to cystinosis, which itself is due to a mutation in the cystinosin gene.

5. The most common risk factor for kidney stone formation in children is:
   a. Hyperuricosuria
   b. Hypocitrituria
   c. Hypercalciuria
   d. Hyperoxaluria

   Answer: c

   Rationale: Although children are more likely to have genetic causes for their disease, like adults the greatest risk factor for kidney stone formation is hypercalciuria and this is most likely idiopathic in nature.

6. Which of the following statements about the pathogenesis of aHUS is correct?
   a. Today, genetic causes (i.e., complement protein mutations) can be identified in more than half of aHUS patients.
   b. Disease causing mutations exclusively affect complement regulators.
   c. MCP/CD46 mutations are the most frequent aHUS-causing mutations.
   d. A subgroup of aHUS patients carries two or more disease causing mutations, a finding without impact on disease manifestation and/or disease severity.
   e. 10% to 15% of aHUS patients carry anti CFH-autoantibodies, which typically recognize N-terminal epitopes and, thus, affect CFH cofactor function.

Answer: a

Rationale: aHUS is caused by dysregulation of the complement system—mainly it's alternative pathway—resulting in exceeding complement activation on the surface of vascular endothelial cells. Complement dysregulation can be caused by genetic mutations in complement regulators (i.e., CFH, CFI, MCP/CD46, THBD/CD141) or activators (i.e., C3, CFB). Today, genetic causes can be identified in more than half of aHUS patients with CFH mutations being the most frequent mutation (20% to 30%). A subgroup of aHUS patients carries more than one disease causing mutation, which correlates with an earlier disease onset and a more severe disease phenotype. 10% to 15% of aHUS patients carry anti CFH-autoantibodies. These antibodies typically recognize the CFH C-terminus, which—amongst others—is responsible for CFH surface binding.

7. A 6-year-old girl presents with nephrotic range proteinuria, mildly elevated creatinine, significantly decreased C3-levels, and moderate hypertension. The patient's history is remarkable for an upper airway infection with sore throat less than 1 month prior to presentation. Which of the following statements is incorrect?
   a. The patient likely suffers PIGN, and no further diagnostic or therapeutic efforts are required as full spontaneous recovery can be expected.
   b. The patient warrants a kidney biopsy.
   c. Both PIGN and C3G/IC-MPGN are possible diagnoses.
   d. PIGN and C3G/IC-MPGN form a disease spectrum with shared underlying pathomechanism, but different mid- and long-term outcome.
   e. Complement dysregulation is central to PIGN and C3G/IC-MPGN pathogenesis, and disease-causing mutations or autoantibodies have been identified in both conditions.

Answer: a

Rationale: While in the past considered different entities, today PIGN and C3G/IC-MPGN are appreciated members of a disease spectrum. Both conditions are characterized by decreased C3 levels and can present with proteinuria (up to nephrotic range), impaired kidney function and elevated blood pressure. While PIGN patients typically present subsequent (1 to 3 weeks) to a streptococcal infection, C3G/IC-MPGN often manifests subsequent to an infectious trigger such as a viral respiratory tract infection. The pathogenesis of both diagnoses involves the complement alternative pathway, and in both conditions complement mutations and/or autoantibodies can be found. PIGN typically takes a benign course with full recovery after 4 to 6 weeks, C3G/IC-MPGN defines a chronic disease with progression to ESRD within 10 to 15 years. Besides clinical presentation, PIGN and C3G/IC-MPGN can typically be differentiated on kidney biopsy.